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**SPECIAL ISSUE:
DESIGN-BASED RESEARCH: CLARIFYING THE TERMS**

Sasha Barab

Guest Editor

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Design-Based Research: Putting a Stake in the Ground

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The emerging field of the *learning sciences* is one that is interdisciplinary, drawing on multiple theoretical perspectives and research paradigms so as to build understandings of the nature and conditions of learning, cognition, and development. Learning sciences researchers investigate cognition in context, at times emphasizing one more than the other but with the broad goal of developing evidence-based claims derived from both laboratory-based and naturalistic investigations that result in knowledge about how people learn. This work can involve the development of technological tools, curriculum, and especially theory that can be used to understand and support learning. A fundamental assumption of many learning scientists is that cognition is not a thing located within the individual thinker but is a process that is distributed across the knower, the environment in which knowing occurs, and the activity in which the learner participates. In other words, learning, cognition, knowing, and context are irreducibly co-constituted and cannot be treated as isolated entities or processes.

If one believes that *context matters* in terms of learning and cognition, research paradigms that simply examine these processes as isolated variables within laboratory or other impoverished contexts of participation will necessarily lead to an incomplete understanding of their relevance in more naturalistic settings (Brown, 1992).¹ Alternatively, simply observing learning and cognition as they naturally

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¹This special issue is dedicated to the memory and intellectual contributions of Ann Brown, who so clearly led the way in illuminating for the field the challenges and opportunities discussed in this issue.

occur in the world is not adequate given that learning scientists frequently have transformative agendas. Education is an applied field, and learning scientists bring agendas to their work, seeking to produce specific results such as engaging students in the making of science, creating online communities for professional development, or creating history classrooms that confront students preexisting beliefs about race, gender, or class. As such, learning scientists have found that they must develop technological tools, curriculum, and especially theories that help them systematically understand and predict how learning occurs. Such design research offers several benefits: research results that consider the role of social context and have better potential for influencing educational practice, tangible products, and programs that can be adopted elsewhere; and research results that are validated through the consequences of their use, providing consequential evidence or validity (Messick, 1992). However, participating in local educational practices places researchers in the role of curriculum designers, and implicitly, curriculum theorists who are directly positioned in social and political contexts of educational practice (both global and local) and who are accountable for the social and political consequences of their research programs.

Increasingly, learning scientists are finding themselves developing contexts, frameworks, tools and pedagogical models consistent with and to better understand emerging pedagogical theories or ontological commitments (see diSessa & Cobb, this issue). In these contexts, the research moves beyond simply observing and actually involves systematically engineering these contexts in ways that allow us to improve and generate evidence-based claims about learning. The commitment to examining learning in naturalistic contexts, many of which are designed and systematically changed by the researcher, necessitates the development of a methodological toolkit for deriving evidence-based claims from these contexts. One such methodology that has grown in application is that of design experimentation or design-based research, frequently traced back to the work of Ann Brown (1992) and Alan Collins (1992).

Design-based research is not so much *an* approach as it is a series of approaches, with the intent of producing new theories, artifacts, and practices that account for and potentially impact learning and teaching in naturalistic settings. Cobb, diSessa, Lehrer, & Schauble (2003) stated:

Prototypically, design experiments entail both “engineering” particular forms of learning and systematically studying those forms of learning within the context defined by the means of supporting them. This designed context is subject to test and revision, and the successive iterations that result play a role similar to that of systematic variation in experiment. (p. 9)

They further suggested that design-based research has a number of common features, including the fact that they result in the production of theories on learning

and teaching, are interventionist (involving some sort of design), take place in naturalistic contexts, and are iterative. Design-based research is not simply a type of formative evaluation that allows learning scientists to better understand the ecological validity of theoretical claims generated in the laboratory. Design-based research, as conceived by Ann Brown (1992), was introduced with the expectation that researchers would systemically adjust various aspects of the designed context so that each adjustment served as a type of experimentation that allowed the researchers to test and generate theory in naturalistic contexts.

Although design-based research has the potential to offer a useful methodological toolkit to those researchers committed to understanding variables within naturalistic contexts, there are many unresolved questions that we as a community must address if our assertions are going to be deemed credible and trustworthy to others. Some questions are: What are the core foci of design-based research and what delineates it from other forms of research? What counts as reasonable and useful warrants for advancing assertions investigated through this type of research? What are the boundaries of a naturalistic context? How do we control researcher bias in selecting evidence, in reporting observations, and in developing trustworthy claims? How do we understand the contextuality of research claims generated *in situ* and use them to inform broader practice? In the following, we begin the process of responding to these questions, a process that is taken up in greater detail through the core articles and commentaries that comprise this special issue and that we hope will be taken up over the next decade by our colleagues.

CHARACTERIZING DESIGN-BASED RESEARCH

In understanding the need for new methods, it is important that one clarifies the distinction between existing methods for understanding learning and cognition and those central to design-based research. Collins, Joseph, and Bielaczyc (this issue) undertake this issue in their manuscript, contrasting several different methodologies with design-based research. They posit seven major differences between traditional psychological methods and the design-experiment methodology (see Table 1 for an abbreviated list). Central to this distinction is that design-based research focuses on understanding the messiness of real-world practice, with context being a core part of the story and not an extraneous variable to be trivialized. Further, design-based research involves flexible design revision, multiple dependent variables, and capturing social interaction. In addition, participants are not “subjects” assigned to treatments but instead are treated as co-participants in both the design and even the analysis. Last, given the focus on characterizing situations (as opposed to controlling variables), the focus of design-based research may be on

TABLE 1
Comparing Psychological Experimentation and Design-Based Research
Methods

Category	Psychological Experimentation	Design-Based Research
Location of research	Conducted in laboratory settings	Occurs in the buzzing, blooming confusion of real-life settings where most learning actually occurs
Complexity of variables	Frequently involves a single or a couple of dependent variables	Involves multiple dependent variables, including climate variables (e.g., collaboration among learners, available resources), outcome variables (e.g., learning of content, transfer), and system variables (e.g., dissemination, sustainability)
Focus of research	Focuses on identifying a few variables and holding them constant	Focuses on characterizing the situation in all its complexity, much of which is not now <i>a priori</i>
Unfolding of procedures	Uses fixed procedures	Involves flexible design revision in which there is a tentative initial set that are revised depending on their success in practice
Amount of social interaction	Isolates learners to control interaction	Frequently involves complex social interactions with participants sharing ideas, distracting each other, and so on
Characterizing the findings	Focuses on testing hypothesis	Involves looking at multiple aspects of the design and developing a profile that characterizes the design in practice
Role of participants	Treats participants as subjects	Involves different participants in the design so as to bring their differing expertise into producing and analyzing the design

Note. Adapted from Collins (1999).

developing a profile or theory that characterizes the design in practice (as opposed to simply testing hypotheses).²

One challenging component of doing educational research on design-based interventions is to characterize the complexity, fragility, messiness, and eventual solidity of the design and doing so in a way that will be valuable to others. This latter criterion implies that design-based research requires more than understanding the

²It is important to note that this contrast is not meant to deride the importance of traditional psychological methods. In fact, it is our belief that design-based research and other methods should be viewed as complementary and supportive—allowing researchers to understand more completely their claims. For example, laboratory-based researchers should ask themselves how their laboratory-based claims would benefit from further testing in naturalistic contexts and design-based researchers should be asking how their claims would benefit from more rigorous testing within laboratory-based contexts.

happenings of one particular context, but also requires showing the relevance of the findings derived from the context of intervention to other contexts. Stake (1995) referred to this type of generalization as a *petite generalization*. The anthropologist Clifford Geertz (1976, 1983) discussed the importance of work having both experience-near significance and experience-distant relevance, stating:

Confinement to experience-near concepts leaves the ethnographer awash in immediacies, as well as entangled in vernacular. Confinement to experience-distant ones leaves him stranded in abstractions and smothered in jargon. (p. 57)

Translating this perspective to design-based research, the validation of a particular design framework is not simply intended to show the value of a particular curriculum. Instead, design-based research strives to generate and advance a particular set of theoretical constructs that transcends the environmental particulars of the contexts in which they were generated, selected, or refined. This focus on advancing theory grounded in naturalistic contexts sets design-based research apart from laboratory experiments or evaluation research.

At first glance, design-based research closely resembles formative evaluation methodologies. Both are naturalistic, process-oriented, iterative, and involve creating a tangible design that works in complex social settings. The process of conducting a formative evaluation—articulating goals, operationalizing measures, examining a phenomena and understanding the consequences of its use (both intended and unintended)—is quite similar to many design-based research studies. Indeed, design-based research has been justifiably criticized for being little more than formative evaluation, even ignoring some of the recent advances made in formative evaluation. What separates design-based research in the learning sciences from formative evaluation is (a) a constant impulse toward connecting design interventions with existing theory, (b) the fact that design-based research may generate new theories (not simply testing existing theories), and (c) that for some research questions the context in which the design-based research is being carried out is the *minimal ontology* for which the variables can be adequately investigated (implying that we cannot return to the laboratory to further test the theoretical claims).

Fundamentally, formative evaluation methodologies (or even instructional design models) are about improving the value of a particular designed artifact (or a process for carrying out design), whereas design-based research is concerned with using design in the service of developing broad models of how humans think, know, act and learn; that is, a critical component of design-based research is that the design is conceived not just to meet local needs, but to advance a theoretical agenda, to uncover, explore, and confirm theoretical relationships. Although providing credible evidence for local gains as a result of a particular design may be *necessary*, it is not *sufficient*. *Design-based research requires more than simply*

showing a particular design works but demands that the researcher (move beyond a particular design exemplar to) generate evidence-based claims about learning that address contemporary theoretical issues and further the theoretical knowledge of the field.

We further illuminate this point by looking at our own work as an example. In Barab's work over the last 5 years, he and his colleagues have worked to develop various designs to support the learning of interdisciplinary content (Barab & Landa, 1997; Barab, 1999), the learning of science concepts (Barab, Hay, Barnett, & Keating, 2000), an appreciation for the practices of scientists (Barab & Hay, 2001), the emergence of web-supported communities to support teacher professional development (Barab, MaKinster, Moore, Cunningham, & the ILF Design Team, 2001), and multi-user virtual environments to support children ages 9–13 in developing their own sense of purpose as individuals, as members of their communities, and as knowledgeable citizens of the world (Barab, Thomas, Dodge, Carteaux, & Tuzun, in press). Whereas this joint commitment to research and service meant meeting the immediate needs of participants, there also existed a complementary goal of producing theory that would have relevance for researchers not involved in the initial design research (see Table 2). Table 2 reveals both the local impact of the work as well as the resultant theoretical contributions, with our argument being that design-based research requires providing local warrants for the effectiveness of the design work while simultaneously attempting to contribute to a larger body of theory. As design-based researchers, we have found it both critical and challenging to continually make both types of arguments, arguments that have both experience-near significance and experience-distant relevance. For example, in Barab's Quest Atlantis project, documenting the learning gains was a necessary part of the research agenda because it provided warrants for the broader theoretical assertions about the relationship among the constructs of playing, helping, and learning, the importance of context in learning, and the value of socially responsive design (Barab et al., in press).

In contrast to other methods focused on producing theory, the most radical shift proposed by design researchers may be the requirement that inquiry involves producing demonstrable changes at the local level. Design-based researchers not only recognize the importance of local contexts but also treat changes in these contexts as necessary evidence for the viability of a theory. Design-based research that advances theory but does not demonstrate the value of the design in creating an impact on learning in the local context of study has not adequately justified the value of the theory. As such, design-based research suggests a pragmatic philosophical underpinning, one in which the value of a theory lies in its ability to produce changes in the world. Such a system of inquiry might draw less from traditional positivist science or ethnographic traditions of inquiry, and more from pragmatic lines of inquiry where theories are judged not by their claims to truth, but by their ability to do work in the world (Dewey, 1938). Researchers looking to add method-

TABLE 2
Characterizing Design-Based Research in Terms of the Designed Artifact
and Resultant Theory

<i>Research Study</i>	<i>Local Impact</i>	<i>Theoretical Work</i>
Hartford Middle School project	Teacher professional development, produced curricular units, student learning and attitudinal gains	Interdisciplinary anchors as useful construct for teacher conceptualization and development of interdisciplinary units
Student apprenticeship camp	Pre–post learning gains, student attitudinal changes, positive attitude of scientists	Richer understanding of apprenticeship-type learning Characterization of differences between <i>constructivist</i> and <i>apprenticeship</i> learning environments
Virtual solar system project	Student learning gains, student collaboration	Claims about project-based learning and cognition as situated, particularly relations among learners’ intentions, tools, and meaning making
Inquiry learning forum project	Participation rates, learning gains, control–experimental group attitudinal differences, interview testimonials	Richer understanding of the challenges of designing for Web-supported communities in the service of learning; Claims about the tenability of designing community
Quest Atlantis project	Pre–post learning gains, control–experimental group quality of work and attitudinal differences; interview testimonials; participation rates; chat and classroom observations	Demonstrating the interrelations of learning, playing, and helping; Richer understanding of the symbolic relations of power and play in multiuser virtual environments; Potential value of socially responsive design

ological rigor and coherent epistemological systems to design-based methodologies might benefit from further grounding in the pragmatic philosophies of Dewey and Peirce, both of whom have provided systems of inquiry rooted *not* in claims of truth, but rather in the viability of theories to explain phenomena and produce change in the world.

ADVANCING CREDIBLE ASSERTIONS

An important question that must be answered by design-based researchers is what counts as credible evidence. Schoenfeld (1992) argued that a sound methodological argument in the social sciences should touch on issues of trustworthiness, credibility, and usefulness as well as the range of contexts in which the researcher believes the assertions should extend. Whereas the first two criteria (trustworthiness and

credibility) are akin to reliability and validity but do not necessarily require the use of objective and quantitative methods for demonstrating they have been met, and the last criterion (usefulness) is somewhat akin to generalizability and external validity, we view the term usefulness as less commonly invoked when determining the strength of a researcher's claims (Dewey, 1938; Messick, 1992). It is one thing to demonstrate learning gains or show that statistical differences have been achieved; it is quite another thing to demonstrate the usefulness or consequentiality of the work. With respect to design-based research, and learning sciences more generally, this consequentiality is an essential criterion for determining the significance of a particular study. Our goal, as applied researchers engaged in doing design work, is to directly impact practice while advancing theory that will be of use to others.

The emphasis on understanding the value of a theory through its consequences on naturalistic systems also borrows from Messick's (1992) notion of evidence of consequential validity for testing. His argument is that the validity of a claim is based on the changes it produces in a given system. These changes or consequences can then be considered evidence in support of validity. Messick's original formulation of consequential validity argues that inquiry is a social enterprise and evidence for the validity of an assertion can be gathered by examining the effects of that assertion on a system; a classic violation of this principle is when standardized tests result in undesirable practices in schools and routinized, shallow learning, suggesting that perhaps standardized tests are a poor instrument for generating assertions about student achievement (Linn, 1998). Design-based research offers a mode of inquiry that embraces this notion of consequential validity, but design researchers need to be clearer about the kinds of claims they make from design experiments and the limitations of their findings. At the expense of being redundant in our argument, we believe that while demonstrating local consequence and utility is necessary it is not sufficient—*design scientists must draw connections to theoretical assertions and claims that transcend the local context.*

One of the central ideas in the scientific paradigm is replicability; however, because design-based researchers cannot (and may not want to) manipulate cultural contexts, it becomes difficult to replicate others' findings (Hoadley, 2002). Therefore, the goal of design-based research is to lay open and problematize the completed design and resultant implementation in a way that provides insight into the local dynamics. This involves not simply sharing the designed artifact, but providing rich descriptions of context, guiding and emerging theory, design features of the intervention, and the impact of these features on participation and learning. Narrative, as one way of making sense of design-based research, is a historical method that involves conveying a series of related plots and describing the temporal unfolding of the design over time (Abbott, 1992; Mink, Fay, Golob, & Vann, 1987). A core challenge in building narrative is what historiographers refer to as the "central subject problem" in which the boundaries of the case itself are delimited (Hull, 1975). Although in some instances the case may have clear boundaries,

more often than not a crucial difficulty lies in “drawing boundaries around the central subject given the continuous character of the social manifold” (Abbott, 1992, p. 63). These boundaries are always fuzzy, with the properties of the cases and the design work passing through multiple transformations over time.

It is the unpacking of these transformations, describing what the case endures, and relating these changes to underlying theory that philosophers refer to as the “colligation” problem. The important and somewhat disheartening point with respect to this problem is an appreciation of each event being complex, enduring multiple transformations, having multiple antecedents, and resulting in a myriad of consequences (Isaac, 1997). This led Abbott (1992) to discuss a case as a sequence of major turning points (kernels) and sets of situational consequences flowing from these kernels. As such, a fundamental challenge in presenting design narratives lies in uncovering these events so that the reader understands their complexity but doing so in a way that lends itself global relevance while at the same time meaningfully capturing the dynamic unfolding of the phenomena.

Design-based research involves more than simply describing the design and the conditions under which it changed. Cobb et al. (2003, p. 10) suggests that “design experiments are conducted to develop theories, not merely to empirically tune ‘what works.’” diSessa and Cobb (this issue), along similar lines, argue that design-based research should involve theory work, treating the design platforms as contexts through which theory may be advanced. This type of work is iterative in nature, with the long-term commitment being to continually refine theoretical claims so as to produce what diSessa and Cobb refer to as “ontological innovations.” They suggest that design-based research allows for the production and testing of theory that can be used to generate, select, and validate specific design alternatives; revealing how various designs predicated on different theoretical assumptions are differentially consequential for learning. In this way, the validation of a particular design framework is not simply intended to show the value of a particular curriculum but results in the advancement of a particular set of theoretical constructs.

Another core challenge in carrying out design-based research arises given the joint role of the researchers as designer and researcher. Design-based researchers are not simply observing interactions but are actually “causing” the very same interactions they are making claims about. Barab and Kirshner (2001) wrote:

The goal of these researchers/educators/designers moves beyond offering explanations *of*, to designing interventions *for*. In fact, and consistent with pragmatists such as Dewey, Pierce, and James, to some degree it is the latter functional constraint that constitutes what is a useful explanation *of*. (p. 4)

This pragmatic approach to research may be valuable in that it creates theory that serves local practice, but it also produces challenges for design-based researchers.

How do we account for the role of the researcher in the design experiments and the associated threats to validity that they bring with them? If a researcher is intimately involved in the conceptualization, design, development, implementation, and researching of a pedagogical approach, then ensuring that researchers can make credible and trustworthy assertions is a challenge. Researchers working in schools often face difficult ethical choices. Do they stand idly by and watch a teacher struggle to use their curricula, or do they intervene providing additional support? Do researchers share stories of struggling students with teachers and allow them to change instruction accordingly, or do they play a “hands-off” role, minimizing their impact on classroom practices? Ironically, although Brown (1992) introduced design experiments in part as a method for developing a richer appreciation of variables as they occurred in naturalistic contexts, her role as context manipulator may have undermined the credibility of her claims. In other words, each systematic alteration of the designed context potentially contributes to the findings and claims being more artificial and less naturalistic.

In Cobb et al.’s (1999) “teaching experiment” approach, this problem of intervention is turned on its head so that issues that arise in the environment are to be accounted for and integrated into existing theory. It is through understanding the recursive patterns of researchers’ framing questions, developing goals, implementing interventions, and analyzing resultant activity that knowledge is produced. Rather than remain detached from the research context, researchers are implored to intervene where possible, using interventions as opportunities to examine core theoretical issues and explore learning. Critics will observe that such interventions “taint” the research context. Cobb argued that effective instructional models develop through these interventions, and it is through subsequent refining and testing that effective models are developed and deployed in other contexts. In this way, each new application is an extension of the theory as its specific characteristics are situated in local dynamics.

It seems that the basic methodological concern for the independence of the researcher and the learning environment is not operative in design-based research, fundamentally challenging the credibility of assertions generated through design research. However, such systematic experimentation is what makes design-based research a potentially useful methodological approach for learning scientists. Therefore, it is the responsibility of the researcher to draw on methodological practices consistent with other qualitative methods (e.g., see Glaser & Strauss, 1967; Lincoln & Guba, 1985) to convince others of the trustworthiness and credibility of claims being advanced. It is also the responsibility of the design-based researcher to remember that claims are based on researcher influenced contexts and, as such, may not be generalizable to other contexts of implementation where the researcher does not so directly influence the context.

Fishman et al. (this issue) further problematized the issue by suggesting that any classroom context, even without the manipulations of a design researcher, is

impacted by the systemic constraints in which it is nested, thereby making the generalizability of any naturalistic findings highly suspect. We believe that contexts are never without agency; there are always teachers, administrators, students, and community members creating context and, therefore, local adaptability must be allowed for in the theory. The goal is not to “sterilize” naturalistic contexts from all confounding variables so the generated theory is more valid and reliable. Instead, the challenge is to develop flexibly adaptive theories that remain useful even when applied to new local contexts. This potential of flexibly adaptive theory does not result because the theory was somehow generated in a context that was free of confounding situational variables, but rather, because the theory is supple enough to maintain its robustness even in the context of changing situational variables. Theory generated from design-based research, from this perspective, must strike a balance between refinement and adaptability.

DEFINING THE NATURALISTIC CONTEXT

A core part of design-based research as applied work involves situating the work in “naturalistic contexts.” Whereas most learning science researchers would agree with this commitment, the boundaries of *context* and what constitutes *naturalistic* may prove elusive. For example, Barab, Hay, Barnett, and Keating (2000) conducted design-based research in a university classroom context, iteratively refining course materials each semester so as to advance a participatory learning framework that was conceptually rich and theoretically grounded. However, their focus was on the individual classroom and not the larger system through which university courses occur. As such, although the work resulted in meaningful pedagogical practices, the theory did not adequately take into account the constraints that shaped the local context of the intervention and, therefore, resulted in theory with limited experience-distant relevance. In fact, even in the initial context of innovation where they demonstrated learning gains and had interested stake holders, the course was discontinued because it was not a profitable model (courses of $n = 20$ as opposed to the more traditional laboratory courses of $n = 100\text{--}150$) and required a certain type of instructor to succeed.

One might argue that within this large university context, doing design research on a single class without simultaneously addressing issues in the broader College of Arts and Sciences and University context constituted an impoverished and fundamentally flawed definition of design research. It is this critique that underlies the final core article of this special issue. Fishman et al. (this issue) argue that most design-based research does not explicitly address systemic issues of usability, scalability and sustainability and that “this limitation must be overcome if research is to create usable knowledge that addresses the challenges confronting technology innovations when implemented in real-world school contexts” (p. 43). They suggest a conception of design-based research that includes research on innovations in the context of systemic reform and that explores usability in terms of “gaps” between the

culture, capability, and policy/management structures. This work pushes us to reconsider the boundaries of context when carrying out design-based research, pushing beyond that which can be designed to a greater appreciation for the constraints of those real-world contexts through which our contexts of implementation are nested. Just as many researchers in the learning sciences have found the overly constrained laboratory contexts limited in terms of generating and advancing applied theory, Fishman et al. would critique designs that did not adequately consider the larger systemic constraints in which the context of intervention is a part. Consistent with these observations, the Cognition and Technology Group at Vanderbilt (1990) developed various learning scenarios to examine the role of context in cognition and to advance what became known as anchored instruction. These learning scenarios were packaged as videodiscs as the Jasper series and made available to classroom teachers. In spite of the theoretical sophistication and the research on its effectiveness, the adoption of the Jasper series by classroom teachers was limited. This was in part due to systemic constraints that have little to do with the theory of anchored instruction but impacts greatly on its utility for real-world contexts.

More generally, as a field we have over-theorized the role of context, and at the same time we have done little to characterize the role of context in ways that can usefully inform our design work. When we leave the relatively impoverished context of the classroom and inquire into phenomenon in more naturalistic contexts, boundaries become less defined and more problematic. Just as we create boundaries for the sake of control and explanation, we need to remember that the world does not divide itself at researcher-defined seams. These seams, rather than being black-boxed or ignored, must be problematized and examined as part of design work, helping to lend both ecological and consequential validity to our work. Ignoring or limiting the fundamental role of context will lead to both impoverished designs as well as under-specified theories that lack generalizable power. As such, much of the design-based research results in boutique projects that have little impact beyond the researcher's vita. As a community we must work to conceptualize and inquire about the material, social, and cultural contexts through which our work takes on meaning.

PARTING THOUGHTS

Over 1 decade ago, Alan Collins (1992) and the late Ann Brown (1992) began conducting what they referred to as "design experiments" because of the belief that many of the questions that were important to them could not be adequately addressed by laboratory-based examinations. Since then, design-based research as a term has grown in popularity and significance. However, we are still at our infancy in terms of having agreement on what constitutes design-based research, why it is important, and methods for carrying it out. The core articles of this special issue re-

spond to these questions, putting forward particular assertions while at the same time grounding these claims in actual examples of their own and their colleagues' work. We are also fortunate to have two strong commentaries that situate this work and that challenge the community with new questions and issues that must be answered if design-based research is going to help us advance our work in ways that others will judge as worthwhile and significant.

In this introduction, we have highlighted and problematized some of the challenges of carrying out design-based research. Over the next 2 years, the *Journal of the Learning Sciences* is especially interested in research that falls under the umbrella of design-based research—as these articles are accepted we will place them on the *JLS* website so that others can continue the dialogue. It is our hope that we as a community can provide the important methodological grounding so that we can uncover useful constructs and advance new theory with respect to how people learn. At the same time, we must work to have an impact on those individuals directly engaged as participants/collaborators in our research partnerships. It is essential that we as a research community take on this challenge, grounding our specific work in credible, trustworthy, and useful studies while contributing more generally to the development of new methodological processes suited to meet the needs of learning scientists. It is only through the rigorous development of a new methodological toolkit that we as design scientists in the learning sciences will be able to put our stake in the ground, thereby gaining the supportive ear of others and contributing to a richer understanding of how people learn.

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Design Research: Theoretical and Methodological Issues

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The term “design experiments” was introduced in 1992, in articles by Ann Brown (1992) and Allan Collins (1992). Design experiments were developed as a way to carry out formative research to test and refine educational designs based on principles derived from prior research. More recently the term design research has been applied to this kind of work. In this article, we outline the goals of design research and how it is related to other methodologies. We illustrate how design research is carried out with two very different examples. And we provide guidelines for how design research can best be carried out in the future.

In the 1990s there has been a movement to develop a new methodology for carrying out studies of educational interventions under the labels “design experiments” or “design research,” which will be used interchangeably in this article. Ann Brown (1992) was a leader in this movement and this article is an attempt to carry forward her work to specify for the educational research community the basis for this movement and the research methods it entails.

Design research was developed to address several issues central to the study of learning, including the following:

- The need to address theoretical questions about the nature of learning in context.
- The need for approaches to the study of learning phenomena in the real world rather than the laboratory.
- The need to go beyond narrow measures of learning.
- The need to derive research findings from formative evaluation.

Although design research is a powerful tool for addressing these needs, this kind of work brings with it serious challenges, including the following:

- Difficulties arising from the complexity of real-world situations and their resistance to experimental control.
- Large amounts of data arising from a need to combine ethnographic and quantitative analysis.
- Comparing across designs.

The notion of design research is very much in vogue in the learning-sciences community, but faces obstacles in the broader research community that make publication and tenure difficult for learning scientists. Whereas some of these obstacles arise simply from introducing a new methodological approach to the research world, some might be mitigated by actions we take as a field. If design research is to become accepted as a serious scholarly endeavor, the learning-sciences community needs to take responsibility for creating standards that make design experiments recognizable and accessible to other researchers, and for developing an infrastructure that can support summative evaluation.

The first section of this article describes where design research falls within the landscape of experimental methods. This section characterizes design research as a particular kind of formative research, and discusses what it can tell you and what it cannot tell you. The second section describes two examples of design research to show the range of methods and scales involved, in particular: Brown and Campione's work in the Oakland Schools around the Fostering a Community of Learners project, and the work of Diana Joseph in designing a passion curriculum in an afterschool setting at a Chicago school. The third section addresses questions about the conventional elements of a design experiment, including the variables to consider, the methods to use, and how to report findings. Finally, the fourth section discusses how the methods of design research might be applied to summative research.

INTRODUCTION

Design Sciences Versus Analytic Sciences

Herbert Simon (1969) in his classic book *The Sciences of the Artificial* makes a distinction between natural sciences and sciences of the artificial, by which he means design sciences. He identifies the various professions, such as architecture, engineering, computer science, medicine, and education, with the sciences of the artificial, and outlines what a curriculum in the engineering sciences might include (e.g., utility theory, control theory, etc.). In Simon's view the design sciences have been neglected, because of the lack of rigorous theories. He argues that recent developments in engineering and computer science have begun to provide the theoretical underpinnings that the sciences of the artificial need. Although he identified critical bodies of theory for the engineering sciences, his analysis does not provide the theoretical foundations for a design science of education.

In one of the first articles on design experiments, Collins (1992) framed his work as heading toward a design science of education. He discussed similar issues to those raised by Simon in terms of a distinction between analytic (or natural) sciences and design sciences. He viewed physics, biology, and anthropology as analytic sciences, where the effort is to understand how phenomena in the world can be explained. He viewed aeronautics, artificial intelligence, and acoustics as design sciences, where the goal is to determine how designed artifacts (e.g., airplanes, robots, or concert halls) behave under different conditions. Just as in aeronautics, where researchers look at how different designs affect lift, drag, and other dependent variables, he argued that we need to develop a design science of education, where we investigate how different learning-environment designs affect dependent variables in teaching and learning.

Design As Intention Versus Design As Implementation

There is a fundamental challenge in developing a design science of education in that the enacted design is often quite different from what the designers intended. Brown and Campione (1996) referred to this problem in terms of "lethal mutations," where the goals and principles underlying the design are undermined by the way the design is enacted. More generally, any implementation of a design requires many decisions that go beyond the design itself. This occurs because no design can specify all the details, and because the actions of participants in the implementation (e.g., students, parents, teachers, and administrators) require constant decisions about how to proceed at every level. Designs in education can be more or less specific, but can never be completely specified. Evaluation of designs can only be made in terms of particular implementations, and these can vary widely depending on the participants' needs, interests, abilities, interpretations, interactions, and goals.

Hence in evaluating any design, it is important to keep in mind the limitations of the evaluation. The effectiveness of a design in one setting is no guarantee of its effectiveness in other settings. This is a fundamental problem pervading all education research, which we return to later when we discuss the limitations of different methodologies for carrying out education research.

The Problem of Narrow Measures

A related problem that research in education faces is the fact that most experiments treat a single variable, such as learning of content and skills, as the only measure of worth. This problem reveals itself most clearly in the notion that a multiple-choice test of content and skills can be used as the “bottom line” measure of how a student, teacher, or school is doing. In fact, employers care much more about a students’ dispositions than about the particular content and skills that students acquire in school (Rosenbaum, 2001; Stasz, 2001). They care whether potential employees show up on time, take responsibility for getting a job done, work well with others, put in their best effort to succeed, and so forth. They would probably regard a school that instilled these virtues as much more valuable than a school that simply instilled a lot of specific knowledge. A similar argument can be made that these dispositions are key to success in college and any other endeavor that students are likely to undertake.

But this is just one aspect of the narrowness with which we evaluate interventions. A central concern of teachers is whether they motivate their students to want to learn. If students come out of school as proficient test takers who hate school, math, or Shakespeare, then the school will have failed. Education needs to produce “expert learners” (Brown, Ellery, & Campione, 1998), who love learning and who know how to find things out for themselves. Educators need to create environments where students are not afraid to put forth new ideas, share what they learn, and produce products they can show to the world. If we do not evaluate our educational environments in these terms then we will misjudge them badly and worse, we will lead them to emphasize the wrong goals (Frederiksen & Collins, 1989; Shepard, 2000).

Characteristics of Design Research

Design experiments were developed as a way to carry out formative research to test and refine educational designs based on theoretical principles derived from prior research. This approach of *progressive refinement* in design involves putting a first version of a design into the world to see how it works. Then, the design is constantly revised based on experience, until all the bugs are worked out. Progressive refinement in the car industry was pioneered by the Japanese, who unlike American car manufacturers, would update their designs frequently, rather than waiting years for a model changeover to improve upon past designs. The approach also is

the basis of Japanese lesson study (Stigler & Hiebert, 1999) where groups of teachers meet together to refine their teaching practices. By studying a design in practice with an eye toward progressive refinement, it is possible to develop more robust designs over time. In a later section of the article we discuss how the design-research methodology might be applied to summative research.

Design research is not aimed simply at refining practice. It should also address theoretical questions and issues if it is to be effective. For example, Diana Joseph (2000), in work described later in this article, assessed a motivational theory developed with Daniel Edelson (Edelson & Joseph, in revision; Joseph & Edelson, 2002). She analyzed the motivational patterns among different participants and how they played out in their work in an afterschool setting. This analysis led to refinements in the design, but also fostered refinements in the theory. Design research should always have the dual goals of refining both theory and practice.

Because design experiments are set in learning environments, there are many variables that cannot be controlled. Instead, design researchers try to optimize as much of the design as possible and to observe carefully how the different elements are working out. Such observation entails both qualitative and quantitative observations, just as Consumer Reports evaluates products in terms of both qualitative and quantitative measures. When some aspect of the design is not working, the design team, including the teacher or facilitator, should consider different options to improve the design in practice, and institute design changes as frequently as necessary. Any changes to one aspect of the design need to be considered with respect to how well they fit with other aspects of the design, since as Brown and Campione (1996) have emphasized, the design needs to be thought of as an integrated system. Thus the evaluation of the design is an ongoing process that changes as the design changes.

Design experiments have some fundamental limitations. Because they are carried out in the messy situations of actual learning environments, such as classrooms or afterschool settings, there are many variables that affect the success of the design, and many of these variables cannot be controlled. Design researchers usually end up collecting large amounts of data, such as video records of the intervention and outputs of the students' work, in order to understand what is happening in detail. Hence, they usually are swamped with data, and given the data reduction problems, there is usually not enough time or resources to analyze much of the data collected. It also takes resources to collect so much data, and so design experiments tend to be large endeavors with many different participants, all of whose work needs to be coordinated. All these factors make design experiments difficult to carry out and the conclusions uncertain.

Characteristics of Other Research Methods

All methodologies have their advantages and disadvantages. Despite the limitations of design experiments, they have an important role to play among the differ-

ent methodologies for studying educational practice. In order to place design experiments in the landscape of different methods we will compare them to three general types of educational evaluation approaches.

Laboratory and training studies attempt to control variables in order to determine how particular independent variables affect a few dependent variables such as the learning of content and skills. Ann Brown (1992) felt that training studies neglect important variables that affect the success of a design. In her view they also tended to be carried on over such a short period of time that there was no way they could identify what were the major effects of the design on learning.

Collins (1999) compared laboratory studies of learning to design experiments in terms of seven contrasting aspects of their methodology:

1. *Laboratory settings vs. messy situations.* Experiments conducted in laboratories avoid contaminating effects. Learners concentrate on the task without any distractions or interruptions. The materials to be learned are well defined and are presented in a standardized manner. Design experiments are set in the messy situations that characterize real life learning, in order to avoid the distortions of laboratory experiments.

2. *A single dependent variable vs. multiple dependent variables.* In most psychological experiments there is one dependent variable, such as the number of items recalled or the percent correct on a test of some kind. In design experiments there are many dependent variables that matter, though the researchers may not pay attention to them all.

3. *Controlling variables vs. characterizing the situation.* Psychological experiments use a methodology of controlling variables, where the goal is to identify a few independent and dependent variables, and hold all the other variables in the situation constant. In design experiments, there is no attempt to hold variables constant, but instead the goal is to identify all the variables, or characteristics of the situation, that affect any dependent variables of interest.

4. *Fixed procedures vs. flexible design revision.* Psychological experiments follow a fixed procedure that is carefully documented, so that it can be replicated by other experimenters. Design experiments, in contrast, start with planned procedures and materials, which are not completely defined, and which are revised depending on their success in practice.

5. *Social isolation vs. social interaction.* In most psychological experiments, the subjects are learning in isolation. There is no interaction with other learners and usually no interaction with a teacher or expert; the material to be learned is simply presented by text or video. By contrast, design experiments are set in complex social situations, such as a classroom.

6. *Testing hypotheses vs. developing a profile.* In psychological experiments the experimenter has one or more hypotheses, which are being tested by systematically varying the conditions of learning. In design experiments the goal is to look at

many different aspects of the design and develop a qualitative and quantitative profile that characterizes the design in practice.

7. *Experimenter vs. co-participant design and analysis.* In psychological experiments the experimenter makes all decisions about the design and analysis of the data, in order to maintain control of what happens and how it is analyzed. In design experiments, there is an effort to involve different participants in the design, in order to bring their different expertise into producing and analyzing the design.

Laboratory studies are effective for identifying effects of particular variables, but they often neglect variables critical to the success of any intervention. Ann Brown (1992) valued them for their role in developing a design, but to test and refine the design requires going into complex settings.

Ethnographic research also is set in the contexts of natural learning environments, but it has some of the characteristics of the analytic sciences, in that it does not attempt to study interventions. In general, ethnographic research (e.g., Eckert, 1989) attempts to characterize relationships and events that occur in different educational settings. There is no attempt to change educational practice, as in design experiments, but ethnographic research produces rich descriptions that make it possible to understand what is happening and why.

Large-scale studies of educational interventions use a variety of measures to determine the effects of a program or intervention. The methods usually emphasize standardized measures and surveys of critical participants, not tied to any particular design. These studies can be used to identify critical variables and to evaluate program effectiveness in terms of test scores, but they do not provide the kind of detailed picture needed to guide the refinement of a design. They are crucial however for summative research, and in our discussion of how design experiments methodology might be extended to summative research, we borrow from this kind of methodology.

Why Design Research?

Design experiments bring together two critical pieces in order to guide us to better educational refinement: a design focus and assessment of critical design elements. Ethnography provides qualitative methods for looking carefully at how a design plays out in practice, and how social and contextual variables interact with cognitive variables. Large-scale studies provide quantitative methods for evaluating the effects of independent variables on the dependent variables. Design experiments are contextualized in educational settings, but with a focus on generalizing from those settings to guide the design process. They fill a niche in the array of experimental methods that is needed to improve educational practices.

EXAMPLES OF DESIGN RESEARCH

To show some of the variety of design experiments carried out in recent years, we will describe two case studies. The first case is Ann Brown and Joseph Campione's *Fostering a Community of Learners (FCL)*, which was carried out in a variety of elementary schools across the country. This illustrates design research that studies how a classroom intervention plays out in a number of classrooms at different levels in an elementary school. The second case we describe is Diana Joseph's development and investigation of the *passion curriculum* approach, a framework for developing interest-centered learning environments. This work was conducted through a video-making club, which served as a prototype passion curriculum, enacted in an urban public school classroom and afterschool program. This case illustrates a small-scale design experiment, where a single researcher conducted a study in order to refine the design parameters for a new type of curriculum.

Brown and Campione's FCL Classrooms

Brown and Campione (1994, 1996; Brown, 1992) developed a model they call *Fostering a Community of Learners (FCL)* for grades 1–8. The model provides what is termed a “developmental corridor,” where the learning community extends not only horizontally across a classroom, but also vertically across grades. This makes it possible for learning topics to be revisited at increasing levels of disciplinary sophistication.

The FCL approach promotes a diversity of interests and talents, in order to enrich the knowledge base of the classroom community as a whole. The focus of FCL classrooms is on the subject areas of biology and ecology, with central topics such as endangered species and food chains and webs. There is an overall structure of students (1) carrying out research on the central topics in small groups where each student specializes in a particular subtopic area, (2) sharing what they learn with other students in their research group and in other groups, and (3) preparing for and participating in some “consequential task” (Scardamalia, Bereiter, & Fillion, 1981) that requires students to combine their individual learning, so that all members in the group come to a deeper understanding of the main topic and subtopics. Teachers orchestrate students' work, and support students when they need help.

There are three research cycles per year. A cycle begins with a set of shared activities and materials meant to build a common knowledge base. Students then break into research groups that focus on a specific research topic related to the central topic. For example, if the class is studying food chains, then the class may break into five or six research groups that each focus on a specific aspect of food chains, such as photosynthesis, consumers, energy exchange, and so forth. Students research their subtopic as a group, with individuals “majoring” by following their own research agendas within the limits of the subtopic. Students also engage

in “crosstalk,” talking across subtopic groups to explain, ask questions, and refine their understanding. The research activities include reciprocal teaching (Palincsar & Brown, 1984), guided writing and composing, consultation with subject matter experts outside the classroom, and cross-age tutoring. In the final part of the cycle, a member from each of the subtopic groups come together to form a “jigsaw” group (Aronson, 1978) in order to share learning on the various subtopics and to work together on some consequential task. Thus, in the jigsaw, all pieces of the puzzle come together to form a complete understanding.

The consequential task requires the different subtopics to be used together to form a common product or common understanding. The choice of consequential tasks is ideally made by the teacher and students together. The consequential task might be a bulletin board display, the design of a bio-park to protect an endangered species, a presentation to the community at large, or in some cases a test of students’ knowledge. These tasks “bring the research cycle to an end, force students to share knowledge across groups, and act as occasions for exhibition and reflection” (Brown & Campione, 1996, p. 303).

We can trace the evolution of FCL as a series of design experiments that culminated in the design described above. The story of the evolution of FCL is a story about successive refinements of a design, so that it better addressed the goals and principles that Brown and Campione (1994, 1996) have developed. As Campione (2000) made clear in the Scribner Award talk at the American Educational Research Association meeting, he and Ann Brown saw their work as going from laboratory research to learning principles to the design of a learning environment. Then based on their analysis of the learning environment, they would make modifications and additions to the learning principles, which in turn led to modifications of the learning environment and new laboratory experiments.

We begin the story of the evolution of FCL with the experiments by Palincsar and Brown (1984) on “reciprocal teaching” of reading. In reciprocal teaching of reading, students are expected to

1. Ask a question about the text.
2. Identify anything that is unclear.
3. Summarize what they have just read.
4. Make a prediction about what is coming next, after each paragraph of text.

The first experiments were done with individual children in the laboratory, and showed enormous gains in their ability to answer questions on reading comprehension tests. Given a procedure that was so successful with individual children, Palincsar and Brown extended the design to groups of students in classroom settings, since this is how reading is generally taught in American schools. To do this they had each child take a turn at being “teacher,” where they asked other students to carry out the four reading comprehension tasks listed previously. Again there

were very large gains in reading comprehension scores for the groups of students engaged in reciprocal teaching.

Phase 1. Brown and Campione (personal communication) found that reciprocal teaching was limited in that the student discussions never were very deep. The discussions were restricted to the text at hand, and so the students tended not to bring in knowledge from other sources. Brown and Campione wanted to create a context where students would be integrating knowledge across multiple texts. In such a context students would build theories about a domain that they could bring to bear in their discussions. It was this attempt to enrich the content of what students were discussing that led to the design of FCL.

In the first implementation of FCL Brown and Campione had students study interdependence and adaptation in biological systems, working in three Urbana Illinois classrooms. Reciprocal teaching was woven throughout the design, whenever students were reading materials. The students were broken into groups to study different aspects of interdependence and adaptation, such as camouflage, animal communication, and predator-prey relationships. The students studied books and other materials the project provided, and then they wrote up what they learned on computers. When each research group had prepared a document describing what they had learned, the students broke up into jigsaw groups, where one student from each of the research groups met with students from the other groups. That student would play the role of the teacher when the jigsaw group read the material that his or her research group had written. The task of each jigsaw group was to put together a booklet on biological systems, which was comprehensible to everyone in the group. In this first version of FCL there was a new emphasis on students gaining deeper understanding by writing explanations for other students and engaging in jigsaw with other students, where each shared their knowledge with students who had worked on other topics.

There were two important theoretical ideas that emerged out of this first phase of FCL: the notion of *diverse expertise* and the notion of a *community of learners*. The initial goal of the design was to encourage students to deeply explore content and to develop their own understanding by generating explanations of what they had learned, both orally and in writing. But Brown and Campione found that more important social goals had also been achieved by the design. Students came to value the expertise of other students; not just content expertise, but sometimes expertise in using computers or in keeping the group working effectively toward their goal. It became clear that students worked together better when they appreciated others' contributions. This is how the idea of diverse expertise took hold, with its emphasis on respect and listening to others. From this developed the view of a community of learners, which is radically different from the emphasis on individual learning that permeates schooling. The idea of a community of learners emphasizes students working to understand different aspects of a domain and sharing

their knowledge with everyone, rather than the traditional notion of all students learning the same thing at the same time.

Phase 2. Despite the initial success of the FCL design, Brown and Campione found from the work students produced that there were many misconceptions about biological systems, which the students were developing. For example, their ideas about evolution were mostly Lamarckian rather than Darwinian. Hence, when Brown and Campione moved to California, they revised the design of FCL to put more emphasis on biological content. To do this they added benchmark lessons and hands-on activities. In the benchmark lessons at the beginning of a unit, a teacher or outside expert introduced key ideas about biology in order to provide the students with the background for their research. At the same time they introduced laboratory activities where students could carry out hands-on experiments related to the topics they were researching. This dramatically expanded the kind of research that the students could carry out.

In this setting computers were still only being used as word processors, which was a waste of a major resource. Hence, Brown and Campione further redesigned FCL to have students go out on the Web in order to find information relevant to their research topics. They also set up telementoring relationships with biology experts (Brown, Ellery, & Campione, 1998). This gave the students access to real expertise from outside the classroom and acted to validate the importance of their research to the students. For example, one class of students became interested in the question of whether AIDS could be transmitted by mosquitoes, and so they investigated this topic thoroughly, calling upon resources and experts from outside the classroom, since the adults in the classroom did not have the information they were seeking. Students clearly gained the ability to address challenging questions of their own making.

These changes extended the notion of community beyond the classroom in order to bring in more expertise from outside the classroom: they were developing a wider *community of practice*, where students had *multiple ways in* to participate. They were no longer limited by the ideas within the classroom, but could go outside to address questions that arose from their research. At the same time Brown and Campione found that the hands-on activities did not integrate well with the other research activities the students were carrying out, and so they decided to discontinue the hands-on activities. Out of this phase they also developed the notion of a *developmental corridor*, where students would cycle through related topics in biology over the years of elementary school, but each time in greater depth, reflecting the developmental level that students were ready to achieve. This last idea required moving to a new elementary school in Oakland, where teachers at different grade levels were willing to participate in implementing FCL in their classrooms.

Phase 3. In the third phase of the development of FCL, Brown and Campione worked with teachers from second to eighth grade among a largely minority population. This allowed them to implement a developmental corridor that led children “gradually toward deep principles of the discipline, such as interdependence, biodiversity, evolution, and adaptation” (Brown & Campione, 1996, p. 306). “By second grade we begin to address animal/habitat mutuality and interdependence. Sixth graders examine biodiversity and the effect of broad versus narrow niches on endangerment. By eighth grade the effect of variation in the gene pool on adaptation and survival is not too complex a topic.” (Brown & Campione, 1996, pp. 307–308). Implementing the developmental corridor also allowed them to set up cross-age tutoring where students in the higher grades tutored second and third graders on the topics they were investigating (e.g., see Brown, Ellery, & Campione, 1998).

A final refinement to the FCL design came from the students themselves. They pointed out that the formal sharing of knowledge came at the end, when the jigsaw groups got together. The students suggested therefore that they needed to have “*crosstalk*” sessions, where students from the different research groups present their intermediate findings to the whole class. Sometimes probing questions from other students leads students down new research paths to resolve the questions that arise during the crosstalk. It is also a forum for raising issues that the groups are worried about. Crosstalk has become a major vehicle for sharing knowledge and building a learning community.

This third phase of their work put together all of the pieces that make up the FCL design outlined in Brown and Campione’s 1996 article. Table 1, taken from Brown and Campione (1996), captures the design principles that evolved from the various phases of the FCL design, together with their previous work on metacognition, analogy, dynamic assessment, the zone of proximal development, and reciprocal teaching. The FCL design, in fact, brings together 30 years of development work as to how best to structure a learning environment for students in schools.

Brown and Campione (1994, 1996; Brown, 1992) have collected a wide variety of data about students and teachers in the FCL classrooms. As Brown (1992) said:

In addition to relatively standard outcome measures involving reading, writing, content knowledge, and computer competence, all of which improve significantly (Brown & Campione, 1994), this project generates a vast amount of information that is not readily subjected to standard measurement devices. We collect transcripts of children’s planning, revising and teaching sessions. We collect observations of teachers’ coaching and responsive teaching, as well as their direct instruction. We have records of student portfolios, including individual and group long-term projects that require them to exploit accumulating knowledge in novel ways. We score electronic mail queries to peers, teachers, and collaborators in the university community. Ethnographic observations of cooperative, or not so cooperative, interactions, such as

TABLE 1
Principles of Learning to Support Fostering a Community of Learners^a

1. Systems and cycles

Seasonal cycles of research–share–perform activities
Supported by repetitive participant structures
Participant structures can be replaced if and only if the replacement serves the simple RSP system
Constancy at level of deep structure, variability at level of surface ritual
All activities for a purpose, a purpose for all activities

2. Metacognitive environment, reflective environment

Active, strategic nature of learning
Self-regulation and other-regulation for common good
Autocriticism, comprehension monitoring
Effort after meaning, search for understanding
Atmosphere of wondering, querying, worrying knowledge
Reflective practices

3. Discourse

Dialogic base
Shared discourse, common knowledge
Seeding, migration, and appropriation of ideas
Mutual appropriation
An interpretive community

4. Deep content knowledge

Developmental sensitivity
Intellectually honest and demanding
Developmental corridors from children's intuitive knowledge to deep principles of a discipline of inquiry
Intermediate goals and levels of abstraction
Support, sharing
Enriched by diversity

5. Distributed expertise

Sharing for a purpose
Collaboration not just nice but necessary
Major, identity, and respect
Multiple ways in, multiple intelligences
Legitimization of differences
Community building

6. Instruction and assessment

Deliberately aligned
Based on same theory, loosely Vygotskian
Guided practice, guided participation
Multiple zones of proximal development
Transparent, authentic, and purposeful

7. Community features

Community of practice
Communities of practice with multiple overlapping roles
Link between current practice and expert practice emphasized
Elements of ownership and choice
Community beyond the classroom wall

Note. RSP = research–share–perform.

^aFrom Brown and Campione (1996).

group discussions, planning sessions, help seeking, peer tutoring and so forth are taken routinely, together with extensive video and audio taping of individuals, groups, and whole classroom settings. In fact, we have no room to store all the data, let alone time to score it. (pp. 151–152)

This wide variety of data enables Brown and Campione to study the FCL classrooms with many different lenses. As Brown (1992) pointed out:

I find that in the interest of converging operations, and because of the multifaceted nature of my data base, I prefer a mixed approach, suiting the method to the particular data. I mix and match qualitative and quantitative methodologies in order to describe the phenomena, a mixture that is becoming commonplace in the journals, reflecting the increasingly complex issues that psychologists now address. In my own work I routinely combine a concentration on large-scale databases with in-depth microgenetic analyses of a few children or perhaps a group.... Our routine practice is to take fairly traditional pretest and posttest data from all the experimental and control students and combine that with a few selected case studies. (p. 156)

Brown (1992) worried about whether by selecting cases to illustrate her points, she may have distorted the data. She cites the example of how Bartlett (1932) misrepresented his data somewhat in his well-known studies of how people's memories for what they have read become distorted over time. As she says, "This problem of the theorist selecting those segments that prove his or her point is endemic in research that depends on transcripts or protocols culled from a large data base." (p. 162).

She also raised the issue of whether the FCL work is just the result of a Hawthorne effect. In the Hawthorne experiments the investigators found that worker productivity increased when they increased the lighting, but also when they decreased it, from which most conclude that any intervention has positive effects. She thinks that because she is showing specific learning effects one would expect to find given the intervention, that this is not a serious criticism. But she quotes the original investigators, Roethlisberger and Dickson (1939) as to why they were disappointed in their results:

The difficulty, however, went much deeper than the personal feelings of failure of the investigators. They were entertaining two incompatible points of view. On the one hand, they were trying to maintain a controlled experiment in which they could test for the effects of *single variables* while holding all other factors constant. On the other hand, they were trying to create a human situation, which remained unaffected by their own activities. It became evident that in human situations not only was it practically impossible to keep all other factors constant, but trying to do so in itself introduced the biggest change of all; in other words, the investigators had not been

studying an ordinary shop situation but a socially contrived situation of their own making.

With this realization, the inquiry changed its character. No longer were the investigators interested in testing for the effects of single variables. In the place of a controlled experiment, they substituted the notion of a social situation, which needed to be described and understood as a system of interdependent elements. (p. 185)

This is exactly the kind of thinking that has led to the methodology of design research. Throughout their work, Brown and Campione (1996) have stressed the importance of thinking systemically about the interdependence of the elements of a design as one tries to assess its impact or modify its elements.

Diana Joseph's Passion Curriculum

The passion school concept is a design for comprehensive progressive education that uses deep learner interests to drive work on serious learning objectives. The idea was conceived initially by Roger Schank and Allan Collins as a natural synthesis of cognitive apprenticeship (Collins, Brown, & Newman, 1989) and goal-based scenario theory (Schank, Fano, Bell, & Jona, 1994). The passion school model is meant to provide specific guidance for structuring learning environments centered in classrooms organized as communities of common interests, rather than age-based communities. The defining principles for passion schools are as follows:

1. That students are assigned to curricula on the basis of their interests.
2. That students learn through active engagement in meaningful works by interacting with expert adults and more and less advanced students.
3. That through this work, learners grapple with important ideas, including adult-defined core competencies, such as those found in state and national education standards.

Diana Joseph's work takes the first step toward the development of the passion school model by shaping activity, so that learners engage in work that is meaningful to them, while at the same time grappling with important ideas. This work is being conducted through an extended, multi-phase design experiment in the context of students learning to create films as part of a "video crew." The passion curriculum design research focuses on the development of design principles, curricular structures, and strategies for activity design in the passion school model. In this section we describe the major phases in this work, to illustrate the ways in which data drives refinement and generalization in design research.

Phase 1. The first version of the video crew was carried out in an urban fifth-grade classroom from February to June 1996. The goal of the video crew was

to develop videos on topics that the students were most interested in. The students worked for one hour a day three to five times a week. There were 33 students in the classroom and Joseph divided them into groups of 4 to 7 students depending on their particular interests. Their chosen video topics included fables, talk shows, sports, dance, singing, and street violence. The students in the groups were assigned different roles, with responsibilities for storyboarding, script writing, sets and props, acting, and camera operation. Because many groups had difficulties working together, one role that became critical was that of the facilitator to resolve conflicts. The curriculum provided students with guidance about their roles through structured materials and scheduled lessons with the instructor. The roles and associated guidance were designed to motivate students to work in areas of strength and weakness, depending on need as determined by the regular classroom teacher. As a culminating product, Joseph put together a video from the various pieces the different groups produced and gave copies to each of the children to take home. Joseph and several undergraduate assistants documented this process through daily field notes and videotape documentation. They interviewed participants prior to and after the Video Crew enactment.

There were several major lessons from this first attempt at the Video Crew that impacted the subsequent design. The major redesign goal was to find a means to focus the students on serious learning goals and to assess what they were learning. In order to address this issue, Joseph designed a certification system, based on the Scout merit badge system, whereby students could demonstrate their mastery of the camera, scriptwriting, acting, etc. Each certificate required students to carry out a set of activities that encompass the skills and knowledge required for mastery of video skills, as well as certain academic skills. When a certificate was acquired there were accompanying rights and responsibilities. For example, students who had earned the camera certificate had the right to take the camera out of the building for the purpose of shooting outside, but they also had the responsibility to teach others how to use the camera properly. The certification system was installed in Phase 2.

Joseph also found extensive differences across students in terms of the degree to which video production was an interest for them. This exposed a problem in the choice of setting for the study — the design research was intended to investigate how a curriculum could serve a group of students with a common interest, but video did not represent a common interest for these students. Furthermore, there was very little difference among students in terms of their experience making videos, collaborating, or creating projects. It was difficult to manage 33 students at a time, when none of the students had developed expertise, so that more expert students could help to manage the whole enterprise. Hence, it was decided to work in an afterschool setting in the next phase, so that it was possible to select students who were particularly interested in video and where the numbers were more manageable. In addition, the work called for an explicit design for training students to mentor others.

Phase 2. The next major phase of the passion curriculum design experiment recreated the video crew on a smaller scale, over a much longer period of time. The video crew prototype curriculum ran from early spring of 1997 until late spring 1998, omitting school vacations. The new design included three major categories of student activity: (1) projects which in this case produced student-made videos; (2) certifications, which linked adult-defined learning objectives with related rights and responsibilities; and (3) community life, which included student-run aspects of the classroom community. Ten students in grades four through six created a large number of complete videos, as well as a large number of plans, scripts, storyboards, treatments, raw footage, props, costumes and other video work components. Genres included original fiction, adaptation of fiction, performance, documentary, mockumentary, animation, weather reports, and rituals. They also worked on nonvideo projects such as a Video Crew website and a number of planned social events. Their work culminated in a student-organized film festival, attended by families, friends, and teachers.

These projects were largely student-developed, though there was some teacher assignment to roles within the projects. The work was conducted with some adult guidance and support for the technical and planning aspects of video production, with content and execution strategies generally left up to the students. In addition to their project work, students worked to demonstrate expertise on certain categories of skills, through certifications. Having developed a certain level of expertise, the senior students worked to mentor novices in spring 1998. Students also participated in the community life of the Video Crew — both in terms of the traditions of video-production organizations and in terms of helping to define classroom norms. Students planned regular off-site visits to a video-editing studio, and occasional location shoots at sites such as the public library and the city zoo. In addition, they persuaded the school principal to permit weekend access to the school building for a site-specific horror-genre video.

A key area of research at this stage was student interest. Data from this phase suggested that strong student interest was unreliable. Even the most passionate students shifted in their engagement, and many students seemed to be motivated by issues other than interest in the activities. Joseph needed a more detailed theory of learner motivation in passion curricula. In collaboration with Daniel Edelson, Joseph developed a framework that used theories and findings from motivation research to organize the design of learning environments (Edelson & Joseph, in revision; Joseph & Edelson, 2002). This theory created a foundation for analysis of the Video Crew data with regard to learner motivation.

Joseph (2000) focused on field notes and artifacts to uncover patterns of learner engagement. Analytical lenses based on Edelson and Joseph's theory indicated that some students exhibited fairly consistent motivations across different activities—some were passionate about video, some pursued other interests, and some appeared to be motivated by other considerations, such as a desire to connect with

friends. Artifacts produced by students with these patterns were analyzed for indications that students had grappled with ideas in the process of creating these artifacts. These findings were of interest both from the point of view of the direct work of design in the passion curriculum project, and from a more general point of view. With respect to the passion curriculum model, these findings pointed at a need for the framework to provide motivational support for students with different tendencies, through role assignment and through refinements to the certification structure. At the same time, the findings suggested the possibility that learners in general, not just in passion curricula, exhibit motivational tendencies that are fairly characteristic over time. In this sense, the passion curriculum work raises general issues, and posits frameworks, such as the Edelson and Joseph theory, that are of potential use in other contexts. This is an example of data from a design experiment raising new questions for basic research.

Phase 3. Between 2000 and 2002, the Video Crew operated in a Chicago charter school with a different cohort of students. In this afterschool setting, students chose to join video crew over other possible choices, so they were more likely to bring a strong interest in video than in the previous settings. Students attended Video Crew up to three days a week for 1–3 hours. Each day had a special focus, such as working on video projects, working on certifications and analysis of others videos. In the first year, students were trained as mentors for new students joining in the second year. Another design change was to establish the intermediate goal of showing videos at different times to other students in the school, in addition to the film festival at the end. An important innovation in this phase was to separate the roles of teacher and researcher. By working in collaboration with a teacher, Joseph separated the passion curriculum model from her own teaching practice. The core questions in this phase focused on needs for more articulated instructional strategies to complement the design strategies (Joseph, Edwards, & Harris, 2002).

The passion curriculum design framework has evolved through this design research to the point where it has a specific framework, a library of case studies describing implementation, and some evidence supporting the conjecture that the framework organizes learner motivation in the intended ways. This foundation has become the seed for the next phase of work—developing and enacting other passion curricula through other designers and teachers. A new passion curriculum, Multi-Media Design Studio, was enacted in fall of 2002, supporting further investigation of the design framework and of the motivational affordances of particular technologies (Joseph & Nacu, submitted). As an engine for the development of a learning environment, the design-research model has guided the evolution of the passion curriculum framework over time. At the same time, by demanding real-world enactment of the design, design experimentation created a setting where a key learning feature—motivation—was far more salient than it is in traditional classrooms, permitting more effective study of this phenomenon.

METHODOLOGY OF DESIGN RESEARCH

In this section we attempt to provide guidance to people who are planning to carry out design experiments. Our approach to design research requires much more effort than any one human can carry out. We put forward these ideas not because we expect each and every design experiment to embody them, but to give an overview of all the things the design-research community is responsible for. In our ideal world, design research will move in the direction of embodying many of the practices we outline here. But it will take teams of researchers and accessible archives documenting design experiments, as we discuss in the last section, to make these dreams at all possible. The guidelines we develop in this section are summarized in Table 2.

TABLE 2
Guidelines for Carrying out Design Research

Implementing a design
Identify the critical elements of the design and how they interact
Characterize how each was addressed in the implementation
Modifying a design
If elements of a design are not working, modify the design
Each modification starts a new phase
Characterize the critical elements for each phase
Describe the reasons for making the modifications
Multiple ways of analyzing the design
Cognitive
Resources
Interpersonal
Group or classroom
School or institution
Measuring dependent variables
Climate variables (e.g., engagement, cooperation, and risk taking)
Learning variables (e.g., dispositions, metacognitive, and learning strategies)
System variables (e.g., ease of adoption, sustainability, spread)
Measuring independent variables
Setting
Nature of learners
Technical support
Financial support
Professional development
Implementation path
Reporting on design research
Goals and elements of the design
Settings where implemented
Description of each phase
Outcomes found
Lessons learned
Multimedia documentation

Implementing a Design

Each implementation of an education design is different. Therefore it is important to identify the critical elements of the design and how they fit together. In order to evaluate any implementation, one needs to analyze each particular case in terms of these key elements and their interactions. Some elements will be implemented more or less as the designers intended, some will be changed to fit the circumstances, and some will not be implemented at all. What is needed is a profile for each implementation as to how each of the critical elements were implemented and how well the elements in the implementation worked together toward the designer's goals.

Brown and Campione (1996) made the argument for the FCL design that it is not important which particular activities they have designed are implemented, as long as the principles that they outline in their article are realized. Their argument implies that the principles they enunciate are the key elements of the design, and that any implementation should be evaluated as to how well these principles were implemented.

Modifying Designs as You Proceed

A goal of design research is to improve the way a design operates in practice. The teacher or researchers may see that an element of the design is not working in the course of the experiment. It is important to analyze why it is not working, and take steps to fix whatever problems appear to be the reasons for failure. In this way we collect information about failures, plus information gathered from the attempted repairs to the design, and whether they succeed or fail. It is critical to document the failures and revisions, as well as the overall results of the experiment.

The experimental methods inherited from psychology assume a fixed procedure is used throughout the experiment. Design research assumes continuous refinement. This difference has deep ramifications and requires changes in the way we analyze and report what is done. We suggest that researchers document their designs in detail, recording all major changes in design. These design changes mark the borders between phases. The goal then becomes to characterize the design elements that are in place in each phase and the reasons for the transitions from each phase to the next. Data relevant to research questions should be collected in each phase. For example, if there were 4 phases in a particular implementation, then it would be good if there were an intermediate assessment of learning outcomes between phases 2 and 3, as well as pretests and posttests. A detailed design history of this kind allows research audiences to evaluate the credibility of design decisions, and the quality of lessons learned from the research.

Multiple Ways of Looking

Rogoff (1995) called for analysts of learning environments to attend to three critical aspects: the personal layer (the experience of the individual), the interpersonal layer (one-on-one interactions), and the community layer. In the context of design experiments, researchers must additionally attend to interactions of learners with elements of the environment. There are many different aspects of what makes for an effective design, and so both designers and evaluators need to wear many hats in order to design and assess educational interventions. Consider some of the different aspects that are relevant to educational designs:

- *Cognitive level:* What do learners understand before they enter a particular learning environment, and how does that understanding change over time? Some of the tools for analysis at this level include observations of thinking through learners' representations and explanations. Through visual and verbal descriptions of ideas, researchers ask learners to expose their thinking. Are the explanations clear? Do representations capture important relationships?
- *Interpersonal level:* This viewpoint addresses how well teachers and students interact personally. Is there sharing of knowledge? Have the students bonded with each other so that they respect and help each other? Researchers use ethnographic techniques to observe these kinds of interactions.
- *Group or classroom level:* This viewpoint addresses issues of participant structure, group identity, and authority relationships. Is everyone participating? Is there a sense of the goals and identity of the group? Again, ethnography is an effective approach to analysis.
- *Resource level:* This level deals with what resources are available to learners and if they are easy to understand and use. How accessible are the resources? How well are they integrated into the activities?
- *Institutional or school level:* At this level issues arise as to communication with outside parties and support from the entire institution. Are parents happy with the design? Do administrators support it strongly? What are the micro-political issues that impact the design?

These levels are very much intertwined. To design and assess these different issues requires many different kinds of expertise: teachers, administrators, psychologists, anthropologists, media designers, etc. Conceivably one person can address all these different perspectives, but it helps to have them all represented explicitly.

Characterizing Dependent Variables

Success or failure of an innovation cannot simply be evaluated in terms of how much students learn on some criterion measure. Different kinds of evaluation are

necessary for addressing questions such as: how sustainable the design is after the researchers leave, how much the design emphasizes reasoning as opposed to rote learning, how the design affects the attitudes of students, etc. To evaluate different variables, it is necessary to use a variety of evaluation techniques, including standardized pretests and posttests, survey and interview techniques, and systematic scoring of observations of the classrooms. Both qualitative and quantitative evaluations are essential parts of design-research methodology.

At least three types of dependent variables are important to assess:

1. Climate variables such as engagement, cooperation, risk taking, student control.
2. Learning variables, such as content knowledge, skills, dispositions, metacognitive strategies, learning strategies.
3. Systemic variables, such as sustainability, spread, scalability, ease of adoption, and costs.

Evaluating climate variables requires observational techniques, either by producing field notes while observing the intervention in practice, or collecting video records of the intervention and scoring those records subsequently. For example, these techniques might be used to evaluate three kinds of climate variables: the degree of engagement of students in learning in the classroom, the degree of cooperation among students in the classroom, and the degree of effort students are making to understand the curriculum topic. To evaluate these variables one might collect videos of different classes spread out over the time the teacher is carrying out the designed intervention. These videos can be scored systematically by two raters with respect to the three dimensions using a five-point scale for each 5-minute interval in the lesson. Raters would be trained using benchmark lessons for which scores have been calibrated with experts.

Learning variables are best assessed by collecting pretest and posttest measures. For example, pretests and posttests can be used to evaluate three kinds of learning variables: content, reasoning, and dispositions. To evaluate learning of content and reasoning, it is possible to use short answer or essay questions, oral interviews, or multiple-choice items. By using items from standardized tests, it is possible to compare performance to national norms for the items. To evaluate learning of dispositions, one might apply instruments developed by Dweck (1986) to assess whether there are changes in students' beliefs reflecting a move from performance goals to learning goals. There have been such changes reported in a design experiment carried out by Lamon and her colleagues (Lamon, Chan, Scardamalia, Burtis, & Brett, 1993; Scardamalia, Bereiter, and Lamon, 1994).

Systemic variables are best evaluated by interviews and surveys. For example, one might evaluate systemic variables, such as the ease of adoption of a design into the curriculum, the degree to which it is sustained in subsequent years, and the

spread of use to other teachers and students. These can be measured by surveys and structured interviews with teachers and students. It is possible to develop a questionnaire that addresses advantages and difficulties teachers encountered in adopting a design in their classroom. The other variables can be evaluated by surveys administered to both teachers and students at regular intervals. The surveys will ask about what aspects of the design are being sustained and are spreading, and which aspects are not.

Characterizing Independent Variables

In evaluating any design there are a large number of independent variables that may affect the success of the design in practice. It is a matter of art to determine what aspects of the implementation situation may affect the success of the design. Our goal here is to say what general aspects of the situation researchers need to consider in order to decide what is affecting the success of the design. The contextual variables that can determine the success of an innovation include:

- *Setting.* The setting of the learning environment is a critical variable in how any design fares. The setting might vary over homes, workplaces, museums, schools, or colleges; elementary, middle or high schools; public or private schools; urban, suburban, or rural schools; elite or community colleges; and so forth. How broadly applicable an innovation is can only be determined by trying it out in many different settings.
- *Nature of the learners.* Critical variables about the learners include such things as their age, socioeconomic status, turnover rate, attendance rate, and so forth. For example, some innovations may work with weaker students and some with gifted students. So it is important to determine for which type of learners the design is effective, and in what ways.
- *Required resources and support for implementation.* In order to carry out any design, there will be a need for resources and supports of various kinds, including materials, technical support, administrative support, and parent support. If a design requires teachers to gather materials, spend time in preparation or other activities, enlist administrators or parents to make the design succeed, then these requirements need to be identified.
- *Professional development.* Often in order for a design to be successful, teachers (and perhaps others) need to be provided with professional development of various kinds. These can encompass workshops, design meetings, courses, videos of exemplary practice of the design, guided practice with expert practitioners, reflective meetings with colleagues, and so forth. Identifying what teachers need to implement the design successfully is an important aspect of designing an innovation.

- *Financial requirements.* Any intervention adds costs that need to be tracked, including equipment costs, service costs, professional support and development costs, replacement costs, and so forth. Very often substantial costs, such as technical support and replacement costs, are ignored when calculating the cost of a technological innovation.
- *Implementation path.* This term covers the variables involved in implementing a design, such as how the innovation is introduced, the time devoted to it, the duration of its usefulness, and so forth. There is a structure to the introduction and evolution of a design that needs to be characterized in analyzing any implementation.

There is a web of interrelations between independent and dependent variables. The division between the two depends on what outcomes one is interested in. But changes in any variable can have effects on other variables through complex feedback loops. Hence, changes in a dependent variable may lead to changes in an independent variable, as when increases in engagement lead to increases in attendance. The language of dependent and independent variables is only meant to capture the distinction between outcomes we should consider and those variables that may affect the outcomes.

Reporting on Design Research

The experimental literature developed a conventional structure for reporting on experiments that evolved over time. The structure consisted of four parts: Background to the problem, experimental method, results, and discussion. Because design research reconceives the experimental process, there needs to evolve a different structure for reporting. Tentatively, we propose that there should be five sections in reporting on design experiments:

- *Goals and elements of the design.* An important aspect of reporting on design experiments is to identify the critical elements of the design and how they fit together to accomplish the goals of the design. The critical elements of a design may be the materials, the activities, a set of principles, or some combination of all these. It is equally important to describe the goals of the design and how all the elements are meant to work together to attain those goals. Goals, critical elements, and their interactions need to be described in enough detail, so that it is possible to evaluate how well the design was implemented in different settings.
- *Settings where implemented.* The description of the settings needs to include all the information relevant to the success of the design outlined in the 'Characterizing independent variables' section. Differences between how the design was implemented in each setting should be detailed, so that readers can evaluate how faithfully the design was carried out in each setting.

- *Description of each phase.* The design is likely to go through a different evolution in each setting, so it is necessary to describe each phase in each setting. When changes are made in a setting, the reasons for the changes should be specified along with the effects of making the changes. It also makes sense to describe how the critical elements of the redesign accomplish the goals of the original design or how the goals have changed.
- *Outcomes found.* The outcomes should be reported in terms of a profile of values on the dependent variables in the different settings, much like qualitative and quantitative data are reported about different products in Consumer Reports. To the degree intermediate data were collected describing the different phases, these should be included.
- *Lessons learned.* Considering what happened in the different implementations, the report should attempt to pull together all the findings into a coherent picture of how the design evolved in the different settings. It is important to describe the limitations and failings of the design, as well as the successes, both in implementation and outcomes.

DESIGN RESEARCH AS SUMMATIVE EVALUATION

Although design experiments were conceived as a formative evaluation strategy, the principles involved do have implications for summative evaluation. We would argue that any assessment of educational innovations must carry out both quantitative and qualitative assessments, using comparative analysis, as does Consumer Reports. For example, if we wanted to compare how effective the Waterford and Peabody reading programs are, we would need to carry out comparative analyses in a variety of different settings, such as urban, suburban, and rural schools, and perhaps even homes, workplaces, and military settings. In such studies there must be a fixed experimental procedure, unlike the flexible design revision we recommend for formative evaluation. The assessment should produce a profile that shows the strengths and weaknesses of the designs being compared. Hence, different designs might be found to be more effective in some settings or with regard to some outcomes.

To have a sound assessment process, design-based researchers as a community should develop a consensus process to determine what variables to look at and how to assess them. The assessment should address the multiple concerns of different stakeholders, including developers, and so they should be included in the consensus process. The design-research methodology argues that we need to look at multiple contextual and dependent variables. At least three types of dependent variables are important:

1. Climate variables such as engagement, cooperation, risk taking, and student control.
2. Learning variables, such as content knowledge, skills, dispositions, metacognitive strategies, and learning strategies.
3. Systemic variables, such as sustainability, spread, scalability, ease of adoption, and costs.

As suggested previously, there are standard ways to assess each of these types of variables.

To carry out investigations fairly, the evaluators must be independent. To carry out such evaluations effectively, the country would need to invest in an independent agency, in the style of Consumers Union, with the expertise to carry out comparative evaluation. Such an agency could develop the expertise and methods for looking in a cost-effective manner at innovations in use, in a way that best informs the many different stakeholders.

CONCLUSION

Ann Brown (1992) felt that laboratory experiments, ethnographies, and large-scale studies are all valuable methodologies to study learning, but that design experiments fill a niche these methodologies do not address. It is clear from the spread of these kinds of research methods (Barab & Kirshner, 2001; Edelson, 2001; Design-based Research Collective, 2003) that design research is here to stay. But design experiments often lead to the collection of large amounts of data that go unanalyzed. Hence, it makes sense for the design-research community to establish an infrastructure that would allow researchers at other institutions to analyze the data collected in design studies, in order to address their own questions about learning and teaching. This would require the community to honor such reanalysis of data with the same status as original research and it would require research journals and tenure committees to take such work seriously. Other fields, such as child language (MacWhinney, 1995), have developed widely available archives of data, enabling researchers to discuss and analyze the same data from many different perspectives. As a community, we should strive to set up an infrastructure that can support researchers at different sites in analyzing the large data sets that design experiments are now producing.

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Creating a Framework for Research on Systemic Technology Innovations

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This article examines why cognitively oriented technology innovations, designed to foster deep thinking and learning, have not become widespread in K–12 schools. We argue a key reason is that most design-based research does not explicitly address systemic issues of usability, scalability and sustainability. This limitation must be overcome if research is to create usable knowledge that addresses the challenges confronting technology innovations when implemented in real-world school contexts. This is especially important in an era when political forces push schools away from the cognitively rich, inquiry-oriented approaches espoused by the Learning Sciences. We suggest expanding our conception of design-based research to include research on innovations in the context of systemic reform as a potential solution to the problem. To that end, we introduce research questions and issues arising from our own experiences with a technology-rich innovation in the context of a systemic reform initiative as a starting point in the creation of an expanded design-based research agenda. These questions and issues have important implications for both the continued viability of research on technologies for learning and on the future of technology use in schools that stems from such research.

Over the past decade, the Learning Sciences have built on knowledge of how people learn (Bransford, Brown, & Cocking, 1999) and made major investments in the design and development of learning environments that employ technology to foster

thinking and understanding with demonstrated positive effects on learning (Roschelle, Pea, Hoadley, Gordin, & Means, 2000). Despite the fact that technology is now considered commonplace in K–12 education (Becker, 1999), most innovations derived from Learning Sciences research, which we refer to as cognitively oriented technology innovations, have not found their way into widespread classroom use. Instead, for a variety of reasons including teacher capabilities (CEO Forum on Education and Technology, 1999), technology infrastructure (Carvin, 2000), school culture (Cuban, 1986) and organizational constraints (Cohen, 1988), the primary uses of technology in schools remain drill and practice, word processing, and web surfing (R. E. Anderson & Ronnkqvist, 1999). These uses of technology may be important initial steps for schools, but they fall short of the tremendous potential of technology to support the rich, inquiry-oriented learning called for in national standards documents (e.g., American Association for the Advancement of Science, 1993), and embodied in Learning Sciences research (Bransford et al., 1999).

Appropriately, the public demands a great deal from its education investment. Technology is an expensive and therefore popular target for criticism, particularly because demonstrating widespread benefits from its use in schools is a challenging problem for research (Means, Wagner, Haertel, & Javitz, 2000). Researchers have long understood that for technology to contribute to learning requires much more than simply installing it in schools. As one research group puts it, in order to be effective, “technology needs to be part of a coordinated approach to improving curriculum, pedagogy, assessment, teacher development, and other aspects of school structure” (Roschelle et al., 2000, p. 78). In other words, the most effective uses of technology are interwoven with the challenges and problems of school reform itself. However, much of the design-based research that is targeted toward the development of cognitively oriented technological innovations has focused on classroom-level or multi-classroom testbed implementations (Gomez, Fishman, & Pea, 1998) and not the larger contexts in which innovations are, or are not, actualized when used across entire school systems.

This article explores the question of why there is not more widespread use of technologies to foster deep learning aligned with national standards, and how we, as a research community, can learn more about how to foster, sustain, and scale these uses of technology. In particular, we explore the use of cognitively oriented technologies within systemic reform contexts, using our own experiences to frame new research questions and issues for exploration that shed light on the problems of creating scaleable, sustainable, and usable technology innovations. Additionally, we shed light on interrelating system variables that should be considered when conducting design-based research.

Why are cognitively oriented technology innovations not widely used in schools? Why aren't they scaleable or sustainable? We believe an underlying explanation to be that we, as a scholarly community, have not focused our research on the development and use of cognitively oriented technologies in a way that ad-

dresses the fundamental needs of school *systems*. Instead, research on cognitively oriented learning technologies has focused primarily on students, teachers, and classrooms as the primary unit(s) of analysis. Though we recognize the need to link technology and reform, the field lacks a bridge between focused research and development of learning technologies and the broad-based systemic use of these innovations in schools. Shepard (2000) recognized this as problem for the broader educational research community in her AERA Presidential Address, when she advised researchers to develop methodologies that embrace “dilemmas of practice.” Such work “would advance fundamental understandings at the same time that they would work to solve practical problems in real-world settings” (p. 13). This focus would lead to the production of more readily “usable knowledge” (Lagemann, 2002). As researchers, we have developed rich understandings of how technology can foster learning in specialized situations; we now need to develop knowledge about widespread appropriation and use of cognitively oriented technologies by schools and school systems as part of real-world reform efforts. Ultimately, this calls for an augmented research agenda designed to enhance the usability of technology innovations developed by the research community, with positive consequences for scalability and sustainability. To address this issue requires that we introduce new questions and ways of thinking about problems into our research agendas. The framework that guides these questions must combine the best of what we currently understand about learning and teaching with technology with what is already known about the challenges of creating systemic reform and the implementation of innovations in reform contexts.

We begin by clarifying the nature of cognitively oriented technology innovations in school and research contexts. Next we reflect on what is known about fostering the widespread use of technology innovations in schools, and provide a framework for considering the usability of innovations. We then turn to research questions arising from our own experiences in working in systemic reform to frame a systemic research agenda that addresses issues of sustainability, scalability, and usability of cognitively oriented technology innovations, concluding with reflection on issues for the research community that are related to carrying out our proposed research agenda.

COGNITIVELY ORIENTED TECHNOLOGY INNOVATIONS

In our thinking, cognitively oriented technology innovations focus on inquiry and approaches to learning as embodied in national standards documents (e.g., National Council of Teachers of Mathematics, 1989; National Research Council, 1996). These innovations include technology as a core component, but are rooted in cognitive and constructivist learning theories (Bransford et al., 1999; J. S. Brown, Collins, & Duguid, 1989). Cognitively oriented technology innovations

range from intelligent tutoring systems that help students learn mathematics (e.g., J. R. Anderson, Corbett, Koedinger, & Pelletier, 1995) to environments that foster communal knowledge-building and support for writing (e.g., Scardamalia & Bereiter, 1991) to tools that scaffold deep explorations in science (e.g., Linn & Hsi, 2000). In these innovations, technology is employed as a tool to support teaching and learning, as opposed to the object of learning. These innovations often use technology to scaffold teaching and learning practices that would be difficult to achieve otherwise, such as making complex causal modeling accessible to students (e.g., Jackson, Stratford, Krajcik, & Soloway, 1994).

There is a continuum of ways that technology is employed in cognitively oriented technology innovations. Instruction can be delivered via computer, as is the case with intelligent tutoring systems, or computers can be used as resources and “learning partners” in classrooms where much learning takes place “off line.” Cognitively oriented technology innovations may be designed to cover a relatively short period of time, or they may be comprehensive, intended to be used throughout an entire year or across multiple years of instruction. Our focus, however, is on cognitively oriented technology innovations that are closely tied to the regular curriculum and tightly integrated with teaching and learning practices. In instances where this is not the case, such as after-school computer clubs (e.g., Zhao, Mishra, & Girod, 2000), there may be high-quality learning and excellent uses of technology, but we do not include these instances in our consideration of cognitively oriented technology innovations for teaching and learning because such extracurricular uses of technology side-step the challenges of systemic reform and are not designed or intended to influence teaching and learning by teachers in regular school subjects as part of the school day.

The Knowledge Integration Environment (KIE; Bell, Davis, & Linn, 1995; Linn & Hsi, 2000) is an example of a cognitively oriented technology innovation. KIE combined a range of networked software tools with constructivist pedagogical principles in order to foster use of evidence and argument in middle school science. In KIE, learning is organized around generative questions in science, such as whether light travels forever or dies out. Students conduct research collaboratively on the Internet to gather evidence or view evidence developed expressly for KIE, and use argument-support software to organize their evidence and make supportable claims. Students then debate their claims and use of evidence in face-to-face classroom discussion as well as through the use of asynchronous on-line discussion tools. The technological aspects of the learning environment in KIE are designed to complement the face-to-face learning facilitated by the classroom teacher, who must orchestrate collaboration, guide students in their learning, and provide assessment and feedback to students. KIE is challenging for teachers in that they need to understand not only the content embedded in this project (which is just one of several in a broader KIE-enabled science curriculum), but also how to help students use the technology, how to foster collaboration, and how to conduct appropriate assessments. As with

many cognitively oriented technology innovations, access to the Internet is crucial. If the Internet connections or computers are not working, it is difficult to use KIE curriculum. In this way, cognitively oriented technology innovations are often demanding of both the instructional and technology infrastructures of schools. Recently, research on KIE has evolved into a larger effort called the Web-based Inquiry Science Environment (WISE; Linn, Clark, & Slotta, 2003), that is attempting to address issues of scalability and sustainability.

Design-Based Research

Many cognitively oriented technology innovations are developed using design-based methodologies in which researchers work closely with teachers and students to design, develop, implement, and evaluate an innovation in real classroom settings (A. L. Brown, 1992; Collins, 1990). Design-based research has great potential for creating “usable knowledge” (Lagemann, 2002) principally because it is intertwined with practice and makes an attempt to study the complex influences of context on teaching and learning (The Design-Based Research Collective, 2003). Design-based research combines inductive qualitative approaches with quantitative and quasi-experimental approaches, varying the method to suit research questions that present themselves over the life of the collaboration. Research staff generally establish a regular presence in the classroom to support the use of the innovation, sometimes modeling or co-leading instruction with the teacher (Cobb, Confrey, diSessa, Lehrer, & Schauble, 2003). In part, this serves to temporarily establish conditions that are favorable to the innovation’s success. Without these conditions, it would not be possible to study the phenomena or ideas of interest. However, if the conditions depend heavily upon an infusion of extra support from researchers, this may pose a challenge to scalability and sustainability.

Another characteristic of design-based research on cognitively oriented technology innovations is the nature of the participants. As Means (1998) put it,

In the majority of cases (but not always), teachers are voluntary participants, and hence likely both to buy-in to the philosophy of the project and to see the connection of the technology used in the project to something they want to do with their students The disadvantage innovative technology-supported projects often face is the fact that they may not be a good match to priorities with an individual school or district. (p. 7)

Design-based research, which has been a major methodology within the Learning Sciences, has the advantage of grounding the lessons of research on cognitively oriented innovations firmly in dilemmas of practice (Shepard, 2000), but in a specialized way that does not necessarily lead to the sustainability or scalability of the innovations. Work to date, and design-based efforts in particular, have helped re-

searchers to better understand the constraints and contexts for classroom uses of learning technologies. But what happens when the innovation is used by dozens or hundreds of teachers who do not share co-ownership of the design with the researchers and may lack specialized knowledge generated from the collaborative research process?

Design-based research focuses on studying an innovation's use within a classroom or several classrooms, and not necessarily on "external" factors that are necessary for the innovation's support. Writing about methodological issues in design research, Collins, Joseph, and Bielaczyc (this issue) point to the importance of including a school or institutional analysis among the ways that a particular design can be studied. They also argue that system variables, such as ease of adoption, sustainability, and spread, are key dependent variables that should be measured in design-based research. We agree, but believe that if design-based research is going to provide guidance for systemic reform, such variables need to be treated as more than outcome measures, but as a central part of the intervention. If we are to foster truly sustainable innovations, there is a pressing need for an extension of classroom-based design research that focuses on schools and school systems as the primary units of analysis (Snipes, Doolittle, & Herlihy, 2002). As we work to build upon the lessons learned from classroom-oriented design-based research, we need to define questions that explicitly address issues of scalability and sustainability, if we hope for innovations to enter into widespread use beyond their original research contexts. To this end, we now turn our attention in this article to the constraints and needs of school *systems*.

SUSTAINABILITY, SCALABILITY, AND SYSTEMIC REFORM

Developers of cognitively oriented technology innovations want their innovations to be sustainable and scalable. The Learning Sciences is a field rooted in cognitive science, but with practice-oriented objectives. Ideally, the use of innovations will extend beyond the time that researchers are directly involved in the classroom. This is the essence of sustainability. Ideally, the pedagogical ideas and uses of technology that are encompassed in the innovation will spread to teachers' general repertoires. If school systems are able to support the practices embodied in the innovation, then their use will also spread to other teachers within or across schools. This is the essence of scalability.

There are various mechanisms employed in attempts to create sustainability and scale. We have chosen to work in the context of urban systemic reform, in which an innovation is intended to reach many teachers and students within a single school system. A fundamental challenge of work in systemic reform contexts is creating alignment across the components of school systems, such as administra-

tion and management, curriculum and instruction, assessment, policy, and technology (Smith & O'Day, 1991). If the challenge of alignment can be met, an innovation has a better chance of being both sustained and scaled because the alignment of the system creates a stable structure and provides needed support. Moreover, cognitively oriented technology innovations that are well matched to systemic reform goals of school systems, such as standards-based instruction, are more likely to be sustainable, and more likely to be scalable to widespread use. These challenges are not new to the domain of cognitively oriented technology innovations, but have been studied in the context of, for example, the more general problem of curriculum implementation (Snyder, Bolin, & Zumwalt, 1996).

Much has been written about the difficulties of creating systemic instructional innovations in general, such as the nested complexities of school organization, family and community concerns, and professional and regulatory agencies (Cohen & Ball, 1999), and problems related to the adaptation of innovations as they are adopted in new contexts (McLaughlin, 1990). Technology introduces additional issues and challenges that are not as well documented, but which we propose need to become the focus of research. Cognitively oriented technology innovations force researchers, schools, and districts to think across conceptual and organizational boundaries in ways that other innovations typically do not. This is because technology used for curricular purposes is often maintained by organizations in the school district that traditionally have not been involved in classroom learning, and new technologies, such as the Internet, require a high level of coordination between classroom users and central network managers (Fishman & Gomez, 2000; Fishman, Lento, Gomez, & Despenza-Green, 1997; Schofield & Davidson, 2002). Because they draw upon resources that are uncommon in nontechnology curricular innovations, cognitively oriented technology innovations can place stress on the alignment of a district's systemic reform agenda, uncovering areas where alignment is insufficient for the demands of the innovation. We elaborate on these issues below.

Systemic Reform and Technology Innovations: What Works

In their study of successful technology implementation efforts, Means et al. (1993) identified the following six common features: Ready technology access and technical support; instructional vision and a rationale linking the vision to technology use; a critical mass of teachers in technology activities; a high degree of collaboration among teachers; strong leaders; and support for teacher time for planning, collaboration, and reporting technology use. Though Means and her colleagues were not explicitly studying implementations of technology in systemic reform, their description of the conditions for success resonates with systemic notions of alignment. In later work, Means found that cognitively oriented technology projects were more successful at a small scale. In trying to explain why this should be, she concluded that, "many of the features associated with successful technology-sup-

ported reform efforts within individual schools are often missing from large-scale technology implementations” (Means, 1998, p. 11). This is likely also a function of the volunteer nature of the participants, who are able to work as independent (and idiosyncratic) actors within the larger system, and the effect of the intensive but highly localized investment in capacity by researchers. In widespread use, it becomes an order of magnitude more difficult to maintain alignment of all the needed organizational components and participants.

Knapp (1997) argued that a key to successful reform is that the major ideas be grasped by participants at all levels of the school system. This requires investment in professional development, teacher leaders, curriculum, and logistical support. Reform efforts are not successful when they are implemented “top down,” as a function of policy implementation, largely because this perspective does not value organizational learning (Fullan & Miles, 1992). The most successful systemic reform efforts are where the local organization either invents or assumes ownership of the core ideas in the reform (Honey & McMillan-Culp, 2000). This was the case in Union City, NJ, a successful urban systemic reform effort involving cognitively oriented technologies (Center for Children and Technology, 2000). In Union City, the school system itself designed the core of the reform initiative in response to state policies that would have led to the takeover of the district. Union City school administrators invited researchers to join them as consultants, but they never ceded control of the core ideals of the reform. This was also the case in the Hanau Model Schools Project (McNamara, Grant, & Wasser, 1998), where researchers worked with a DoDEA school in Germany to integrate technology throughout the curriculum, developing a comprehensive approach to professional development for the district as a key component of the reform.

A FRAMEWORK TO INSPECT THE SCALABILITY, SUSTAINABILITY, AND USABILITY OF INNOVATIONS

Some of the conditions under which reform and technology within reform become successful have been identified (e.g., Knapp, 1997; Means, 1998; Means et al., 1993), and so have many of the problems that are encountered when trying to use technology at scale in schools (Means, Penuel, & Padilla, 2001). But there is little guidance in the literature for those interested in establishing those conditions or resolving those problems. In order to take this critical next step, it is necessary to conduct systematic research on cognitively oriented technology innovations in a variety of settings. Research on such innovations, as a design-oriented enterprise, attempts to understand the interaction between innovations and their contexts of use in order to increase the *usability* of the innovation. We take the term “usability” from human-computer interaction (Nielsen, 1993), where it connotes a measure of the extent to which those who take up innovations or products are able to employ

them to accomplish their work. If a reform innovation is not usable, it is unlikely to be adopted, and will be neither sustained nor scaled. To understand how to enhance the usability of cognitively oriented technology innovations, it is necessary to frame questions that directly address issues of scalability and sustainability. Our goal is to develop research that enables the linkage of current knowledge to dilemmas of practice (Shepard, 2000) in school systems.

A feature of research on systemic reform and large-scale studies of technology is that researchers normally have an outsider or *etic* perspective on the object(s) of study. The researchers or evaluators are almost never the same people who have worked to create the reform. By contrast, a feature of research on cognitively oriented technology innovations is that the researchers are usually the developers of the innovation (ideally codevelopers with the school participants), and thus have an insider or *emic* perspective. Being “on the inside” fundamentally changes one’s perspective on the reform process. Research to date has told us much about where to look for problems in the systemic uses of technology or reform more generally, but it has not provided guidance about how to *create* or *enable* change. *Etic* approaches attempt to understand and explain, but *emic* perspectives, such as design-based research, struggle to influence and shape the change process in order to create a desired outcome as part of the understanding process. Developing a research agenda that fosters the creation of cognitively oriented technology innovations that are usable in systemic reform contexts is, we believe, the key contribution of this work, building upon earlier research that has described problems and conditions for success.

A core question that needs to be addressed is just what is the definition of a “usable” cognitively oriented technology innovation? A basic answer is that an innovation is usable if a school organization can adapt the innovation to local context, enact the innovation “successfully” (as jointly defined by the school and the developer), and sustain the innovation. This question may also be turned on its head, and answered by examining what aspects of innovations or school systems create challenges for usability.

In earlier work (Blumenfeld et al., 2000), we have argued that there are three dimensions within school systems to which one must attend in order to create usable innovations: school culture, capability, and policy/management. We arrayed these dimensions in the form of three axes originating from a common point (the origin, which represents the current capacity of the district) to form a three-dimensional space we call the “usability cube” (see Figure 1). An innovation can be placed in the space created by these three axes, where the “distance” between the innovation and the origin represents the gap that exists between the capacity required to successfully use the innovation and the current capacity of the district. The creation of usable innovations (and successful reform), conceptualized in this manner, is a process of working to “close the gaps” that exist. This might involve increasing district capacity through professional development or changes in policy, or re-

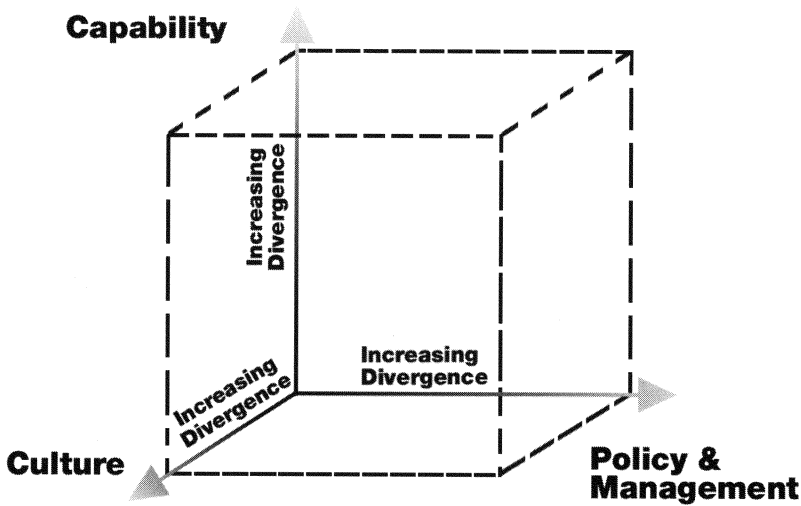


FIGURE 1 Framework to evaluate the usability of innovations (Blumenfeld et al., 2000).

working the innovation so that it better matches existing capacity. We argue that both types of changes are ultimately required in the creation of usable innovations that foster reform. Understanding the opportunities for and processes by which one closes gaps is the area for research that we propose in this article.

To illustrate how this “usability cube” model can be used to describe and even predict the difficulty any particular innovation faces in the adoption process, we have selected two examples from research on cognitively oriented technology innovations that demonstrate two different approaches to becoming scalable and sustainable. The first of these is the work of the Pittsburgh Advanced Cognitive Tutor (PACT) Center work with Cognitive Tutors in mathematics (Corbett, Koedinger, & Hadley, 2001), which have become widely used by schools nationally. The second is the reform effort of the Union City, NJ, public schools (Center for Children and Technology, 2000), which represents a pedagogical, curricular, and technological innovation that has been scaled throughout an entire school district.

Example: Cognitive Tutors

Cognitive Tutors, which were developed at Carnegie Mellon University using Anderson’s theory of human and machine learning (J. R. Anderson, 1983), have been used successfully to teach high school mathematics to students in a variety of set-

tings (Corbett et al., 2001). These systems have been commercialized by an organization called Carnegie Learning Corporation, and are currently being used in hundreds of school districts throughout the country, though not necessarily system-wide within individual school districts (Corbett et al., 2001). By most measures, the Cognitive Tutors can be said to have successfully reached scale.

Viewed from the perspective of the usability cube (Blumenfeld et al., 2000), we would say that Cognitive Tutors are a likely candidate for such success. For example, knowledge of the common errors that students make in a domain, and how to correct them, is an instance of crucial pedagogical content knowledge for teachers (Shulman, 1987). Because Cognitive Tutors are constructed around a model of learners' common errors and paths to understanding, they provide support to teachers in this difficult area, thus reducing a potentially challenging capability gap for teaching mathematics. Many teachers express concern about their ability to facilitate students moving at their own pace through material when using tutors, as opposed to in unison as is typical in more familiar didactic approaches to instruction. This potential capability gap is also reduced by the software, as its design "provides just the support students need to move successfully at their own pace" (Corbett et al., 2001, pp. 241–242), thus allowing teachers the flexibility to focus on other issues, such as spending more time with students who are struggling. It is important that schools have capability in the form of technical support to keep the computers that run the tutors working, and Carnegie Learning Corporation has focused some efforts in this direction (Corbett et al., 2001). However, because the tutors utilize standard computers available in most schools, and do not require access to networks, the range of technical problems is reduced. In early stages of development, the tutors required much greater computational power than was typically available in schools, but recognizing this as a problem, the designers worked to close this particular capability gap (Corbett et al., 2001). Cognitive Tutors are a good fit with school culture, because they supplement but do not replace mathematics teachers, and are flexible with respect to whether computers are located in the math classroom or in a separate computer lab. Furthermore, Carnegie Learning sells the Cognitive Tutors in conjunction with complete mathematics curricula that are designed to fit directly into existing categories of high school math subjects (e.g., Algebra, Geometry, etc.). In terms of the policy/management dimension, Cognitive Tutors speak directly to a felt need of school administrators, as they have demonstrated success at helping students achieve in mathematics, an area that many school districts find challenging and which is highlighted in the current state and federal policy environment by the No Child Left Behind Act (U.S. Department of Education, 2001).

Because the demands of Cognitive Tutors do not overly stress schools' capability, policy/management structures, or culture, they do not present a large usability challenge to schools, and are therefore good candidates for scalability and sustainability. At this point, one might ask why all technology innovations are not created in the mold of Cognitive Tutors? The reason is that, while the learning the-

ory upon which Cognitive Tutors are based (J. R. Anderson, 1983) can be readily applied to domains such as mathematics, computer programming, or other relatively well-bounded problem spaces where designers can anticipate likely student errors, the theory has a much harder time with open-ended domains, such as humanities. For the same reason, Cognitive Tutors are not currently a good fit with open-ended, inquiry-oriented pedagogies because the content and direction of student learning cannot usually be predicted in advance, though some progress is being made in this direction (e.g., Koedinger, Suthers, & Forbus, 1999).

Example: Union City, NJ

The cognitively oriented innovation developed by the Union City, NJ, public schools as part of their decade-long reform effort (Center for Children and Technology, 2000) represents almost complete adoption of an innovation within a single school district, as opposed to broad adoption by schools spread across the country as was the case with Cognitive Tutors. Union City transformed itself from a district on the verge of a state takeover due to low test scores into the highest-performing urban district in their state. The core of the Union City innovation is a redesign of their curriculum with a focus on second-language literacy (they serve a mostly Spanish-speaking population), with complementary pedagogical reforms to create a student-focused and literacy-based approach with an emphasis on collaboration and communication. Technology plays a central role in the Union City reforms, as students create a wide variety of computer-based writing and multimedia projects as part of the communication objectives of the district, and rely heavily on the Internet as a source of texts for use in all subject areas (Honey, Carrigg, & Hawkins, 1998). When Union City embarked on their innovation, they faced tremendous challenges in terms of all three dimensions of the usability cube: capability, culture, and policy/management. The process of reform used by Union City is a good example of how these dimensions interrelate, and how solutions on any particular dimension can require that one simultaneously address other dimensions.

An initial challenge faced by Union City was to address gaps in teachers' capabilities, both with respect to the new pedagogy and the new technologies. The district made use of extensive professional development, a typical approach to reduce capability gaps. However, they simultaneously made changes in policy/management by focusing the reform on early grades first, and allowing teachers who were not willing to participate to either move up grades (where the reforms were not yet installed) or to switch to schools in the district that were not in the first wave of reforms. The district also focused on changing the culture of both the classroom and professional development by inviting teachers to bring students with them to workshops. Students were asked to focus on the technologies, and then act as classroom aides to help alleviate teachers' capability concerns. By having students integrally involved in the enactment of the innovation, and by starting with younger grades,

Union City created a situation where the students *themselves* acted as change agents, expecting their middle school instruction to be similar to what they experienced in the early grades, thus exerting an internal force for change that affected school culture (Center for Children and Technology, 2000). This is in stark contrast to projects that inadvertently work against the students' pedagogical expectations by altering a single "experimental" classroom in the older grades (e.g., Gomez et al., 1998). At this point, the district made it possible for teachers who were *still* resistant to the changes to either retire or leave the district, a strong show of policy/management support with implications for school culture. Because the Union City innovation was "home grown," and led by district administrators, it was never much of an issue to reduce gaps in policy/management. By focusing on that dimension, the district was able to create conditions for closing the gaps in capability and culture. What worked in Union City is extremely specific to its context, but it provides a telling example of working to reduce gaps between district capacity and the demands of an innovation in order make the innovation usable, achieving both district-wide scale and sustainability.

In the remainder of this article, we turn to an examination of our own experiences in working collaboratively with schools to create scalable and sustainable cognitively oriented technology innovations within the context of systemic reform. We draw upon these experiences to outline a proposed research agenda. We build upon findings from earlier research that identify places where the demands of innovations and the capacity of school contexts are likely to diverge, and suggest new questions that help us understand how to close the gaps. We examine issues that pertain to the capacity of school organizations with respect to creating usable cognitively oriented technology innovations in systemic reform, and questions that pertain to the organization of the research community. An additional challenge for any such research agenda is that, working from an inside or *emic* perspective, the questions asked should have utility for *both* the research and practitioner communities. Our hope is that, in pursuing this proposed agenda, the field will develop a body of examples of how cognitively oriented technology innovations come to be usable in a range of systemic contexts. This research will ideally reveal a range of strategies employed by researchers and practitioners to reduce gaps between the capabilities of schools and the demands of innovations, yielding information that others can employ to create the needed conditions for the successful scaling and sustainability of cognitively oriented technology innovations.

A RESEARCH AGENDA FOR EXPLORING SCALABILITY, SUSTAINABILITY, AND USABILITY OF INNOVATIONS

Perhaps because the success of any instructional innovation is closely linked to teaching practices, much research in the Learning Sciences has focused on class-

room teaching and student learning. But educators at all levels of the system are key to the eventual success of classroom-level innovations. Thus, a focus on the capacity of the larger school organization is critical when seeking to increase the usability of cognitively oriented technology innovations. We argue that the following areas are critical components of a research agenda on the usability of cognitively oriented technology innovations in systemic reform contexts: teacher learning, assessment, technology planning, and organizational structure and leadership. We explore each area below in the context of our own experiences in developing a cognitively oriented technology innovation with an urban school district.

Our work in the Center for Learning Technologies in Urban Schools¹ provides a window into the challenges of conducting design-based research on creating usable cognitively oriented technology innovations as part of systemic reform. Our goal is to develop and integrate inquiry-based science curricula with embedded technology in the Detroit Public Schools. This involves extensive curriculum development (Singer, Marx, Krajcik, & Clay-Chambers, 2000), design and integration of technologies to support student and teacher learning (Marx, Blumenfeld, Krajcik, & Soloway, 1998), broad-based professional development (Fishman, Best, Marx, & Tal, 2001), and collaboration with teachers and school and district administrators (Murray, Fishman, Gomez, Williams, & Marx, 2001).

Our work began in the fashion of most cognitively oriented technology innovations. We developed materials using focused design-based approaches (Krajcik, Blumenfeld, Marx, & Soloway, 1994). When we believed that our innovation was working well (Krajcik et al., 1998), we attempted to expand our work to include many settings as part of Detroit's systemic reform initiative. In the process, we encountered multiple challenges to the usability of our innovation. These included (aside from challenges related to technology use, which we will discuss in depth below), the educative design of the curriculum (Schneider & Krajcik, 2000), the effectiveness of the professional development (Fishman, Marx, Best, & Tal, 2003), and the relationship between school and community (Moje, Collazo, Carrillo, & Marx, 2001). In working to resolve these challenges, we developed and began to employ our "usability cube" model to gauge the "fit" of our innovation to the capacity of the school district's culture, capabilities, and policy/management structures (Blumenfeld et al., 2000). Through this process, we uncovered a broad range of "gaps" between the capacity of the district and the demands of the innovation, which when taken together comprise the dimensions that a research agenda on cognitively oriented technology innovations needs to address. Below, we consider the major gaps that we encountered and propose specific research questions to address, and hopefully narrow, each gap. The four issues we choose to highlight (see

¹More information about LeTUS can be found at <http://www.letus.org/>. More information on hi-ce can be found at <http://www.hi-ce.org/>.

Table 1) are likely to be relevant to cognitively oriented technology innovations in a broad range of systemic reform contexts.

Creating Scalable and Sustainable Teacher Learning

There is little doubt that professional development is a key component of successful systemic reform initiatives (Supovitz, 2001). Gaps in teacher capabilities in terms of content knowledge and pedagogical knowledge present a serious hurdle to reform. Recent research has provided evidence on the value of focusing professional devel-

TABLE 1
Key Issues to Explore in Design-Based Research on Cognitively Oriented
Technology Innovations in Systemic Reform Contexts

<i>Issue</i>	<i>Areas for Research</i>
1. Teacher learning	How can PD to enhance teachers' capability be made usable for teachers with a broad range of motivation, skill, and working contexts? How do district policies and cultures shape the participation of teachers in PD? How can PD be scaled to reach thousands of teachers and maintain high quality standards? How can teacher learning be linked to student learning and standards-based accountability systems?
2. Assessment	How can instructionally useful forms of assessment inform standards-based accountability systems? How can urban school systems balance the time demands of high-stakes testing with the demands of cognitively oriented innovations?
3. Planning for technology	What are the characteristics of successful technology planning practices? How can district policies for acquiring, distributing, and maintaining technology be aligned with instructional needs for technology? How can different innovations effectively leverage and share technology resources within a district?
4. Organizational structure and leadership	How can schools examine their organizational structures to promote realignment around the demands of cognitively oriented technology innovations? How should school organizations be realigned to better support innovations? How can distributed leadership be fostered in school organizations? What forms of professional learning help school leaders build capacity to support reform and innovations?

Note. PD = professional development.

opment on teacher content knowledge (Wilson & Berne, 1999), on student learning (Kennedy, 1999), on the importance of extended participation, and alignment with broader reform agendas (Garet, Porter, Desimone, Birman, & Yoon, 2001; Supovitz, Mahyer, & Kahle, 2000). Cognitively oriented technology innovations complicate the picture further, adding knowledge of technology and how to teach with it to the list of capabilities teachers must master (Margerum-Leys & Marx, *in press*). Though many projects have focused on professional development and technology, relatively few have focused on technology integrated with core curricular goals (President's Committee of Advisors on Science and Technology, 1997). Notable exceptions that have focused on professional development in systemic settings include the work with the Hanau Schools in Germany (Wasser & McNamara, 1998), the work in Union City, NJ (Center for Children and Technology, 2000), and work to integrate calculators comprehensively throughout mathematics curriculum in Montana (Zucker & Marder, 1998). Cognitively oriented uses of technology often employ challenging pedagogical approaches, such as project-based learning (Blumenfeld et al., 1991). Teachers may need to learn new content or new approaches to managing instruction with technology, thus widening the usability gap in terms of the capabilities needed to employ the innovation. A further challenge comes from the large numbers of teachers in urban districts. How can a successful professional development program be scaled to hundreds or potentially thousands of teachers and still maintain quality and focus? One proposed solution has been to employ on-line learning environments, and there is promising work in that area, though more is needed (Shrader et al., 2002).

Our work in Detroit attempts to provide rich professional development opportunities for teachers that focus on the curriculum instead of on the technology. To do this, we have employed a wide variety of approaches, including extended summer and monthly Saturday workshops, teacher study groups, in-classroom support, educative curriculum materials, and on-line professional development tools. This broad-based effort was intended to allow us to explore the range of opportunities for teacher learning in order to better understand linkages between particular professional development practices and changes in student outcomes (Fishman et al., 2003). We also realized that such an intensive professional development enterprise was ultimately unsustainable, and so sought to document both costs and benefits of different approaches in order to provide guidance to the district for future investment of effort, in particular attempting to link professional development to student outcomes. If these efforts are to be sustainable and scalable, it is important that the district assume responsibility for organizing and operating the professional development, a process which is now underway (Fishman, Fogleman, Kubitskey, Peek-Brown, & Marx, 2003; Margerum-Leys et al., 2003). We are also exploring the use of on-line learning environments as a means of creating sustainable and scalable teacher learning opportunities (Fishman, 2003).

In working to close the gaps between teachers' capabilities and the demands of our innovation, we uncovered a range of challenges to the usability of the profes-

sional development components of our innovation that were related to both the culture and the policy/management dimensions of our usability cube (Blumenfeld et al., 2000). For example, an extended program of professional development must account for teachers who have different (and changing) capability levels and who will therefore be on different learning trajectories. There will be differences in motivation and participation (Supovitz & Zeif, 2000), problems with teacher turnover and mobility, problems created by teachers teaching out of the subject area of their credentials, and the problem of finding time in teachers' work days to engage in teacher learning. All of this must be reconciled against the backdrop of organizational culture that favors one-day or other short term approaches to professional development. What we wish to highlight is that what at first appears to be a usability or design challenge on the dimension of teachers' capability is in fact a multifaceted problem that involves district policy/management practices with respect to teacher learning as well as organizational culture. Below we present areas for further research on teacher learning with respect to cognitively oriented technology innovations that will provide information about how to close the gaps that arise, and potentially lead to design-based research that on innovations intended to be used beyond those schools where the researchers are guiding the implementation process.

Proposed areas for further research. There is much needed research in the area of policy/management challenges that arise in attempting to develop systemic cognitively oriented technology innovations. How can teachers who do not have ready access to computers in their own classrooms gain confidence and skill in using computers as part of their teaching? How can teachers' time constraints be overcome in order to create sufficient time for extended participation in professional development? How can district and building administrators create sufficient incentives for teachers to participate in professional development beyond normal working hours? How can districts create professional development that reduces gaps in teachers capability to employ technology as part of cognitively oriented curricular innovations, integrating knowledge of technology with knowledge of content and pedagogy? How can high-quality professional development be extended to potentially thousands of teachers district-wide while maintaining quality and focus? What role does or should on-line professional development play in addressing these questions?

Linking Cognitively Oriented Technology Innovations to High-Stakes Assessment

A key "feature" of systemic reform is new assessment and accountability measures that place tremendous pressure on schools and present a critical alignment challenge for systemic reform programs (Goertz, 2001), creating gaps between the policy/management structures of the district and the innovation. This pressure has in-

creased with the introduction of the “No Child Left Behind Act” (U.S. Department of Education, 2001), which increases the amount of testing and raises the stakes for underperforming schools. Developers of cognitively oriented technology innovations must understand how their innovations fit with federal, state, and local policy initiatives, and make such linkages transparent for schools. New technologies present both new forms of student work and new means for conducting assessment (Pellegrino, Chudowsky, & Glaser, 2001). What kinds of assessments are both consistent with the goals of cognitively oriented technology innovations *and* with the demands of standards-based accountability? Goldman (2002), in her AERA Division C Vice-Presidential address, argued that it is the responsibility of the cognitively oriented technology research community to create new forms of assessment that bridge this gap.

We have attempted to address this issue in our work by explicitly connecting our curricular materials to district and state frameworks for science education (Singer et al., 2000). However, this approach does not resolve all of the potential problems, as state testing itself can be out of alignment with the standards (Goertz, 2001; Porter & Smithson, 2001). This is because standardized testing is a distal measure with respect to local teaching and learning in schools (Ruiz-Primo, Shavelson, Hamilton, & Klein, 2002). Paradoxically, the time districts devote to standardized assessment activities cuts into the amount of extended time available for focus on inquiry-oriented learning as called for in the standards (e.g., National Research Council, 1996). In our experience in Detroit, as political pressures and public opinion created a press for more accountability through the use of standardized assessment, increasingly more classroom time was given over to preparation for and taking a range of assessments such as the state mandated tests and comprehensive batteries, which created both a policy/management and a culture challenge to the usability of cognitively oriented technology innovations. As more of the school year in Detroit was devoted to external assessment (in noncontiguous blocks), our cognitively oriented technology innovation had to compete for the limited time remaining. In such an environment, it became increasingly difficult for teachers to carry out extended inquiry-projects during certain months of the school year. Further research in this area is crucial. If we do not help schools find ways to address assessment needs productively, cognitively oriented technology innovations become less usable in light of the culture of standards-based assessment to which all urban schools must respond.

Proposed areas for further research. How we might shape or align the instructionally useful forms of alternative assessment employed in many cognitively oriented technology innovations to meet the accountability needs of districts, thus reducing a major policy/management gap in the usability of cognitively oriented technology innovations? Many cognitively oriented technology innovations have developed rich means of portfolio or artifact assessment as part of their research

(Pellegrino et al., 2001), but how might these approaches be applied across an entire school district? This is a key scalability issue for assessment. Can we find ways to conduct rigorous and meaningful assessment tasks, aligned with cognitively oriented innovations, in ways that are scalable within and comparable across schools and districts? How can the content and learning strategies embodied in the technology use(s) be represented in high stakes tests? From a policy/management perspective, how can school organizations balance the time demands of high-stakes testing with the time demands of inquiry-oriented teaching and learning?

Planning Effectively for Cognitively Oriented Technology Innovations

Teachers require ready access to technology in order to use cognitively oriented technology innovations. This is a usability issue for innovations that spans school culture and policy/management issues. If school and district technology plans are not aligned with goals for teaching and learning, teachers and students will not be able to gain necessary access to technology. Unfortunately, this is frequently the case, as technology plans rarely focus on teaching and learning and instead read like shopping lists for hardware and software, functioning more as a bureaucratic placeholder than a real plan for action (Fishman & Pinkard, 2001). Research on effective technology planning processes emphasizes the importance of forming a planning group with representatives from all constituencies within the school community, and of having a strong focus on curricular goals (Regional Technology in Education Consortia, 1996). However, much of the work in this area still approaches the technology planning process independently from the particulars of school reform (L. S. Anderson, 1999). Furthermore, the planning processes advocated in much of the literature are designed to fit within existing organizational structures, not to challenge them or seek reciprocal changes between the capabilities of extant structures and desired innovations (Fishman & Pinkard, 2001).

Technology must be acquired before it can be used, but the purchasing and distribution of technology is a hazy area of school policy at best. In Detroit, for example, curriculum decisions are centralized, but technology acquisition, although guided by district standards, is decentralized and building principals make individual decisions about technology purchases. The result is that computer capabilities are not uniform, and machines are difficult to maintain and upgrade. Issues such as these are normally overlooked by researchers, but they are serious impediments to the use of cognitively oriented technologies, which typically place high demands on computers.

Teachers and students need access to technology. Administrators need help in understanding the implications of placing computers in labs versus classrooms. Either solution has problems, because computers distributed throughout classrooms are difficult to secure and maintain, while centralized computer labs create sched-

uling conflicts between teachers that make it difficult to utilize technology as a tool to be used throughout the curriculum. Technology breaks down frequently, and few schools have computer technicians on staff to provide maintenance. Labor rules in urban school districts are tricky regarding personnel who are not certified to teach, especially when those positions demand high salaries. The result is that teachers cannot get the help they need when they need it, which becomes a strong disincentive to using technology. A national survey of technology use in schools found that more than two-thirds of teachers nationally reported not being able to get timely technology help (Ronnkvist, Dexter, & Anderson, 2000). Furthermore, the survey indicated that support was less available in low than in high-SES districts, reflecting yet another dimension of the “digital divide.” Everyone seems to recognize this problem, but the cost of solving it is so high that no solution is forthcoming. Temporary solutions such as the “e-rate” telecommunications discount program (Carvin, 2000) can provide some relief to urban districts, but these programs are temporary fixes, and do not represent mature thinking about technology whereby ongoing costs become line items in a school budgets, not special expenses supported by external grants and awards.

In our work with Detroit, we attempted to help schools think strategically about their technology acquisitions, both at the building and at the district level. This proved challenging in that there were several different organizations with responsibility for technology (see next section), but no clear chain of authority that connected them together. Another challenge was that our particular cognitively oriented technology innovation had to coexist with ongoing uses of technology in the district (many of which were drill-and-practice type of activities), and there was frequent competition for resources. This was in part due to the lack of a unified district-level plan for technology that assigned clear priorities to different initiatives, leaving principals on their own to make decisions in this regard. Finally, even in situations where everything was properly aligned for the use of our cognitively oriented technology innovation, there were still serious maintenance problems that could prevent activities from proceeding. Our intermediate solution to this challenge was to place our own technology support personnel in the schools to ensure that technology would be in working condition when teachers needed it, temporarily (but not sustainably) reducing the gap in capability for using the technology associated with our innovation.

Proposed areas for further research. Researchers working with cognitively oriented technology innovations need to conduct research to determine the characteristics of successful technology planning practices. What planning approaches create opportunities to evaluate gaps in existing capacity and the demands of cognitively oriented technology innovations? How can districts align their plans for acquisition of, access to, and support of technology with their teaching and learning goals? How can different technology-related innovations effec-

tively share resources within a school or district? How can school budgets and contracts be realigned to support ongoing infrastructure costs for technology? What conditions are needed in order for a technology plan to be implemented effectively, and how are those conditions fostered?

District Organization, Leadership, and the Usability of Technology Innovations

As alluded to in the previous section, we have found that the structure of the district organization itself can facilitate or impede the usability of cognitively oriented technology innovations. Much classroom technology is designed to be “stand alone,” to be used by students working in isolation on tasks such as drill and practice or working with multimedia presentations. Cognitively oriented technology innovations, however, often make use of networks to facilitate collaboration and data sharing among students and teachers. The introduction of networks brings to the surface many areas where district organization does not align well with the goals of school reform or cognitively oriented technology innovations.

The Internet is both one of the most promising and at the same time the most challenging technologies to be employed in classrooms (Fishman & Gomez, 2000; Schofield & Davidson, 2002), making organizational communication problems apparent. This is because the Internet connection to the classroom is dependent upon cooperation and coordination with multiple organizations within the school system. A teacher can plan a lesson using stand-alone software and, assuming that the computers are working and available, be confident that the tools will work when called upon. There is no way, however, for a classroom teacher to predict whether or not the Internet will be available when needed. In K–12 settings, especially urban settings, the Internet is “down” more than it is “up,” making such planning a gamble. This introduces challenges to usability, in that the classroom can no longer operate in isolation from its surrounding school or district context. In order to use the cognitively oriented technology innovation, teachers are dependent upon a broad range of district personnel beyond the classroom.

In many large school districts, responsibility for technology management is assigned to centralized management and information services (MIS) departments. These departments were responsible for computers and networks for district business functions long before the classroom use of computers became popular (Hodas, 1993), and were obvious candidates for overseeing the instructional uses of these networks. The key problem is that MIS departments generally do not have “support for learning” as a central part of their mission. In our work, we discovered that curriculum and technology administrators had not previously had occasion to discuss how their two organizations should interact. The introduction of our cognitively oriented technology innovation provided the impetus for these discussions, in which it was discovered that in fact the two parts of the organization had

divergent goals and missions. In large part, what is needed is a “tightening” of the “loose coupling” for which educational organizations are well known (Weick, 1976), such that different elements of the organization (central administration, school administration, and classroom teacher) are accountable to one another and have clear chains for reporting problems and seeking resolutions (Elmore, 2000).

Because cognitively oriented technology innovations require support, input, and knowledge from so many diverse components of school organizations, they stand to benefit from an emerging view of school leadership called “distributed leadership” (Spillane, Halverson, & Diamond, 2001). This view posits leadership as an emergent property of how organizations operate, instead of as traits held by individual actors. In an environment that values distributed leadership, knowledge, feedback, and most importantly, decision making authority can be shared among those who are most integrally involved in enacting the innovation, thus decreasing usability issues by dealing with potential barriers as they emerge. In making this argument, we are not calling for “site-based management” or other forms of decentralizing school organizations. Indeed, research on what works in school reform among large urban districts indicates that a strong central voice and direction for reform is a key to success (Snipes et al., 2002). However, having a strong centralized vision for district change is not the same thing as consolidating the necessary leadership and knowledge to drive the reform in a centralized authority. Indeed, reforms on the whole are more likely to succeed when people at all levels of the organization share ownership of that core vision and feel empowered to act on its goals (Elmore, 2000).

In our work with Detroit, the need for a distributed view of leadership became apparent in efforts to establish classroom Internet connectivity. Central office personnel with decision-making authority for Internet access had no established avenues for learning about how the Internet was to be deployed in individual school buildings, and building-level administrators were often unaware of the specific network-connectivity demands of teachers using the LeTUS curriculum materials. This led to situations where school personnel believed that they had done what was required to enable classroom use of the Internet, but had no way of knowing whether their solutions matched with the needs of those below them in the organization. In an attempt to resolve these disconnects and encourage a distributed approach to leadership within the district, we worked to create contexts where administrators could meet with their peers and others in the system, including teachers, to uncover and discuss the issues they were facing related to the use of cognitively oriented technology innovations (Murray et al., 2001).

Proposed areas for further research. Organizational structure issues create serious usability gaps for innovations along dimensions of both district culture and policy/management. How can districts best be helped to examine their organizational structures in light of the demands of cognitively oriented technology innovations? How should organizations be realigned to increase the usability of

cognitively oriented technology innovations? How can schools foster distributed leadership for technology innovations that spans the organization? What forms of professional learning help school leaders build their capacity to support reform?

ISSUES RELATED TO THE RESEARCH COMMUNITY'S CAPACITY

The challenges for schools and school organizations that we describe above will sound familiar to anyone who has studied the integration of technology into K–12 classrooms, and probably also to those who have worked with instructional innovations in general. What makes research on the use of cognitively oriented technology innovations in systemic reform different is that one is forced to consider issues of school capability, culture, and policy/management simultaneously in order to come to an understanding of potential challenges to usability. Thus far, our discussion may seem as if we place the burden for closing usability “gaps” on schools and school personnel. To the contrary, we wish to highlight a point we made in describing our “usability cube” framework (Blumenfeld et al., 2000), which is that reducing barriers to usability is a two-way street. It is as critical for us, as researchers, to learn how to adjust the demands of our cognitively oriented technology innovations as it is for school organizations to embark upon changes to meet innovations’ demands, while preserving the core principles of the innovation. It is in this way that the boundaries of design-based research go beyond the technical structures of the innovation. There are also issues that, while perhaps not areas for research in themselves, are important with respect to the ability of the research community to carry out systemic research on cognitively oriented technology innovations. Below we discuss two of these issues: How researchers consider the nature of an “innovation,” and the critical issue of collaborative relationships between schools and academic research organizations.

The Nature of Innovations

To make cognitively oriented innovations more usable, researchers must reconsider just what, exactly, *is* an innovation in a systemic reform environment. The research literature on curriculum innovations describes two primary perspectives: “fidelity” and “mutual adaptation” (Snyder et al., 1996). The fidelity perspective seeks to keep innovations intact, and measures success of implementation in terms of divergence from the intended innovation. The essence of this perspective is captured in the very name of the “technology transfer” offices in many universities. The mutual adaptation perspective, which was first highlighted by the Rand “Change Agent” study (Berman & McLaughlin, 1975) recognizes the importance of local re-invention of innovations in order to better match the norms (and capac-

ity) of the adopting organization. We entered into our collaboration with Detroit knowing that we did not want to employ a “fidelity” perspective towards our innovation, but we were unprepared for the number and range of changes to our innovation that became necessary as we negotiated to reduce the capability challenges involved in making our innovation usable. We began with a view of diffusion for our innovation where we hoped to work with early adopters in order to build a critical mass for widespread adoption throughout the district (Rogers, 1995). As we realized that our initial attempts to create capacity within the district were not going to be successful, we shifted our perspective towards mutual adaptation. In doing so, we faced an inherent challenge of this approach – variations that cause innovations to become very different than originally envisioned, potentially weakening the impact of the innovation (McLaughlin, 1990). A key challenge in systemic research on cognitively oriented technology innovations is to understand when modifications violate core principals of the innovation, and to make sure that these changes are avoided. Schools will “push” on innovations to conform to business-as-usual (Cuban, 1986). At the same time, researchers need to push back on the school to preserve the core principles of their innovations. For example, administrators may be reluctant to grant science teachers greater or more flexible access to the computer lab than other teachers. Innovation designers may be reluctant to alter the demands their innovation make on access to computers. Finding a compromise is likely to engage both policy/management structures and school culture issues.

How can researchers create flexibility in their innovations to allow for mutual adaptation that preserves core principles? An example of how *not* to do this comes from past efforts to create “teacher proof” innovations. This is done either by attempting to direct the teacher’s every action, or through the use of technology, remove the teacher from the role of instructor altogether. More recently, developers of cognitively oriented technology innovations have moved to create flexibly adaptive materials, in order to enhance their fit with divergent classroom contexts (Squire, MaKinster, Barnett, Leuhmann, & Barab, 2003) and encourage mutual adaptation. But precisely how to do this is still an open question. An approach taken by many projects is to under-specify the curriculum, perhaps presenting only a loose framework of goals and suggested activities, but leaving it to teachers to define the details. This approach caters only to a minority of teachers who are able to invest effort in such development (Cohen & Ball, 1999). Teachers are unlikely to be willing to develop a whole curriculum around an underspecified innovation, as Cuban and colleagues found in their examination of why technology in schools is not well utilized even when broadly available (Cuban, Kirkpatrick, & Peck, 2001). More specification around how the innovation “should” be used is necessary, but within broad parameters. In addition, for many teachers in systemic reform contexts, the curriculum materials themselves can carry crucial information about how to use the technology, as in the vision of “educative” curriculum materials called for by Ball and Cohen (1996).

In our work, we attempted to create materials that were educative, highly developed, and specific about how teaching might proceed (Schneider & Krajcik, 2000). In professional development, we attempted to stress with teachers that our materials *could* serve as a complete guide if teachers wished, but if they wanted to adjust activities to better suit their own particular goals and contexts (especially for scheduling constraints), they should feel free to do so. In practice, novice teachers often felt uncomfortable deviating from the printed curriculum, and as a consequence were often unable to complete it. Alternatively, when teachers *did* feel comfortable enough to make modifications, the changes sometimes undercut the intention of the innovation, such as when teachers would omit technology use altogether due to lack of comfort with or availability of technology (Blumenfeld et al., 2000).

How Can Collaborative Partnerships Enhance the Usability of Innovations?

Strong collaboration between school systems and the developers of cognitively oriented technology innovations is critical for addressing gaps of culture, capability, and policy/management. All parties must own and be committed to the innovation (Honey & McMillan-Culp, 2000). This goes further than simple endorsement by the central office. It involves creating a common vision and plans to achieve it. It means that researchers need to work with educators to specify and develop plans for enactment that account for divergences between the capacity of the school system and demands of the innovation. Plans for enactment must be developed at all levels of the system with mechanisms aimed at integration and coordination so that everyone sees the innovation as part of district rather than outside efforts and strives to make it work. Researchers need to explore the dimensions of these partnerships. What are the components of a successful partnership? How are such partnerships fostered? Confrey and her colleagues found that failure to establish partnerships and create a shared understanding of the goals and research approach was a proximal cause of a project being forced out of its school setting (Confrey, Bell, & Carrejo, 2001). The Schools for Thought project in Nashville encountered a similar issue with respect to the local school board (Goldman & Cottom, 2001), where local politics created unforeseen challenges to a project that was otherwise successful. The lack of stability among top school administrators, particularly in urban districts, presents a challenge to any attempt to create educational innovations that by their very nature are intended to be implemented over an extended period of time. In situations where leadership shifts, an external research partner might in fact be a source of continuity, helping to ensure that activities at the level of classrooms, schools, and middle levels of district management find an audience with new top leadership, in order to promote continued effort for promising innovations. This was our experience in Detroit, where the LeTUS partnership helped to provide a consistent goal for curricular reform that has, so far, lasted through

three changes in district leadership. In part, this was because the partnership was seen as a source of continued external visibility and funding for the districts' ongoing efforts at reform.

In our experience with Detroit, we found that our extended collaboration was important for establishing trust with schools, for bringing multiple viewpoints to inform strategy, for opening doors to solve problems such as making the Internet available to specific classrooms, and for creating specialized professional development programs to serve teachers involved in our cognitively oriented technology innovation. Tenacity is a key element of our successful collaboration with Detroit. That we work as partners with schools for periods of years is viewed positively by teachers and administrators; they know that we will be there when they need us. To be successful, these partnerships must outlast the typical duration of current funding cycles or the notoriously short tenure of urban school administrators. But as academics, there will come a point where we will move on to other sets of issues, perhaps in another location. For this reason, researchers and schools need to work in collaboration with entities that can take stable innovations and support their ongoing use in school systems. A key to our partnership was the presence of a champion within the district, who helped to legitimize our innovation within the district. Are the presence of such individuals a matter of happenstance? Or can research uncover strategies for identifying and working with champions? How can innovations be made sustainable when champions leave the district or are replaced?

In addition to collaboration between researchers and schools, there also needs to be collaborations among researchers, and between researchers and commercial enterprises. The layers of challenge inherent in systemic research on innovations that we describe are daunting; the broad variety of expertise and effort needed to address these challenges will undoubtedly be overwhelming for researchers working by themselves. For that reason, researchers who employ technology for learning would benefit from establishing linkages to other researchers working in school reform to build larger organizations that *do* have the capacity to address the varied needs of reform. Our own research group combines expertise in computer science, educational technology, psychology, science education and language and literacy. Other research groups have reached out across projects to form innovative collaborations, such as Schools For Thought, which combines three different cognitively oriented technology innovations into a district-wide reform effort (Williams et al., 1998). How can these kinds of multiorganizational collaborations be fostered?

Can multiple cognitively oriented technology innovations exist in a shared school context? At the moment, this is problematic due to the piecemeal nature of much technology design. The technologies that are most used in schools are general-purpose tools that were developed for businesses, such as word processors and web browsers (Becker, 2000). Software tools developed by the research community, on the other hand, are usually developed for a specific purpose. Each tool has a unique interface, requiring teachers and students to learn new commands, skills,

and metaphors. The learning curve is tremendous, particularly for a tool that students might use once, and only briefly. Principles of software usability (e.g., Nielsen, 1993) and learner centered design (Soloway, Guzdial, & Hay, 1994) predict that such hurdles are strong disincentives to the adoption of technology. In our work, we have found greater success by developing tools that are used in many curriculum units across grades. How can the field develop shared frameworks for the design of common technology interfaces? This is a serious challenge that can only be resolved through collaboration.

The issues of collaboration and the manner in which we conceptualize cognitively oriented technology innovations are potentially significant barriers to the usability of those innovations. If we are to make progress on closing usability gaps, it is essential that we reexamine the assumptions inherent in our own research organizations that may impede our ability to help schools make effective use of our innovations.

CONCLUSION

Why have so few cognitively oriented learning technologies found a place in the everyday practice of teaching and learning in K–12 schools? We argue that a primary reason is that research to date has not focused on issues of how such innovations function at the level of school systems. This results, in part, from the fact that much design-based research focuses on a designed product or resultant theory and not the system variables that impact the scaling potential of the work beyond the sites where the research was carried out. We have argued for extending or conception of design-based research to include research on system-level issues that impact the scalability, sustainability, and ultimately the usability of innovations. We also described a framework for examining the problem space for such research that explores usability in terms of “gaps” between the culture, capability, and policy/management structures that exist within schools and the demands of cognitively oriented technology innovations. Our experiences working in the context of systemic reform reveal that solutions to these problems do not result from focusing on any one of these dimensions to the exclusion of others. To the contrary, it is necessary to conceptualize research that addresses multiple dimensions in a way that reflects the real-world complexities of the settings in which cognitively oriented technology innovations are to be used. It clear to us that no technology innovation, no matter how compelling or well conceived, will become broadly used and sustained without attending to the issues surrounding the “dilemmas of practice.” Previous research has uncovered many of these issues, but has not offered solutions. In part, this may be because each issue has multiple components that need to be addressed across areas of capability, culture, and policy/management. The ultimate goal of our proposed extensions to design-based research is to uncover the

means by which cognitively oriented technology innovations can be made highly usable by schools. It is now possible to build upon research on cognitively oriented technologies at the classroom level in order to understand the constraints and contexts at the level of school systems. We have suggested the areas of teacher learning, assessment, technology planning, and organizational structure and leadership as areas that may be particularly valuable for further research related to creating usable cognitively oriented technology innovations. We hope that an expanded body of research in this area will allow schools and researchers seeking to create broadly scalable and sustainable cognitively oriented technology innovations to examine how others in similar contexts “reduced gaps” so that innovations might be employed successfully.

We understand that not all researchers who work with technology will find work in systemic reform contexts appealing or appropriate. However, it is essential that we continue to explore “cutting edge” technologies that may not be ready for widespread use in schools, as well as basic cognitive research on learning and understanding. But the field, as a whole, would benefit from engaging in systemic design-based research on technology innovations. If we do not undertake this challenge, the risk is that cognitively oriented technologies that demonstrate great promise for improving learning and are consistent with the goals of national standards are unlikely to achieve widespread use in schools, because we will not understand how to create conditions that make them usable in real school contexts. If we do not make progress in this area, we will have missed an opportunity to bring some of the most potentially valuable technological innovations from the research community to bear on the ongoing challenge of systemic school reform.

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Ontological Innovation and the Role of Theory in Design Experiments

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The motivation for this article is our belief that theory is critically important but currently underplayed in design research studies. We seek to characterize and illustrate a genre of theorizing that seems to us strongly synergistic with design-based research. We begin by drawing contrasts with kinds of theory that are relevant but, we contend, by themselves inadequate. A central element of the type of productive design-based theorizing on which we focus is “ontological innovation,” hypothesizing and developing explanatory constructs, new categories of things in the world that help explain how it works. A key criterion to which we adhere when discussing ontological innovations is that theory must do real design work in generating, selecting and validating design alternatives at the level at which they are consequential for learning. Developing and refining an ontological innovation is challenging and requires the kind of extensive, iterative work that characterizes design experiments more generally. However, the pay-off in terms of clarity of focus and explanatory power can be great. We present two case studies that illustrate the development, refinement, extension, and instructional application of ontological innovations.

The nature of theory and its role in educational studies and in design research in particular is a deep topic deserving of extended discussion. It has frequently been argued that the development of theory should be one of the primary goals of design research (e.g., Collins, Joseph, & Bielaczyc, this issue; Confrey & Lachance,

2000; Design-Based Research Collaborative, 2003; Edelson, 2002; Gravemeijer, 1994a; Labato, 2003; Simon, 2000). Despite this recognition of the potential contributions of theory, we contend that design experiments have generally been underdeveloped as contexts for the development of theory. Our primary purpose in this article is to enrich ongoing discussion of the role of theory in design research with some new avenues of consideration.

On the vast background of the nature and function of theories, the goals of this article are modest. We cannot hope for definitive arguments in favor of theory, nor for an extensive characterization of how theories are built (or should be built), tested, and how they work. Instead, we aim only to accomplish the following: We will argue briefly that theory is important. Then, our principal thrust will be to characterize roughly elements of a genre of theory that has been underplayed in design experiments but that can, we believe, be productively developed, tested, revised, and elaborated while conducting such studies. We initially characterize this type of theory negatively, by contrasting other meanings for “theory.” On the positive side, we concentrate on one distinguishing process in developing the kinds of theories in which we are interested, that of “ontological innovation”—the invention of new scientific categories, specifically categories that do useful work in generating, selecting among, and assessing design alternatives. Examples of ontological innovations developed and refined while conducting programs of design research include elements of the theory of quantitative reasoning proposed by Thompson and colleagues (Thompson & Thompson, 1994, 1996) and the constructs of model-of and model-for formulated by Gravemeijer (1994b). Examples of ontological innovations that were developed in other disciplines but that have been used widely in educational research include the constellation of constructs that comprise Lave’s (1988) and Wenger’s (1998) situated theory of learning and the notion of an activity system as formulated by Engestrom (1998, 1999).

Discussions of the nature of science, and of theories in particular, tend to suffer from a high degree of abstractness. Meta-scientific frameworks, which might facilitate description and comparison of theories, are not common currency. This difficulty is exacerbated in design research, where investigators tend to follow their noses, doing the work of science as they understand it, without extended rationale or public explication. Compounding the problem is the fact that design research is a relatively new and still evolving methodology. The community of researchers has not had time to filter and share reflections on the phenomenology of “doing science” in this way. As a consequence, the delineation of agreed upon practices and the development of a grounded language for describing them is very much in progress.

To combat abstractness, we trim our coverage and depth goals further to make room for two case study descriptions of our own practices of theory development and use. Our intent in presenting these case studies is to exemplify and concretize the discussion of ontological innovations in the context of design research.

THE IMPORTANCE OF THEORY

In many corners of science, the importance of theory is completely uncontested. The history of physics, for example, may be viewed profitably as a series of broader and deeper theories, each of which was complemented, refined, or subsumed in successor theories. Newton's theory was substantially modified by Einstein's special relativity. Quantum mechanics located Newtonian mechanics as an approximation relevant to everyday sized objects and everyday motions. Later, field theory integrated special relativity and quantum mechanics.

Theories have always displayed a principal part of the power and elegance of science. They embody generalization, bringing order to a vast array of seemingly disparate phenomena that come to be seen as special cases of some theory. They encapsulate the most secure of our knowledge claims at any stage of scientific advance. They enable us to discriminate between relations that are necessary and those that are contingent. They delineate classes of phenomena that are worthy of inquiry and specify how to look and what to see in order to understand them. This last characteristic—epigrammatically, “teaching us how to see”—is particularly evident in the type of theory on which we focus in this article.

Theory even provides a suggestive framework for understanding sciences' evolution; once a theory is initially established, scientists typically (1) follow up details and special cases, (2) adapt theories to application, and (3) seek to account for anomalies that, sometimes, result in the creation of new alternative, subsuming, or complementary theories.

Education seems a different matter. No psychological or sociocultural theory has had the breadth, precision and detailed validation of any of a number of theories in physical or biological sciences. Theories concerning educational matters seem to replace one another, rather than subsume, extend, or complement other theories. Although the state of the art constantly changes, it is often difficult to tell that progress is being made. Indeed, some have been drawn to argue that human sciences are simply not amenable to theory building as we know it in other sciences (e.g., see the discussion of “theoretical frameworks” and references in Eisenhart, 1991).

We see no reason to believe that the human sciences are essentially different from other sciences with respect to the value of theory. All well-developed current sciences went through early stages where progress was more chaotic and uncertain. When things settled down, theories and theoretical development have always proved central to progress. Indeed, the complexity of extended real-world interventions that characterize design studies highlights the pressing need for theory while simultaneously making the development of useful theories more difficult. When working in the multifaceted or even seemingly chaotic settings in which design studies are conducted, one must have some orientation on central versus peripheral concerns, and one must be very clear on what general results are intended.

In our view, the apparent complexity and messiness of human action and learning in these settings indicate our collective limited theoretical sophistication.

One thing is certain: If we do not pursue theories that enable us to come to grips with the complexity of design experiments, we will not achieve them. The central problem, as we see it, is that of how best to pursue theoretical agendas, befitting what we believe we know so far and acknowledging whatever special circumstances are our lot in education-related sciences.

CHARACTERIZING “THEORY” FOR DESIGN STUDIES

Design studies, or design experiments, are iterative, situated, and theory-based attempts simultaneously to understand and improve educational processes (Brown, 1992; Cobb, Confrey, diSessa, Lehrer, & Schauble, 2003; Collins, 1992; Edelson, 2002). Theory can, however, mean different things to different people. What kinds of theories can be produced by and can serve to further the aims of design experiments? Among the many possible criteria for types of theories, one stands out as critical for design studies. Theory must do real design work in generating, selecting and validating design alternatives at the level at which they are consequential for learning.¹ With this criterion in mind, we identify several types of theory that can be of value when conducting design studies. However, we argue that, by themselves, they constitute an inadequate theoretical basis for design research in the long term.

Grand Theories

“Grand” theories of cognitive or social processes are the educational equivalent of “evolution” or “Newtonian mechanics.” Examples include Piaget’s theory of intellectual development (Gruber & Voneche, 1977), Skinner’s behaviorist theory of learning (Skinner, 1974), or Newell’s “unified theories of cognition” (Newell, 1990). The difficulties with such theories lie typically in some combination of being, as yet, immature (e.g., false as categorical prescriptions of cognitive or social processes), imprecise (so that implications at the level of design decisions are unsure), or simply too high-level to inform the vast majority of consequential decisions in creating good instruction. To take a specific case, Piaget developed his theory to address epistemological issues that concern the nature and growth of knowledge. Nonetheless, his work had a strong educational influence from the 1960s at least through the early 90s. We feel, as many others do, that Piaget’s ideas

¹Learning, per se, is a canonical goal for education-related design studies. However, alternative goals certainly exist, including engaged participation, community development, development of a personal scientific identity, and so on. To simplify discussion, we take learning here as a paradigmatic goal, and support for learning as the relevant design focus.

were overextended into education. Indeed, Piaget felt that way himself. Attempts to apply his ideas have often resulted in unwarranted generalizations of what were actually more focused empirical results (Metz, 1995). Furthermore as Walkerdine (1988) documents, readings of Piaget's ideas were primarily used to rule out approaches to learning and instruction ("young children cannot consider scientific explanation because they are concrete thinkers"), rather than to manage design decisions among possibilities in ultimately effective instruction.² Piaget's theory is powerful and continues to be an important source of insight. However, it was not developed with the intention of informing design and is inadequate, by itself, to do so deeply and effectively.

Orienting Frameworks

"Constructivist theory" or "cultural-historical theory" are often appealed to as the basis for instructional design. On the one hand, commitments at this level are highly consequential in identifying family resemblances and perhaps core values among those who do design research. They define important social affiliations among scientists and name widespread thrusts into thinking about and carrying out instruction. Yet, much that is involved in developing specific designs does not follow from these orientations. What happens when "building on student ideas" seems to fail to work? How, exactly, should one build on student ideas? Communities of learners are sometimes not optimal learning cultures when they are patterned on a limited range of exemplars that emphasize only induction into participation. Some cultural artifacts are much more powerful for learning than others; how do we measure limitations and improve elements that are weak?

Orienting frameworks seldom provide strong constraints or detailed prescriptions. Their value instead resides in the general perspectives that they provide for conceptualizing issues of learning, teaching, and instructional design. Principles of refinement and the rectification of failure, the meat of much design, typically require detailed specification within such a perspective. Orienting frameworks are probably best viewed as meta-theories, presumed general constraints that define general aspects of hoped for and needed more specific theoretical frameworks.

Frameworks for Action

Another common usage of "theory" refers to more or less general prescriptions of pedagogical strategies. Examples include Papert's "constructionism" (Papert &

²A fuller treatment would require attention to recent attempts at instructional design that appeal to grand theory in selecting many specific features of instructional interventions, e.g., Anderson (1990). Without meaning to appear dismissive, for lack of space, we say only that there is no wide-spread acceptance of such theories and their consequences in educational design.

Harel, 1991), Ann Brown and Joe Campione's "fostering a community of learners" (Brown & Campione, 1996) and Ann-Marie Palincsar's "reciprocal teaching" (Palincsar & Brown, 1984). Marcia Linn talks about "scaffolded knowledge integration" (Linn & Hsi, 2000). In some of our own work (diSessa, 1992) we advocate "learning by designing" as a powerful framework for instruction. These frameworks certainly provide some focus and direction to the design of learning environments, and they are often effective heuristics. However, such frameworks themselves typically fail to serve the role of theory in the sense that we believe to be most important for one central reason: They do not cleanly separate their scientific claims and validation from their suggested actions. That is, the theory or theories behind frameworks for action are relatively inexplicit, complex, and often involve multiple very diverse elements that cannot plausibly be brought under a single umbrella. In the case of students' learning through design, we argue that it is attractive because it combines good affective properties with good cognitive ones. The details of the separate theories of cognition or affect are not really at stake.

We introduce the phrase "managing the gap" to name the issue that is behind the failure of most frameworks for action to achieve what we would like accomplished. The "gap" arises from the fact that instruction is the result of many sorts of complex, interacting elements. Instruction depends on the values of the participants; it depends on technological infrastructure; it depends on the nature of classroom discourse; it depends on practicalities such as available time. We also want to *make* instruction both depend on and serve to test theory. And yet, in order to see and assess the impact of underlying theory, we must cleanly separate it from the myriad of other issues that we handle, as best we can, in the management of trade-offs among the multiple constraints impinging on instruction. In the ideal case, then, pedagogical strategies and conjectures are separated by a *carefully considered and articulated gap* from the theory or theories that explain or motivate them. A well-managed gap separates the implications of a particular theoretical claim from other claims and also from atheoretical aspects of design. Attention and effort are necessary to perform this management.

To take a noneducational example, there can be no doubt that there is science in the design of airplanes. However, the shape of a Boeing 747 aircraft does not follow in a direct and simple way from any of this science. Neither does the shape of the aircraft, as a whole, directly test elements of the underlying theory. With sufficient care (corresponding to managing the gap between design and theory), however, a failure attributable to the shape of the aircraft might implicate a failure of a theory of strength of materials, not just to a careless mistake, a failure to anticipate transient loads, or a poor choice of materials.

As with grand theories and orienting frameworks, our point is not to denigrate frameworks for action. They play a critical role in organizing research about instruction given the current state of the art, and they will probably continue to do so for the foreseeable future. Yet, their purpose is more directly practical and less di-

rectly connected to core scientific values, such as precision, exhaustive and explicit formulation, and falsifiability. Frameworks for action specify goals and actions in instruction; the type of theory with which we are concerned typically lies “on the other side of the gap,” where entities and relations are specified, but actions, goals, and instrumentality are much less so.

Domain Specific Instructional Theories

When people refer to theories in the context of design experiments, they are often referring to *domain specific instructional theories*. Theories of this type entail the conceptual analysis of a significant disciplinary idea (e.g., density, sampling distribution) together with the specification of both successive patterns of reasoning and the means of supporting their emergence. Means of support are usually construed broadly and can include the classroom activity structure and discourse as well as instructional tasks and tools. In contrast to frameworks for action, theories of this type are of more than heuristic value and embody testable conjectures about both learning processes and the means of “engineering” them. Furthermore, although they are typically developed within an orienting framework, they reflect the detailed work involved in formulating, testing, and revising a hypothetical learning trajectory.

As we indicated, design studies involve iterative cycles of design and analysis. Domain specific instructional theories represent an advance in the design aspect of design research. In fact, we can view them as constituting the rationale for pedagogical designs; generalization across a number of such theories can result in the development of a framework for action such as that of Realistic Mathematics Education developed at the Freudenthal Institute in the Netherlands (cf. Streefland 1991; Treffers, 1987). However, the fact that domain specific instructional theories are tied to specific designs reflects the emphasis on relatively situated and pragmatic concerns at the expense of managing the gap between explanatory theory and pedagogical conjecture.

Our intent in making these observations is, again, not to disparage theories of this type, particularly as their development constitutes one of the primary reasons why we conduct design studies. Instead, it is to highlight that design research will not be particularly progressive in the long run if the motivation for conducting experiments is restricted to that of producing domain specific instructional theories. It bears noting, for example, that although such theories are informed by repeated cycles of design and analysis, the analyses are typically conducted by using established theoretical constructs. As a consequence of the concerns that motivate their development—including communicating design rationale to teachers—domain specific instructional theories will not by themselves enable us to scrutinize and reconceptualize aspects of competence in mathematics or science, or to better handle the complexity and diversity of settings in which we conduct experiments.

Beyond narrowing our consideration by contrasting the cases of grand theory, orienting frameworks, frameworks for action, and domain specific instructional theories, can we be more affirmative in characterizing a type of theory that can be important for design work? Of course, a huge amount could and should be said, but here we concentrate on one important element, ontological innovation.

ONTOLOGICAL INNOVATION

The idea behind ontological innovation is deceptively simple. Science needs its own set of terms or categories to pursue its work. Again, this has always been true of developed sciences: “force,” “gene,” “natural selection,” “molecule,” “element,” “catalyst.” The process of creating such categories, however, is far more complicated than writing down a definition, or finding a relevant meaning in a dictionary. Instead, defining the technical terms of science is more like finding and validating a new category of existence in the world; hence we use the term *ontological innovation*.

The essential challenge can be expressed simply enough. Scientific terms must “cut nature at its joints.” That is, they must make distinctions that really make a difference, ignore the ones that prove to be inconsequential, and enable us to deepen our explanations of the phenomena of interest. We must develop theoretical constructs that empower us to see order, pattern, and regularity in the complex settings in which we conduct design experiments. Ontological innovations are attributions we make to the world that necessarily participate in our deepest explanatory frameworks.

We use two examples to illustrate the notion of ontological innovation and some intrinsic difficulties (“bootstrapping”) in developing one. A common laboratory experiment has students measure a varying voltage and consequent current in a circuit. The students “discover” Ohm’s law, that current (I) is proportional to voltage (V), “ $I = (1/R) V$ ” (the proportionality constant is the inverse of resistance). The task of a real scientist is much different. The terms current, voltage and resistance must first be conceptualized and operationalized. The bootstrapping process is complex. It turns out that meter mechanisms mainly measure current. But we cannot even know this until we have operationalized current in a way other than “that which proportionally drives meter mechanics”! Further, how do we measure voltage? A modern scientist could use Ohm’s law and a known resistance together with a current-sensitive meter mechanism. The problem is obvious: knowing Ohm’s law is by far the easiest way to know how to measure (in this case, voltage), in order to “discover” Ohm’s law.

Measurement is a key practice in education as well as in the physical sciences, so it is worth pursuing our metaphor just a bit further. Voltage must be measured with a two-wire probe. Why not a single wire? Again, conceptualization appears necessary prior to measurement, which seems prior to validating the conceptualization. Measuring current requires one to break a circuit and insert the meter, and

it requires the meter not add significant resistance to the circuit. How would one know that? The full theory appears necessary in order to measure, in order to test the theory.

All of these bootstrapping problems apply to design studies. The last of our observations—constrained and careful breaking of a circuit in order to tap theoretical properties necessary to understand it—is particularly evocative. Not only do theoretical terms provide us with explanatory constructs that we use to understand phenomena, but they guide us in the tricky interventions necessary even to “see” theory-relevant things. In the student exercise of “discovering Ohm’s law” that started this example, students just insert a probe as directed, and black-box mechanisms read out relevant things that go together exactly as the theory says they should. Scientists in the process of developing theory somehow perform operations to see things that they are just beginning to conceptualize.

Our second example undoes an accidental salience in the first, that of precise, quantitative measures in accomplishing science. This example makes the point that scientific categories do not always need to have quantitative properties to provide for strong implications and falsifiability.

Consider the case of the germ theory of disease: tiny, invisible, biological elements cause human illness. No precise measurements or quantitative laws are at stake, and the exact nature of those biological elements, although certainly within the scientific program, is not relevant to some powerful theory-based predictions, both practical and intrascientific. To wit, if we isolate sick people from well people, the well people will not get sick. If illness truly develops spontaneously, the theory is refuted. Turning to practical implications, we might be able to “kill” those invisible elements, say with heat or some kind of poison. We might Pasteurize, or conceptualize what a mold can do for some diseases.

The germ theory of disease provides a clear *locus for refinement*. What exactly are those tiny elements, and how do they work? Some of them (bacteria), it turns out, are full-fledged organisms susceptible to many killing methods. Some (viruses) have quite peculiar ontological characteristics, and “killing them” or preventing their spread becomes subtle and theoretically intensive. The concept of locus for refinement is important, as, unlike theories that are treated as *faits accomplis*, the development of ontological innovations in design-based research seems inevitably a long process of refinement and extension. This was very much the case for the innovations developed by Thompson and colleagues (Thompson & Thompson, 1994, 1996) as well as in our own work.

In the following two sections, we provide two brief case studies of design experiments that led to, and then developed, ontological innovations. The main elements of our analysis are to sketch (a) the origins and nature of ontological innovation, (b) their impact on the conceptualization of learning and on subsequent educational design, and (c) the processes of validation and refinement, and what that further precipitated in design. When possible, we describe the need for and our

efforts to manage the gap between theory and the details of instruction, and, in addition, we bring out a few relevant and interesting properties of design experiments, per se.

CASE STUDY I: META-REPRESENTATIONAL COMPETENCE

Opportunistic Research, Retrospective Analysis, and Discovery

In 1989 and 1990 the Boxer group at UC Berkeley was in the midst of a design experiment aimed at teaching elementary school students some high school and university physics (diSessa, 1995). We won't recount the motivations or results of that experiment because the focus of our attention here is something new that emerged from that context. In fact, what we observed was quite unexpected. It concerned what we subsequently came to refer to as the students' *meta-representational competence*: their substantial expertise in inventing, evaluating, and refining a variety of representational forms. Although these observations did not fit cleanly within our existing scientific framework, they did fit beautifully within our orientating constructivist framework, and also remarkably well in our framework for action, students' design. The fact that what we discovered was entirely consistent with our orienting and action frameworks, but quite distinct scientifically, exemplifies the contrasts we made in introducing theory-for-design.

Discovery of new foci of attention is actually quite common in design experiments, if it is not absolutely necessary. Failures or surprising successes not infrequently push toward, and sometimes enable, new lines of inquiry, possibly involving new ontologies. In some other cases, typically in initial failures, we manage the gap by patching enough to get by, without pulling the surprising occurrence into the core, scientific program. For the discovery relevant here, we had no means to pursue the new theoretical issues by any particular actions during the project in which it emerged.

What we could do within the initial design experiment was to conduct a *retrospective analysis* of our "surprising occurrence," and, from that, get a head start on follow-up work aimed specifically at clarifying what we had discovered. Retrospective analyses more frequently occur closer to the main scientific line of an experiment, to get at microstructure related to conjectures that might have been roughly validated, or to study unanticipated, theory-relevant failure. For example, a post test might show that students learned something, but one might want to backtrack to see in more detail how that happened, say, in classroom discussion.

How did we come to have the video that allowed a retrospective analysis of an unanticipated event? It was part luck and part typical design-experiment disci-

pline. Design experiments usually encompass extended periods of time and much activity that is not highly scripted and/or it might be outside the focus of principal conjectures (constituting a somewhat uncontrolled part of the “gap” referred to previously). During many of these times, nothing particularly interesting or relevant occurs. It is manifestly impossible to study everything that happens in a design experiment, and trying to do so exhibits lack of scientific focus. Systematic analysis has its virtues, but researchers have finite time and hopefully the wisdom to study mostly issues that are well prepared and tractable.

On the other hand, surprising things essentially always happen, and some data collection beyond core focus is always a good idea, pending trade-offs of time and effort with other tasks. In our experience, much of that data never gets analyzed. Many times we decide in advance on principles of data selection. For example, we have often decided to tape every full-class discussion, even if nothing critical is anticipated.

We did not have that particular principle in action during the case in point—which would have led to a corpus of hundreds of videos. The lucky part of having video of our surprising event was that an alert graduate student noticed some interesting things going on in a warm-up activity, so he suggested we video the full activity.

The task as set was an introduction to graphing. Instead of jumping into instruction, we had time for and decided to implement a more exploratory exercise of seeing what our students knew about graphing. At least, that is what we thought we were about. We asked students to design their own representations of motion, expecting that some might produce graphs and show us how they made them. The orienting framework suggested studying student ideas; one major course-specific frame for action, student design, suggested posing a design task.

An explosion of ideas resulted. Students produced and critiqued dozens of representational forms. None of the first many representations was graphing. Instead, graphing emerged clearly only very late in the activity, about two-thirds through the 4 to 5 day activity (40 minutes per day). What was most salient for us was the almost uniformly intense engagement on the part of the students. Sometimes the teacher couldn’t even find her way into the conversation. Students came to class with ideas, and they went out continuing to discuss possibilities that had arisen.

Had we designed such a rich and enthusiastic exploration, we would have been extraordinarily pleased. Having it presented to us with little preplanning, we were stunned. Although not in our game plan, we felt we had to follow up. We set our agenda to understand how it was possible for this activity to arise. More particularly:

1. What knowledge accounted for the contributions of students?
2. What contributions did the teacher make? To what extent were her actions critical to the outcomes?
3. How did ideas build on one another: For example, did they flow logically, or accumulate in an ad hoc manner? How did students actually get to graphing?

4. What conceptual knowledge of motion (the designated learning target of the course) was implicated in what the students did?

Briefly, we found (diSessa, Hammer, Sherin, & Kolpakowski, 1991):

1. Students seemed to have a wealth of ideas about how to create and judge representations. The logic of the representational forms that were created was, for the most part, scientifically appropriate. Students saw many good and bad properties of the representations they produced, and they were creative in patching problems that were uncovered.

2. We note that teacher interventions constitute a potential competitor to “student knowledge” as an answer to the question “how did students manage this?” It became quickly evident that many student creations were not and could not have been contributed by, or instructed by the teacher. In fact, she did not understand several of the representations initially when they were produced. With respect to the narrower question, did the teacher make graphing appear, our analysis suggested that while she pushed criteria of judgment that would favor graphing, she did not either introduce key ideas, nor coerce student judgments that “graphing is best” in the end. The primary outcome concerning teacher strategies was to note the extraordinary flexibility of this teacher’s strategies, which were applied in a highly contextualized manner.

3. Students combined ideas from other students fluidly and sensibly. With respect to graphing, per se, the trajectory had both elements of inevitability, and elements of idiosyncrasy. Students seemed to have or develop all the basic resources needed to understand graphing, and it matched their criteria of judgment as a “good representation” well. Yet, the particular path of invention involved probably irreproducible specificity and timing of conceptual contributions by students.

4. The study of conceptual knowledge of motion was treated separately in an article that set conceptual development within another framework for action, “benchmark lessons” (diSessa & Minstrell, 1998). The analysis led to some interesting conclusions and speculations about new paths of conceptual development (possibly contributing to a domain specific instructional theory), but, in fact, conceptual development concerning motion seemed nearly orthogonal to the main line of representational design.

Ontological Innovation: Meta-Representational Competence

The major result of our retrospective analysis was the “discovery” of *meta-representational competence* or simply MRC. MRC, as an ontological innovation, implicates a specific body of knowledge that students have, and which can be developed, lying behind students’ abilities to *create*, *critique*, and *adapt* a very wide

range of effective scientific representations. Elements of our retrospective analysis, cited above, suggested that students' prior knowledge, as distinct from "what the teacher taught," lay behind MRC (item 2), and, furthermore, that that knowledge was also distinct from other bodies of knowledge, such as conceptual knowledge of motion (item 4). "Discovering MRC," of course, is a very substantial simplification of the process, which seemed to us consistent with the bootstrapping of conjectures on the nature of the entity, and methods of "seeing" and measuring it, which we introduced with the "discovery" of Ohm's law.

Three points are worth emphasizing about this ontological innovation. First, we feel MRC is a genuinely new construction. Although several bodies of literature relate to it, our formulation is different in some important respects from any of them. Developmental psychology has studied children's representational competence, but treated it as slowly changing and not nearly as differentiated (see below), richly generative, and capable of supporting midduration (several days or weeks) development (contrast both a "lab task" on the short time-scale end, or a "developmental level" on the long time-scale end). Prior instructional work that has highlighted student representational creativity (e.g., Cadwell, 2003) has not led to a fine-grained analysis of the specific knowledge underlying that competence, and it has usually been disjoint from consideration of the acquisition of competence with scientific representations.

Second, we emphasize the analogy with the germ theory of disease. The existence of MRC can be a substantial discovery at a relatively undifferentiated level. So, for example, we believe that essentially any teacher can reproduce at least some aspects of the creativity and energy from students that drew our attention in the first place, given a little help in understanding what MRC is and how one acts to evoke it. But, in addition, MRC sets a locus for refinement that can bring additional science and practical impact. See continuing discussion, later in this article.

Third, like essentially all scientific discoveries, instrumental impact is not easy to lay out. Indeed, we expect a lot of creative design will be needed to add to MRC's purely scientific properties in the service of good instruction. The gap between theory and practical action is complex, and it is not bridged merely with more science. In our case, we conjecture two important functional niches for MRC. First, it is a valid target of instruction in its own right, worthy of its own slot in the curriculum. Making such a judgment implicates values (what should be taught) not just science (what students know or can come to know). Our second functional niche is as a frame and background for the study of the many specific scientific representations that one finds now in conventional curriculum. Simply put, particular representations may be more learnable and sensible to students in the context of understanding what representations do in general, how they work, and what alternatives might exist for given ones (like graphing). We have taken small steps in both of these directions, but the agenda is far from finished.

Refinement, Iteration, and Extension

This section emphasizes MRC as a locus for refinement. Over a roughly 5-year period, we followed up on our initial discovery in several directions. First, we developed a preliminary taxonomy of subsystems of MRC, including the two most obvious parts mentioned above, inventive resources (how do students make new representations?), and critique (how are representations judged?). Inventive resources were amenable to study in additional examples of full-class design, very much paralleling our original miniexperiment; one sees direct evidence of inventive resources in the physical representations students produce (Sherin, 2000). For example, students regularly and easily apply a particular schema for time-based representations, where time is displayed as a sequence of separate icons or images. In contrast, the breadth and nature of students' criteria of judgment were difficult to study in these instructional contexts, so we did a lab experiment with a small number of students, seeking to draw out and study criteria (diSessa, 2002). (We had to cut the fabric of in-class learning to see student criteria, as one needs to cut a circuit to measure current.) Design experiments seem intrinsically to call for multiple methods and naturalistic study is not *sine qua non* (Brown, 1992).

Iteration of the “inventing graphing” classroom activity has been critical in separating accidental from intrinsic properties of MRC. In the years since, we have undertaken about a half-dozen “replications” in various contexts. Typical of design experiment methodology, replication does not mean literally trying to do exactly the same thing again, but exploring variations and what stays the same (and what changes) on redoing an activity. Briefly, we have found:

1. MRC has shown through strongly in every iteration, with a broad range of students across multiple age levels. Indeed, the kinds of representations our first group of students did not nearly exhaust the possibilities.
2. While our first try generated truly exceptional interest and engagement, every iteration also showed substantial student involvement.
3. While our first teacher continued to serve as a benchmark for scaffolding expertise, teachers with very different styles can still find their way effectively in conducting these sessions.

We initiated a study of MRC in a neighboring domain, not concerning representing motion, but with respect to representing spatially distributed data, for example, altitude across some terrain, or brightness across a photograph. This constituted a sort of “transfer” test for the concept of MRC. While we never believed MRC was completely domain-independent, it was important to get some calibration. Is motion a special ground of representational competence?

Briefly, we discovered that MRC does basically carry over to spatially distributed data, with some important caveats. Students were quite competent in produc-

ing a fairly wide range of cogent representation; they were relatively good at judging their value; and they even spontaneously developed approximations of prototypical scientific representations (topographic maps, roughly the equivalent of “graphing” with respect to spatially distributed data). The caveats, however, are illuminating. First, students’ MRC does not support the same kind of enthusiastic, extended engagement with design that we encountered with motion. In recent work, we feel we uncovered some reasons for this (Sherin, Azevedo, & diSessa, *in press*). Second, students’ “representational misconceptions” seem to be more prominent with respect to spatially distributed data (Friedman, 1999; Elby, 2000). As a result, we developed and tested instructional strategies for supporting better competence for dealing with certain representational technologies (image processing systems; see Friedman & diSessa, 1999).

A final note: Because MRC was not anything that we expected to run into, a significant part of our work in following up was locating and positioning our work with respect to various bodies of literature concerning general representational competence. See, particularly, Sherin (2000), which situated our work with respect to developmental studies of representational competence. Scientific work requires such accountability, and it is one reason design researchers need to limit their attention to a limited number of theoretical foci in any given experiment, relegating many of the other issues that arise to a practical managing of the gap.

Synthesis and Extension

Ontological innovation often provides at least as much of a basis for further work as it does “results,” *per se*. MRC proved a solid basis for a progressing paradigm of scientific research: It encompasses a wide and varying domain of phenomenology; it suggested the possibility of new instructional goals (elements of MRC, itself; representational design with respect to different classes of representations), and new measures of success (high-level understanding of representations, not simply learning “additional particular scientific representations”); finally, it interpreted and helped us deal with educational difficulties such as representational misconceptions (Elby, 2000) and how to conceptualize problems we discovered in students’ use of technology (Friedman & diSessa, 1999). A recent article, diSessa (*in press*), attempts to synthesize the major lessons learned in our studies of MRC, and it sets a further agenda for research.

In this section, we briefly highlight productive connections between MRC and other research agendas. Ontological innovation “bleeds over” to related studies, either demanding complementary study (which is more prominent in our second case study, below), or contextualizing and contributing to other research foci.

MRC has proved a good context in which to pursue the study of teacher strategies. Recall, this was one focus of the initial MRC retrospective analysis. MRC is highly generative knowledge, in contrast to strategic or “structural” understanding

such as understanding particular scientific concepts. Getting students to design and critique novel representations is a prototypical task for learning and also a test of advancement. Yet, such tasks have had comparatively little study, in contrast to, for example, lessons focusing on structural scientific understanding (e.g., jigsaw lessons to illuminate function/structure relations in ecology). How does one guide student design without stifling it? The Boxer group produced some relevant studies. Madanes (1997) developed the concept of critical pedagogical moves, moves that provide an essential leap in the focus or flow of inquiry. MRC strongly contextualizes such moves. What counts as an “essential leap” must be calibrated by understanding the development of specific competence, in this case, representational competence. Granados (2000) looked at the problem of establishing an appropriate intersubjective agreement among teacher and students on the range and limits of design tasks. diSessa and Minstrell (1998) looked at MRC-related data to elaborate the meaning of a particular genre of somewhat more teacher-driven lesson (compared to open design), *benchmark lessons*.

Finally, the Boxer group has produced a small set of articles trying to understand interest and engagement, the overt reasons we started our study of “inventing graphing” in the first place. In Sherin, Azevedo, and diSessa (in press), we identify structural differences in MRC related to different domains (e.g., MRC related to motion and MRC related to spatially distributed data), which account, in part, for differences in student engagement and enthusiasm. Azevedo (submitted) attends to student personalization of their efforts in MRC-based learning tasks, implicating “personal excursions” as highly functional in allowing students strong personal links, while maintaining a focus on “what the students are supposed to be learning.” Again, the point of mentioning these is to illustrate how ontological innovation can open up new avenues of pursuing older questions, or how it provides synergistic contextualization of complementary studies.

CASE STUDY 2: SOCIOMATHEMATICAL NORMS

Opportunistic Research, Reorientation, and Retrospective Analysis

In 1986, the research team of Paul Cobb, Erna Yackel, and Terry Wood began a series of year-long design experiments in collaboration with first, second, and third grade teachers. The second classroom session of the very first of these experiments involved the observation of an unexpected phenomenon that precipitated a new strand to our research program that has continued for 17 years. Briefly, we intended to focus on young children’s mathematical learning primarily from a cognitive perspective. However, the teacher with whom we collaborated initiated a shift in discourse during the second whole class discussion from talking with the stu-

dents about their solutions to explicitly talking with them about how they should talk about their solutions. Our investigation of shifts of this type, which we subsequently came to call the *renegotiation of classroom social norms*, led to an explicit focus on classroom discourse and communication. Thus, a single surprising observation led us to reorient our research program on the fly and led to a series of investigations, each of which spawned new issues that required us to extend our prior analyses. Our decision to video-record all classroom sessions made it possible for us to modify our research focus and to pursue issues that arose opportunistically by conducting retrospective analyses, similar to what happened in the meta-representational competence case study.

As part of this research program, Erna Yackel observed that the teachers with whom we worked frequently asked if anyone had solved a task a different way and began analyzing teachers' and students' interactions in these situations. This investigation turned out to be particularly fruitful in that Yackel clarified and illustrated how teachers and their students negotiated the norm of what counts as a different mathematical solution. This analysis also involved a significant advance over our initial work on classroom social norms in that the latter are not specific to mathematics. The social norm that students should make different contributions to discussions is applicable to subject matter areas other than mathematics. In contrast, the norm of what counts as a *mathematically* different solution is specific to mathematics. The analysis of mathematical difference therefore precipitated an ontological innovation—sociomathematical norms, of which mathematical difference is one—that overcame a limitation in our analyses of classroom social norms.

In explicating the norm of mathematical difference, we made a number of observations (Yackel & Cobb, 1996):

1. The norm of mathematical difference established in a particular classroom is interactively constituted by the teacher and students, and is thus a joint accomplishment. This theoretical claim holds for all classrooms including those in which that teacher acts as an overt mathematical authority and has a predetermined view of what will count as mathematically different. Even in these cases, the students have to play their part by coming to act in accord with the teacher's expectations. Consequently, this norm does not correspond to an overlap in the teacher's and students' individual understandings. Instead, it is located within the ontological domain of social interaction and communication.

2. What counts as a mathematical difference can differ markedly from one classroom to another, and this can significantly influence the nature of the learning opportunities that arise for students. For example, in some first grade classrooms, a number of different types of counting methods for finding sums and differences count as different, whereas in other classrooms in which we have worked, counting methods are treated as the same and are differentiated from various thinking or derived-fact solutions.

3. The establishment of the norm of mathematical difference can give rise to learning opportunities for students during whole class discussions that otherwise would not have arisen. To make contributions that count as different, students have to make their own solutions objects of reflection in order to assess similarities and differences with other contributions.

4. The norm of mathematical difference can also give rise to learning opportunities for teachers. This is particularly the case when they do not have prior experience asking students to explain their reasoning and therefore have little basis for anticipating either students' solutions or the mathematically significant distinctions between them.

Ontological Innovation: Sociomathematical Norms

We came to realize that in analyzing the norm of mathematical difference, we were drawing attention to an aspect of the hidden curriculum of mathematics classrooms. This realization led us to focus explicitly on what students had to know in action in order to participate effectively in whole class discussions conducted by the collaborating teachers. In doing so, we identified several other specifically mathematical norms whose negotiation is typically less explicit than that of mathematical difference. These include the norms of what counts as a sophisticated mathematical solution, an acceptable mathematical explanation, and a clear mathematical explanation (Yackel & Cobb, 1996; Bowers, Cobb, & McClain, 1999). To differentiate these norms that constitute part of the hidden curriculum of mathematics classes from general social norms, we proposed to call them sociomathematical norms.

We continued this line of research because of its potential usefulness, a key consideration for design researchers. For example, our analyses pointed to a range of understandings in action that students appear to develop as they contribute to the initial establishment and ongoing regeneration of sociomathematical norms. These include judging what counts as a different solution, a sophisticated solution, an acceptable argument, and a clear argument. In the course of conducting these analyses, we came to reconceptualize one of our primary instructional goals, that of supporting students' development of intellectual autonomy in mathematics. We had previously viewed intellectual autonomy as a characteristic of particular students but now recast it as a characteristic of a student's way of participating in the practices of a classroom community. In particular, students who are intellectually autonomous in mathematics draw on their own intellectual capabilities when making mathematical decisions and judgments as they participate in these activities. The link between the growth of intellectual autonomy and development of classrooms in which mathematics is realized as a form of inquiry is readily apparent given that in such classrooms, the teacher and students together constitute a community of validators. However, students can take over some of the teacher's tradi-

tional responsibilities only to the extent that they can judge in action both when it is appropriate to make a mathematical contribution and what counts as an acceptable mathematical contribution. These are the types of judgments that the teacher and students negotiate when establishing sociomathematical norms. Furthermore, these types of judgments and the specifically mathematical beliefs and values that they entail capture and specify much of what is implied by the notion of mathematical disposition, a major focus of reform recommendations (National Council of Teachers of Mathematics, 1991, 2000). We therefore conjectured in initiating and guiding the negotiation of sociomathematical norms, teachers are simultaneously supporting their students' development of particular mathematical dispositions.

It is important to stress that in developing the theoretical construct of sociomathematical norm, we proposed a way of making sense of what is going on in mathematics classrooms. This proposal involves an ontological innovation in that it is concerned not merely with how to look at things in general, but with particular things to look at in classrooms. As this innovation contributes more directly to analysis than to the design aspect of design research, it does not give rise unproblematically to instructional recommendations or prescriptions. Nonetheless, the gap between theory and practice is relatively narrow because the proposal involves empirically testable conjectures about the relation between social process in which students participate and the understandings that they develop while doing so. We in fact viewed the construct of sociomathematical norm as a conceptual tool that can be used to orient ongoing interpretations of classroom events. These interpretations and the suppositions and assumptions that underpin them have real consequences in that they inform (but do not determine in detail) instructional design and pedagogical decisions that are made both in action and while reflecting on prior classroom events and planning for future instructional sessions.

Intervention, Iteration, and Refinement

Sociomathematical norms provided an excellent locus for refinement. The initial analyses that we conducted when formulating the construct of sociomathematical norm were descriptive and involved the retrospective analysis of video-recordings and other data sources including copies of the students' written work. In subsequent design experiments conducted by Paul Cobb, Kay McClain, Koeno Gravemeijer, Janet Bowers, and Michelle Stephan, we monitored classroom sociomathematical norms on-line and intervened on the basis of these analyses. Our intent in doing so was to improve the quality of whole class discussions as social settings for the students' mathematical learning. For the most part, the conjectures that motivated these interventions proved to be viable. However, we made three unanticipated observations that suggest directions for further inquiry and refinement.

The first observation concerns the way in which the emergence of one sociomathematical norm (e.g., mathematical difference) can provide a basis for the

emergence of subsequent sociomathematical norms (e.g., mathematical sophistication; see McClain & Cobb, 2001). In retrospect, this finding is almost banal. For example, if one solution is judged to be more sophisticated than another, that clearly implicates a judgment of difference. Valuing and being able to judge sophistication automatically carries over into judgments of difference, and likely vice versa. This finding orients us to investigate evolutionary patterns in the development of sociomathematical norms. The second observation concerns the important role that the development of notation schemes played in supporting the emergence of particular sociomathematical norms. This observation indicates the need to investigate the process by which sociomathematical norms emerge in more detail and, in particular, to attend to the material supports involved. The third observation concerns the manner in which the establishment of particular sociomathematical norms could directly influence the mathematical agenda. We found, for example, that the establishment of certain sociomathematical norms could give rise to learning opportunities for students that extended beyond those that we conjectured when intervening (McClain & Cobb, 2001). As an illustration, the distinctions between the sophistication of various types of reasoning made in particular task settings are sometimes adapted and extended spontaneously to other settings by either the teacher or the students. This observation indicates that there is much yet to be learned about the establishment and subsequent evolution of sociomathematical norms on the one hand, and the mathematical learning opportunities that arise for students on the other.

Several researchers have conducted related investigations and have identified a number of additional sociomathematical norms (e.g., Hershkowitz & Schwartz, 1999; Sfard, 2000; Simon & Blume, 1996). For example, Hershkowitz and Schwartz (1999) document the norms of what constitutes a good mathematical hypothesis and what constitutes mathematical evidence when solving open-ended instructional activities. One can imagine how a listing of sociomathematical norms might be extended almost indefinitely. In keeping with the basic tenet that theoretical constructs do useful work, the most pressing concern in our view is not to develop an exhaustive taxonomy of norms. Instead, it is to identify and investigate in greater detail those norms that appear to be particularly relevant to the agenda of supporting students' learning of significant mathematical ideas.

Synthesis and Extension

In the process of developing the notion of sociomathematical norms, we became aware of a limitation of this construct that was significant, given our concerns and interests as design researchers. Although sociomathematical norms are specific to mathematics, they are relatively broad in that they cut across mathematical domains. We therefore found it necessary to develop an additional construct, that of a *classroom mathematical practice*. This construct is concerned with the normative ways of reasoning and arguing that the teacher and students establish while discussing par-

ticular mathematical ideas. In analyzing the evolution of classroom mathematical practices in a particular classroom, we can in effect document the collective mathematical learning of the classroom community. Taken together, the three constructs of social norms, sociomathematical norms, and classroom mathematical practices constitute the core elements of the interpretive framework that we currently use to analyze the classroom microculture and thus the social situation of students' learning (Cobb, 2000). Viewed within the setting of this framework, sociomathematical norms are integral features of a proposed ontology of the mathematics classroom.

Sociomathematical norms and, more generally, the development of an interpretive framework that enables us to come to grips with the messiness and complexity of classrooms has been one of several strands of our research program. The analysis of sociomathematical norms has in fact proved relevant in several other strands that are surprisingly parallel to the case of meta-representational competence. For example, Kay McClain has focused on teachers' decision making and learning. Her analyses point to both the challenges involved for the teacher in initiating and guiding the renegotiation of sociomathematical norms, and the importance of doing so if tasks, tools and inscription are to function as means of supporting an instructional agenda that aims at significant mathematical ideas (McClain, 2002). In contrast to the relatively passive role that is frequently implied by the metaphor of the teacher as a facilitator, her analyses of sociomathematical norms contribute to the development of an empowering vision of the teacher's role in the mathematics classroom.

The construct of sociomathematical norm has also proved useful in understanding both the process by which students' interest in engaging in mathematical activity can be cultivated and the personal identities that students develop in particular classrooms. In the latter case, we draw on the constructs of social and sociomathematical norms to operationalize the idea of the normative identity as a doer of mathematics that students are, in effect, invited to develop in particular classrooms³ (Cobb & Hodge, 2003). Although this strand of our work is still in a formative phase, our hope is that it will contribute to the goal of ensuring that *all* students have access to significant mathematical ideas.

Reflection

The case of sociomathematical norms, parallel to the meta-representational competence case, serves to illustrate that design research can serve as the context for the development of ontological innovations that enable us to specify with some

³Speaking of students being *invited* to develop a particular type of identity as a doer of mathematics is a sloppy way of talking in that it implies that the normative identity is established solely by the teacher, and that the students then accept or reject it. In keeping with the view that norms are joint accomplishments, it is more appropriate to say that students develop their personal classroom identities as they contribute to (or oppose) the ongoing regeneration of the normative identity as a doer of mathematics.

precision the phenomena that we seek to study. Theoretical work of this type can be demanding in that it involves “building the plane while flying it.” However, the payoff is that conceptual tools are developed in the context in which they are designed to be used, that of developing analyses that can feed back to inform the ongoing instructional design and teaching effort.

The case of sociomathematical norms seems particularly analogous to that of germ theory. Just as germs are tiny, invisible, biological elements, so we can no more observe sociomathematical norms directly than we can directly perceive students’ conceptual process. Instead, we have to infer the norms established in a classroom by identifying patterns and regularities in teachers’ and students’ classroom interactions. Just as there are standards of evidence for making inferences about the presence of particular types of germs, so much of our work has involved explicating and scrutinizing the criteria that we use when identifying the sociomathematical norms established in particular classrooms. Further, just as germ theory is an interpretive perspective that enables us to make sense of certain biological phenomena, so sociomathematical norms are constituents of an interpretive framework that enables us to make sense of certain social phenomena. In both cases, the primary motive was to understand while remaining vigilant that proposed theoretical constructs do useful work (Stokes, 1997). Together with Toulmin (1963), we contend that the quest to understand is a primary hallmark of science, with prediction and control as necessary by-products.

CONCLUSION

We have attempted to illustrate the role of a particular class of theories in design experiments—those involving ontological innovation.⁴ We argued that although grand theory, orienting frameworks, frameworks for pedagogical action, and domain specific instructional theories are useful to the conduct of design studies, they do not provide an adequate theoretical basis for design research in the long term. Ontological innovation illustrates a kind of theory building that has been underplayed in design research but that is, we believe, well suited to being developed in and to contributing to design studies. Of course, we do not mean to imply that all theories involving ontological innovation are “good” or useful, nor that useful theories that do not involve ontological innovation are impossible.

⁴As a somewhat ironic historical observation, we note that Collins’ well-known 1992 paper on design experiments seemed to present “determining all the variables in a design experiment” as a first, and possibly not-that-difficult task. He said that, “Because [educational innovations] have been so varied, they should have uncovered most of the variables needed for a design theory.” (p. 20) Whether they count as variables or not, we believe that ontological innovations obviously are critical to consider, difficult to define, and not nearly all already discovered.

In characterizing ontological innovations, it is more feasible to focus on their function in cycles of design and analysis than to try and identify inherent properties apart from the contexts in which they are developed, refined, and used. With regard to design, ontological innovations can delineate new competencies that become a focus on instruction (e.g., meta-representational competence) or can provide new goals for instructional environments (e.g., a productive ecology of sociomathematical norms). In addition, their formulation can lead to the reformulation of previously adopted instructional goals (e.g., intellectual autonomy as a characteristic of a student's way of participating in the practices of a classroom community rather than as a characteristic of a student *per se*). Furthermore, an ontological innovation can orient the designer to scrutinize previously implicit aspects of a design. For example, students' use of inscriptions is a primary means of support in almost all designs. However, different designs support students' developing competence in inventing, evaluating, and refining a variety of representational forms more or less well. The theoretical category of meta-representational competence orients the designer to examine this aspect of a proposed design. Similarly, all designs involve assumptions about the nature of classroom interactions and discourse. The theoretical category of sociomathematical norms orients the designer to both explicate these assumptions, thereby further specifying the design, and to examine the types of mathematical dispositions that the design appears to support. Finally, an ontological innovation can inform the formulation of new benchmarks for assessing either the consequences of instruction (e.g., meta-representational competence) or the classroom learning environment (e.g., sociomathematical norms).

With regard to the analysis aspect of the design research cycle, ontological innovations provide new lenses for making sense of what is happening in the complex, more-or-less real world instructional setting in which a design study is conducted. An ontological innovation is of value to the extent that it enables us to see and account for patterns and regularities that can inform pedagogical and design decisions. In this regard, we note that interpretations of classroom events reflect suppositions and assumptions about learning, teaching, and mathematics as well as a range of issues typically subsumed under the heading of affective factors such as students' dispositions towards and interest in engaging in mathematical activity. These suppositions and assumptions are, in part, embodied in ontological assumptions, and thus ontology is reflected in the interpretations we make of classroom events. Although ontological innovations may not determine particular pedagogical and design decisions in any detail, they can profoundly shape the things we see, and thus the data on which those decisions are based.

Previously, and in this article generally, we emphasized the contributions ontological innovation can make to design experiments. But we also believe design experiments are excellent contexts in which to do the work of ontological innovation. In particular, design experiments

1. Promote *grounding* of theoretical constructs in real-world experiences.
2. Foster the development of *useful* constructs.
3. Provide multiple exposures to empirical test that aid in the difficult and extended work of *refinement*.
4. By the same token, help develop constructs that are *robust* in their application across variations in context.

In presenting the two case studies, we illustrated that the complexity of instructional settings often drives the formulation of new theoretical entities. As the development of the categories of meta-representational competence and of sociomathematical norms were both occasioned by surprising observations, we should clarify that this is a common but by no means necessary story. Work on an ontological innovation might, for example, be initiated by the anticipation of an interpretive challenge in an upcoming design experiment. For example, can we operationalize, know how to see, “a student’s concept” in classroom discussions? In such a case, preparation for the experiment and for the development of an innovation might include exploring potentially relevant theoretical constructs in other disciplines. One might think that one could just import those other-disciplinary constructs, but in our experience that is rare. Instead, they serve as resources and scholarly contextualization for our own work of innovation.

Initial invention or discovery in a narrow sense is a small part of ontological innovation. Both case studies emphasize new theoretical entities as *loci for refinement* of many sorts. Typically, our conceptualization of a new theoretical category needs to be revised and refined as it is used and proves to be unduly simplistic, overly restrictive, or inadequate. There are almost always contextual differences that are instructionally important (e.g., between motion representations and representations of spatially distributed data). Sometimes, new constructs wind up doing part of, but not all of what we had initially hoped (e.g., complementing sociomathematical norms with the idea of mathematical practices). The need for further study is almost a foregone conclusion precisely because the posited existence of new theoretical entities is consequential. While retrospective studies may advance our understanding, perhaps even by beginning an ontological innovation, many further cycles of conjecture and implementation are typically needed both to specify the nature of new entities, and also to develop a rich understanding of details that affect instruction differently in different contexts. It is in the course of this process that the innovation is operationalized and standards of evidence are clarified so that other researchers can both use the category with some fidelity and can monitor analyses in which it has been employed.

In addition to refinement and operationalization, we briefly illustrated how collateral studies can be affected by ontological innovation. Study of teacher strategies and student engagement were stimulated and contextualized by the relevant ontological innovations in both our case studies. These developments serve to fur-

ther emphasize the programmatic nature of design research that involves the formulation of ontological innovations.

Making ontological innovations is plainly difficult. We have refrained from narrow prescriptions about how to develop or validate them in part because that encroaches on some of the deepest issues of “how to do (theoretical) science well.” Even if we had unlimited expository space, that is a task for a larger community.

We have taken pains to acknowledge that an ontological innovation stands at some distance from many elements of instructional design. “Managing the gap” between theory and what we must do to teach is an essential part of the task. It does not advance either science or instruction to pretend that theories are encompassing and highly prescriptive of instruction. As we have argued, theory development in general, and ontological innovations in particular, are nonetheless critical for the long-term scientific health and practical power of design-related educational research.

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COMMENTARIES

If Design-Based Research is the Answer, What is the Question? A Commentary on Collins, Joseph, and Bielaczyc; diSessa and Cobb; and Fishman, Marx, Blumenthal, Krajcik, and Soloway in the *JLS* Special Issue on Design-Based Research

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So that readers can interpret this commentary in light of my overall perspective on design-based research (DBR), I believe that this approach is an important addition to the repertoire of educational scholars, a useful complement to traditional research strategies such as laboratory studies and randomized clinical trials. That said, this commentary is intended as constructive feedback: “tough love” for scholars (including myself) who are proponents of DBR. As discussed below, at this early stage of its evolution less ambitious claims about DBR, more careful delineation of its limited role in the spectrum of experimental methods, the evolution of collective standards for what constitutes quality in DBR, and the focusing of the DBR community’s efforts on issues of concern to practitioners and policymakers all seem appropriate next steps.

WHAT IS DESIGN-BASED RESEARCH: “SEEN THROUGH A GLASS, DARKLY ... ”

Collins et al. (this issue) define DBR thus:

Design experiments bring together two critical pieces in order to guide us to better educational refinement: a design focus and assessment of critical design elements. Ethnography provides qualitative methods for looking carefully at how a design plays out in practice, and how social and contextual variables interact with cognitive variables. Large-scale studies provide quantitative methods for evaluating the effects of independent variables on the dependent variables. Design experiments are contextualized in educational settings, but with a focus on generalizing from those settings to guide the design process. They fill a niche in the array of experimental methods that is needed to improve educational practices. (p. 21)

This is a bounded statement about the role and value of DBR. However, all three articles in this issue—and many of the articles in resources such as the special DBR issue of *Educational Researcher* (Vol. 32, No. 1, January, 2003)—push the edges of this formulation, implying that DBR is a kind of “Swiss Army Knife” for scholars, capable of excelling at a wide range of research purposes. This claim is dubious, because a strong finding from studies of artifacts is that all-purpose designs do a little of everything, but usually do nothing particularly well. The methodological issues discussed below are much of the reason that defining DBR in a realistic way is difficult for its proponents.

METHODOLOGICAL CHALLENGES: LOOKING IN THE DARK ALLEY RATHER THAN UNDER THE STREETLIGHT

The queasiness about DBR felt by many scholars conservative in their research methods stems from the realization that in DBR studies many variables are deliberately and appropriately not controlled, the “treatment” may evolve considerably over time, and even the research methodologies utilized may shift to fit the morphing intervention. Further, to aid with interpretation under these difficult circumstances, in DBR large qualitative and quantitative datasets of various types are often collected by many different participants, introducing substantial problems of alignment, coordination, and analysis. To a methodologist steeped in traditional Campbell and Stanley research strategies, this combination of challenges may seem less a promising new approach to scholarship than a type of study conceived in hell as Sisyphus-like torture for investigators.

As evidence of these problems even in exemplary DBR, Collins et al documents how Brown and Campione's *Fostering a Community of Learners (FCL)* DBR studies generated an unmanageable (and almost unstorable) amount of data. The investigators appropriately worried about selection bias in choosing which cases to highlight and about Hawthorne effects potentially introduced by their unavoidably obtrusive presence in implementing evolving design interventions and in collecting massive mixed-methods data.

DBR advocates respond to these concerns from conservative methodologists with metaphors reminiscent of the apocryphal story of the drunk looking for his lost keys under the streetlight where he can see, rather than in the dark alley where they were dropped. I agree with the aptness of this analogy, but worry about the methods DBR studies use for coping with the darkness of the alley. In research, compensating for chaotic implementations that have high noise-to-signal ratios by collecting huge amounts of data and selectively sifting through it seems off-target, rather like a business model that involves losing money on each sale, but tries to make up the losses on volume.

Further, in much field-based scholarship now conducted (including some DBR studies presented at AERA and ICLS), the research design seems "under-conceptualized and over-methodologized." This is particularly pernicious in students' dissertations from research universities priding themselves on ethnographic and DBR approaches (including, alas, mine own institution). My rationale for the claim of "under-conceptualized" is that, given the title of such a study, most scholars—and practitioners—could write an abstract of the conclusions section without any further information. In other words, the results reported are simple "common sense" for anyone with experience in educational settings.

diSessa and Cobb discuss this issue in detail, arguing that much DBR lacks a strong theoretical foundation and does not attempt to generate findings important for the refinement and evolution of theory. Part of this shortfall may be that the skills of creative designers and the attributes of rigorous scholars have limited overlap. Effective DBR groups have a complex "cognitive ecology" with contradictory tensions: freewheeling, "whatever works" innovation versus controlled, principled variation. People fascinated by artifacts also are often tempted to start with a predetermined "solution" and seek educational problems to which it can be applied, a strategy that frequently leads to under-conceptualized research.

My reason for the claim of "over-methodologized" in these flawed research studies is that only the first five percent or so of the data collected were needed to induce the findings. The remainder of the data gathering and analysis (requiring hundreds or even thousands of hours) was massive overkill. Everything that moved within a 15-foot radius of the phenomenon in question was repeatedly interviewed, videotaped, surveyed, and so forth – and this elephantine effort resulted in the birth of mouse-like insights in their contribution to educational knowledge. Certainly, underconceptualizing and overmethodologizing is not intrinsic to DBR, and some

DBR studies result in valuable findings using elegant collection and analysis strategies. Still, the trap of too little thought and too much method is one into which DBR can easily fall, believing that large amounts of data enable one to build on the shifting sands of uncontrolled variables, morphing interventions, and changing research strategies.

The concept of “design” itself can blur under these conditions. Collins et al. (this issue) reported:

Brown and Campione (1996) made the argument for the FCL design that it is not important which particular activities they have designed are implemented, as long as the principles that they outline in their article are realized. Their argument implies that the principles they enunciate are the key elements of the design, and that any implementation should be evaluated as to how well these principles were implemented. (p. 34)

As discussed later, the epistemological beliefs a DBR group holds strongly influence what those investigators consider to be a “design.” However, I would be hesitant to hire an engineer whose “design” consisted of a statement that the structure built would be based on the principles of Newton’s Laws, gravity, and friction. Such underspecification can easily lead to situations in which DBR presents unfalsifiable propositions, with failures always attributable to defects in implementation rather than flaws in the theory-based design itself.

This raises the interesting issue of when a design should be considered too ineffective to merit further scholarly investment. Investing in designs that show little promise in initial implementations, but nonetheless are the subject of extensive ongoing iterative refinements, is reminiscent of the “epicycles” strategy by which Ptolomaic scholars argued against “Occam’s Razor” Copernican theories for a Sun-centered, rather than Earth-centered, solar system. When is an initial implementation “good enough” to merit further exploration – or an interesting enough failure that the field will benefit from pursuing why a theoretically attractive design strategy produced unexpectedly dismal results? Without standards for determining when to abandon a design approach as unpromising, the DBR field risks being seen as a venue for suboptimal educational strategies endlessly tweaked by their proponents in hopes of an unlikely breakthrough.

In making judgments about the promise of an approach, differentiating a design from its “conditions for success” is important. The effective use of antibiotics illustrates the concept of “conditions for success”: Antibiotics are a powerful “design,” but worshiping the vial that holds them or rubbing the ground-up pills all over one’s body or taking all the pills at once are ineffective strategies for usage – only administering pills at specified intervals works as an implementation strategy. A huge challenge we face in education, and one of the reasons our field makes slower progress than venues like medicine, is the complexity of conditions for success re-

quired in effective interventions; nothing powerful in facilitating learning is as simple as an inoculation in medicine.

As Fishman et al. observes, DBR findings typically show substantial influence of contextual variables in shaping the desirability, practicality, and effectiveness of designs. This raises challenges in longitudinally differentiating a design from its conditions for success. For example, in describing their DBR, Fishman et al. depicts “conditions for success” challenges related to teacher professional development (a common issue in DBR). Resolving implementation problems such as this presents choices about alternative approaches to the iterative evolution of a design. In this particular case, alternative strategies include changing the design so that the intervention is more “teacher-proof,” expanding the design so that extensive teacher professional development is now part of the “treatment,” or abandoning the design as unpromising because its effective use requires a level of knowledge and skill likely unattainable in the typical teaching population for the foreseeable future.

The DBR community frequently responds to “conditions of success” challenges by using the second strategy of expanding the design to include conditions for success that seem problematic. Whatever its tactical merits, this is risky strategically because such “design creep” leads to every study evolving into a full-scale systemic reform initiative rather than bounded research on a particular type of innovation. An expansive approach also can lead to a situation in which an unpromising design is never abandoned because each successive iteration incorporates another “condition for success” attributed as the cause of less than stellar results to date.

Instead, interweaving complementary DBR studies is a better strategy. If teacher professional development (TPD) is a major stumbling block for a particular design intervention, then partnering with a DBR group studying a TPD design is superior from a research perspective to expanding one’s own design. In general, the DBR community needs to discuss the relative merits of alternative strategies for how investigators respond to “conditions for success” challenges that emerge in an implementation.

Answering the various questions about standards and quality I have posed previously requires an evolutionary process of DBR community dialogue about the purpose and processes of DBR; this could include the collective selection of “magic numbers” denoting acceptable scholarship. An example of the latter is the 0.05 level of statistical significance seen as a standard across the research community. A 1-in-20 probability of an outcome attributed to chance rather than to the effects of an intervention is in a sense arbitrary; the level could be 0.10 or 0.01 depending on the ratio of investment to risk the research community chooses to use. So in initial implementations of a design should the DBR community have a “magic number” for effect size (or some comparable metric) to denote which research is promising enough to merit further work? Should we have some similar set of collective arbitrary-but-rational standards for how specified a design intervention must be to be considered “true” DBR? Should we evolve community

guidelines for avoiding “underconceptualized and overmethodologized” studies, indicating what types of educational interventions are worth studying given the level of effort required for DBR and delineating what types of data collection are manageable and appropriate?

As Kuhn’s work on the structure of scientific revolutions indicates, these are questions that any field faces in moving from a preparadigmatic to a paradigmatic stage, becoming more a science and less an art. As Stokes discusses, whether the purpose of the field’s research is seen as curiosity-driven or need-driven also influences the answers to these queries. My sense is that the majority of scholars using DBR want their work to address real educational problems and would like their research seen as legitimate within the larger “scientific” community. This would argue for the field beginning the difficult test of delineating explicit (or at least tacit) standards for potential worth of the knowledge to be gained from a proposed DBR effort, degree to which its design should be specified, effect sizes (or similar metrics) considered promising in initial implementations, effective strategies for data collection and analysis, and so forth.

Developing a collective stance on standards would require more consensus on the epistemology of educational research than currently exists within the DBR community. At the objectivist extreme of the epistemological spectrum, diSessa & Cobb posit that sophisticated research methods could resolve “the apparent complexity and messiness of human interaction” into predictive theories comparable to those in the physical and biological sciences. They see general “orienting frameworks” guiding the development of “grand theories” of cognitive and social processes related to learning, with these theories also informed by targeted “frameworks for action” and “domain-specific instructional theories” that arise from the pragmatic realities of teaching.

At the subjectivist extreme of the epistemological continuum, radical constructivists in education believe that no objective reality exists to be collectively measured and predicted. “Reality” is individually constructed, with that formulative process influenced by the sociocultural setting of the individual. Of course, many DBR investigators are at neither end of this spectrum, but the range of viewpoints “in the middle” is quite broad. For example, most situated learning theorists would place themselves toward the subjectivist side of the continuum, whereas most cognitivists would locate their position closer to the objectivist stance.

This divergence in beliefs is reflected in differing perspectives on what constitutes a “design.” Those who lean toward an objectivist point of view tend to prepare DBR participants to function like astronauts: trained to execute detailed contingency plans that prespecify responses to various situations that may emerge in implementation. In contrast, investigators inclined towards subjectivism see specific individual and social factors overwhelming the contingency plans of any standardized curriculum, instruction, and management strategies, instead preparing

DBR participants to function like physicians who can invent as needed individualized diagnostic protocols to aid learners.

Also, this divergence on what types of epistemological theories underlie the study of human behavior raises another question about standards: how much theory should guide a design for studies of its implementation to merit the term “design-based research”? For example, Fishman et al. cite as exemplary DBR the evolution both of the Carnegie Tutoring Systems and of the Union City, New Jersey systemic reforms. The former is clearly John Anderson’s application of cognitivist theories to a series of iterative designs. However, while Fred Carrigg (the core figure in Union City) is certainly informed by research, my knowledge of this reform suggests that he played a role closer to Edison making systematic trials of light filaments than a scholarly investigator simultaneously deepening theoretical insights and design heuristics. This is another dimension on which the absence of collective standards results in lack of clarity about what is considered effective DBR. Until the DBR community’s divergent perspectives on fundamental issues of epistemology evolve toward common ground, we are unlikely to have a productive dialogue that develops both a common, bounded definition of design-based research and standards for the various dimensions of DBR studies discussed earlier.

INDUCING ONTOLOGICAL INSIGHTS: WHAT QUALIFIES AS “NEW, IMPROVED”?

Collins et al. described DBR as similar to Simon’s design sciences (e.g., aeronautics, acoustics), as opposed to analytically oriented natural sciences such as physics and biology. diSessa and Cobb indicate that ontological insights informing analytic theories can come from well conducted DBR, an inductive bridge in the reverse direction from designers deductively applying theoretical principles in the natural sciences to develop artifacts effective in accomplishing their purposes. “Interventionist ethnography,” in which DBR studies perturb typical learning settings by introducing evocative, theory-influenced designs, then draw out implications for new theoretical ontologies of phenomena, is an intriguing way to frame a potential advantage of DBR.

Certainly, implementing well-formulated designs has led to surprising findings about the quality of thought, motivation, and action that children can accomplish at various developmental levels. For example, NSF-funded projects such as Kaput’s SimCalc, Horwitz’s BioLogica, and Rubin’s TapeMeasure have documented typical middle school students mastering science and mathematics previously thought appropriate to teach only to “majors” in those disciplines at the advanced high school or college levels. Without the technology-based designs that evoked and scaffolded this learning on the part of students, the research community would

likely have continued underestimating learners' potential capacity for thought and action.

Within the DBR community, what standards should we use to judge the likely value of a proposed ontological insight? One can imagine a continuum with one extreme representing discovery of an ontological typology that is novel across all fields of human knowledge (a very rare occurrence), and the other extreme representing a pedestrian transfer of a minor ontological category to a small community of researchers studying a narrow topic (unfortunately, relatively common in the scholarly literature). One can imagine a second continuum representing potential value, with one extreme indicating probable, powerful leverage for improvement the insight might provide across many settings and the other extreme denoting likely minor value in only a few contexts. How in the multi-dimensional space created by these continua—and others representing additional metrics the DBR community might add—should the DBR community define its standards for a “substantial” ontological contribution?

One can ask how diSessa and Cobb would position their own work in such a value-space. To illustrate the process of determination, I think the existence of meta-representational competence and interest on the part of students is a more novel finding than the existence of sociomathematical norms. (In fact, I think determining the absence of the latter would have been surprising). On the other hand, to the extent that insights from studying sociomathematical norms generalize across developmental levels and disciplines, I think this work is potentially more powerful “on balance” than likely applications of students' meta-representational skills—although both topics seem useful and intriguing.

This is the type of grounded dialogue in which we as a community should engage, using case examples of our own work to build collective understandings of standards. For example, would we rank Brown and Campione's work (e.g., communities of practice, developmental corridors) as providing novel examples of ontological innovations, or are these relatively familiar ideas in other fields of knowledge usefully applied in the FCL studies? To put my own stake in the ground, I would argue the latter rather than the former; but others might disagree, especially if more aspects of quality are added to the simplistic two-dimensional evaluative system I have sketched for purposes of illustration.

An important issue in ontologically driven research is how to get beyond the taxonomic stage to the attribution of possible causal interrelationships. That much of educational research is still descriptive is not surprising, given that our field is “pre-paradigmatic” in some fundamental ways. However, the ultimate value of ontologic innovation in education stems from the ability to apply these insights to improve student motivation and learning outcomes, which—as diSessa & Cobb describe—requires evolving taxonomic insights into theoretical formulations and design heuristics.

DESIGN-BASED RESEARCH IN CONTEXT: NO SCIENCE LEFT BEHIND

Two of the three articles in this JLS special issue discuss DBR in a “context independent” manner; in contrast, Fishman et al. relates this form of scholarship to issues of educational utility and policy. What issues in the evolution of DBR are highlighted by the current political situation of the No Child Left Behind (NCLB) federal initiatives, guided by the national Institute for Educational Sciences formulation of “scientifically based research” (SBR)? Much has been written about the defects of NCLB and SBR, including a critique by this author. Reviewing those debates is beyond the scope of this commentary, but some issues for DBR are especially salient in this political/ideological context.

Fishman et al. observes that cognitively oriented technology innovations are not widely used in schools. These authors correctly focus on scalability and sustainability as a major problem for our designs (although easily generalizable and transferable innovations may still face substantial challenges due to ideological issues and popular misconceptions about the nature of learning). A research conference was recently held at Harvard’s Graduate School of Education on these and related topics involved in “scaling up” technology-based educational innovations (<http://www.gse.harvard.edu/scalingup/>), and my discussion is based in part on the dialogues at that event.

How scholars, practitioners, and policymakers view the value of DBR is strongly influenced by NCLB and SBR. First, practitioners struggling with very demanding, narrow criteria to which they are held accountable (e.g., students classified as “mildly mentally retarded” are expected to perform at grade level on high stakes tests) are primarily interested in promising interventions that could reach subpopulations for whom conventional instructional practices are ineffective. They are typically and understandably less interested in contributing to an iterative process of theory-based development for constantly morphing, loosely specified interventions with complex conditions for success, obtrusive and intensive data collection strategies, and no guarantee of effectiveness.

This divergence in motives between DBR investigators and the contexts in which they must operate often leads to problems. Do we as scholars start with what practitioners identify as problems (e.g., how can unqualified reading teachers be trained to use a didactic teacher-proof approach to compensate for their weaknesses), or with what we believe are theoretically promising interventions that would require transformation of the current system to effectively implement? To the extent that we emphasize the latter and refuse to soil our ivory-tower hands with the former, we risk being seen as nuisances more than helpers, as dilettantes studying the arrangement of deck chairs while the Titanic is sinking. Defining a middle ground that starts with practitioners’ issues, but then helps them evolve their thinking towards transformative approaches, requires sophistication and patience; yet this is the climate within which DBR must now function if we expect re-

spect, resources, and valid venues (as opposed to boutique settings) within which to conduct our studies.

Further, how do we choose when faced with the alternatives of modifying a design in ways that pose interesting issues for the refinement of theory, versus altering the intervention to increase its usability, scalability, and sustainability? As Fishman et al describes, these objectives are not necessarily opposite, but neither are they completely congruent. Should a successful DBR project “split” at some point, with a theory-driven subgroup continuing evolution of the intervention in “hot-house” settings, while a subgroup “closer to contractors than engineers or scientists” produces a stable, sustainable, scalable variant for widespread usage? Because we know the “throw the blueprints over the transom to the assembly line foreman” approach to technology transfer is ineffective, how could the research community modify its incentive systems to make the latter role attractive to scholars and to build capacity for the complex skills involved? Simply making funding available is insufficient; of the four quadrants that define research types in NSF’s Research on Learning and Education program (ROLE), quadrant four on large-scale implementations is undersubscribed relative to the more conventional forms of research funded by quadrants one, two, and three.

Put bluntly, neither policymakers nor practitioners want what the DBR community is selling right now. We appropriately don’t match the narrow conceptions of science currently in vogue at the federal level, but have much internal standard-setting to accomplish before we can put forward a defensible alternative. Nor would that development of standards fully satisfy what practitioners are understandably seeking—yet, if we view them as partners with valuable knowledge for co-design rather than as experimental subjects to manipulate, we must respect their viewpoints on the objectives, process, and outcomes of DBR, even if this diverts resources from our exegesis of theory and our push for transformative change.

CONCLUSION: WALKING OUR TALK

This commentary has posed more questions than answers, and yet the aggregated questions implicitly suggest an answer: the DBR community should engage in substantial collective reflection on setting standards that improve the quality of DBR and on refining our innovations so that they matter to the audiences for our research. The process of DBR is itself a design, and its evolution should proceed using the same design and research methods we apply to the innovations this method generates. Reaching consensus in such a dialogue is vital to the success of DBR; but this will not be easy, because differing assumptions, beliefs and values underlie apparently similar efforts in DBR. Still, I am optimistic that, with effort, we can develop a process far better than a Swiss Army Knife to aid us in searching that Dark Alley.

Design Research in Education: Yes, but is it Methodological?

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It is difficult to write a commentary on emerging methods. One wishes to avoid the temptation of the negative critic (Dewey, 1980):

... the critic, guided by personal predilection or more often by partisan conventionalism, will take some one procedure as his criterion or judgment and condemn all deviations from it as departures from art [read: research] itself. He then misses the point of all art, the unity of form and matter, and misses it because he lacks adequate sympathy, in his natural and acquired one-sidedness, with the immense variety of interactions between the live creature and his world. (p. 313)

Equally, one does not want to demonize those who are thoughtful about the use of methods such as randomized trials (see Mosteller & Boruch, 2002), techniques against which design study methods are wrongly pitted.

I hope I make clear that I see design study methods as a protean expression of research “impressionists” whose art will illuminate and enrich the “studio” research art we revered until the fertile contributions of Collins, Brown, Hawkins, and Pea in the early 1990s. The next task is to establish the logos of design research so that we can argue, methodologically, for the scientific warrants for its claims. Thus, although it is important for the health of a set of methods that profusion precede pruning (Kelly, 2003), it is time for the pruning to begin.

This collection of articles provides evidence for the vigor of design studies in education. Each of the articles provides a substantial contribution to prior efforts (e.g., Kelly, 2003), and to articles that have so far primarily appeared in edited books (e.g., Collins, 1999). Other attempts at creating design-based formative re-

search methods exist (Reigeluth, 1989; English, & Reigeluth, 1996; Reigeluth & Frick, 1999). A form of design research in education has been active in Europe for some time, called “developmental research” (e.g., van den Akker, Branch, Gustafson, Nieveen, & Plomp, 1999).

Collins, Joseph, and Bielaczyc (this issue) focus on the later work of Brown (a co-originator of the concept of design experiments in education) and current research to illustrate how insights from design experiments can guide a series of studies. The article by diSessa and Cobb labels an explanatory construct an “ontological innovation” and shows how it can support and propel a number of related studies. The article by Fishman et al. pushes not only design study research, but educational research, generally, to consider the outer reaches of the model of design and research stages described in Bannan-Ritland (2003); viz., diffusion, adoption, and adaptation of innovative practices. If the focus on the “ultimate fate” of adoption by others of the intervention/artifact is absent during a study or across a series of studies, then local, but not broader impact, is likely to be demonstrated.

Below, I begin with a brief discussion on the classification of design studies and then highlight the methodological challenges that need to be addressed if we are to develop design studies from a loose set of methods into a rigorous methodology.

CLASSIFICATION AND SCOPE OF DESIGN STUDIES

Engineering metaphors for design studies are proving useful in science and in mathematics education (diSessa & Cobb, this issue), in the development and sustenance of communities of learners and passion curricula (Collins, et al., this issue), and in cognitively informed technology interventions and their survival in the real world of school systems (Fishman et al., this issue).

One way to classify design studies is by characterizing their “outcomes” or “products.” Some design studies wish to inform us about a model of practice, others about learning, still others about the design and use of a new piece of software or “learning environments.” As Simon (1969) noted, design is not design without some form of designed artifact—even if the goal of the artifact(s) is to advance a different outcome such as new theory. Let me quickly add that the concept of an “artifact” need not be “concrete” such as a computer program, it might describe aspects of “activity structures, institutions, scaffolds and curricula” (Design-Based Research Collective, 2003, p. 6). Nevertheless, design remains a *transitive* verb. In my opinion, design studies should produce an artifact that outlasts the study and can be adopted, adapted, and used by others (e.g., either researchers or teachers); otherwise, the fact that the study used an *iterative* process simply characterizes the procedures that were followed.

Artifacts

Process as Artifact: Teaching Experiments

Some educational researchers work directly with teachers and the teaching process; in fact, the process of engagement between the teacher and student (with the researcher sometimes playing the role of teacher) can be claimed to be the object of study (for a history of this method and its roots in Russian teaching experiments, see Steffe and Thompson, 2000). Note that the word “experiment” is used advisedly in teaching experiment methods. For example, Steffe and Thompson (2000) and Confrey and Lachance (2000) refer explicitly to the testing, respectively, of hypotheses and conjectures, though the methodological implication of these words deserves study (see following paragraphs).

The methods of the teaching experiment may be subdivided into variants such as “multi-tiered teaching experiments” (Lesh & Kelly, 2000); “transformative teaching experiments” (Confrey & Lachance, 2000); and “collaborative teaching experiments” (Cobb, 2000). I value the adjective “teaching” in the label and prefer it to the generic label “design experiment.”

A further reason to retain the adjective “teaching” is that these variants of design studies are centrally concerned with curricular subject matter — in the above cases with the teaching and learning of mathematics. For example, one of the touchstones for judging progress in a teaching experiment on the psychological concept of learning transfer was students’ constructions of mathematical concepts (e.g., Lobato & Siebert, 2002; Lobato & Thanheiser, 2002).

Software as Artifact

At a different pole are studies that have a focus on the design of a model of learning or teaching as reified in a software artifact. The design of such artifacts usually involves engineering a broader “learning environment.” Space limitations preclude more than a listing of a small number of studies that may fall under this umbrella (e.g., the *Progress Portfolio*, Loh, Radinsky, Russell, Gomez, Reiser & Edelson, 1998; *Worldwatcher*, Edelson, Gordin & Pea, 1999; *LiteracyAccess Online*, Bannan-Ritland, 2003; *Genscope/Biologica*, Hickey, Kindfield, Horwitz, & Christie, 1999; *SimCalc*, Roschelle, Kaput & Stroup, 2000; and *Jasper Series*, Cognition and Technology Group at Vanderbilt, 1997).

The point here is not that these studies produce software per se. Rather, that the exploration of research questions about learning (of students, teachers and researchers) are reified, explored, and tested by the design and use of the software/learning environment. Indeed, some claim that some student cognitions may not emerge without a designed environment (e.g., Roschelle et al. 2000).

Centrally, the existence of a software artifact or conceptual framework (e.g., from a teaching experiment) allows for its use and testing elsewhere. Its later adop-

tion elsewhere is not a trivial matter, of course, and this important problem has not yet been the focus of attention of design studies (Fishman et al., this issue).

DESIGN STUDIES: A SET OF METHODS OR A METHODOLOGY?

Proponents of design studies claim that these studies can elicit a generative framework, advance our understanding of disciplinary knowledge, serve as an incubator for new research techniques, advance our skill in designing learning environments or lead to better instrumentation – all features that may prove useful in the same or future studies.

If these goals are going to be more fully realized, however, design studies must develop from a loose set of methods to a methodology. A method is a procedure, a process, a set of steps to follow. Design studies to date have been described primarily using a set of *process* descriptors (e.g., these studies are interventionist, iterative, process focused, collaborative, multileveled, utility oriented, and theory driven, see Shavelson, Phillips, Towne & Feuer, 2003). Unless this set of procedures is under-girded by a conceptual structure that forms the basis for the warrants for their claims, design study methods do not constitute a methodology and will contribute only haphazardly to an aggregative science of learning. Please note that I am not suggesting for a moment that design researchers are unaware of any of the following issues. Rather, I am suggesting that design studies are a new form of research actions that require a rethinking of these standards. A mature methodology will have certain characteristics, including, but not limited to the following:

Argumentative Grammar

An argumentative grammar is the logic that guides the use of a method and that supports reasoning about its data. It supplies the *logos* (reason, rationale) in the methodology (method + logos)¹ and is the basis for the *warrant* for the claims that arise. Thus, for randomized field trials, the *logos* draws from the structure of the arguments provided by R. A. Fisher concerning small sample analysis, statistical reasoning, and randomization procedures (see, Bennett, 1990).

The argumentative grammar of randomized field trials can be described *separately* from its instantiation in any given study so that the logic of a proposed study and its later claims can be criticized. Thus, many reviewers reject studies not on the choice of method (procedure), but on their violation of the underlying logos that one expects to see with that choice of method. Indeed, one may argue that the ele-

¹The same root appears in the names of many sciences (e.g., bio-logos, psyche-logos, socio-logos).

gance of the argumentative grammar provided by Fisher (including, as he does, a homely example about drinking tea) is attractive to policymakers and reassures them that randomized field trials provide a “scientific” method in education.

What, therefore, is the *logos* of design studies in education? What is the grammar that cuts across the series of studies as they occur in different fields? Where is the “separable” structure that justifies collecting certain data and not other data and under what conditions? What guides the reasoning with these data to make a plausible argument? Until we can be clear about their argumentative grammar, design study methods lack a basis for warrant for their claims. Please note that a simple assertion that design studies use “grounded theory” or “thick description” with a mere citation to some complex source does not constitute an acceptable basis for according design studies the status of a methodology. Indeed, in my opinion, these bibliographic sources inadequately address the data-analytic methods and problems of design studies.

Contributions to Problems of Demarcation or Meaningfulness

Problem of Demarcation

This section considers the character of the claims that can be derived from design study methods (see Runkel & McGrath, 1972; Kelly & Sloane, 2003). In order to take their place as a methodology, design studies must clarify how their results contribute primarily to the problem of demarcation or to the problem of meaningfulness or to both. The problem of demarcation describes the application of argumentative techniques to differentiate scientific claims from those of pseudoscience, or from metaphysical claims (Popper, 1965). Scientific arguments backed by empirical methods can, for example, debunk astrology, yet refine astronomy (see also, National Academy of Sciences, 1998).

The question for design study researchers is, “Can this new set of methods establish boundaries (demarcations) between sound and unsound claims about learning and teaching?” If design studies cannot develop in ways similar to randomized field trials, survey methods, or rigorous ethnography (to name some mature methodologies), then their contribution to the problem of demarcation may be limited. I read the following list of descriptors of design studies by Collins (1999) to suggest that these methods are currently unequal to the task. According to Collins (1999), design studies:

1. Conduct research in messy non-laboratory settings.
2. Conduct research involving many dependent variables.
3. Characterize the situation as opposed to controlling variables.
4. Use flexible design revision instead of staying with fixed procedures.
5. Value social interaction over social isolation in learning.

6. Generate profiles rather than test hypotheses.
7. Value participants' input for design and analysis rather than relying solely on the judgments of the researcher.

When one foregoes experimental controls, how can one generalize to other settings regardless of how rich are the local descriptions? When multiple dependent variables are used, how does one tease complex interactions apart, and are causal attributions plausible? While allowing for flexible design revision may make sense in engineering product design, what are the consequences when this method is used in educational settings where the "product" is something more ephemeral, called learning, and the status of the claims are intended to be general? Although it is attractive to consider social interaction during a design experiment, is it not also important to forefront individual cognition and so link to other areas of science (e.g., McCandliss, Kalchman & Bryant, 2003)?

Although it is attractive to develop profiles of the learner along a number of dimensions, what is the basis for selecting these profiles and not others that might appear equally plausible since the data are so noisy? Should the testing of hypotheses be eliminated entirely, or what does "hypothesis testing" mean in such a fluid milieu? Finally, although much can be said in favor of coparticipant involvement during research, what impact does the involvement of many voices have on the inferential character of any findings that result from the study? Whose voice takes precedence in scientific claims and why?

Generalizations Over Actors

This section on generalizations draws heavily on Runkel and McGrath (1972). Since many of design studies focus on just a few subjects (often students), there is a problem of generalization of the descriptions to a larger N (e.g., other students in the same class who are not the focus of study, students in non intervention classes; students in other schools or settings, and so on). The method thus lacks the sampling and descriptive power of a survey, making claims about "actors" (in this case students) weak.

The generalization-to-actors problem is confounded by the fact that the sample of cognitive behavior of students is in response to a targeted perturbation (i.e., as part of a "design experiment"), which weakens generalization to *cognitions* in unstudied classmates, cognitions in the larger student body, and to the normal cognitions of children outside of school settings. In other words, the sampling problem over actors is compounded.

Generalizations Over Behaviors

In design studies, the notion of experimental control is often consciously violated so as to serve local emerging needs. Thus, since there is little attempt at insti-

tuting either structural or statistical controls in design studies (e.g., the use of some form of experimental design or covariation analyses), then the warrant for making causal claims about behavior (in the face of untested alternative hypotheses) is weak (Shavelson et al., 2003).²

Generalizations Over Contexts

It might be argued that claims about *context* are legitimate for design studies, but notice, again, that these contextual descriptions are not about “normal life in classrooms;” rather, the “context” is a unique, engineered, acted-upon designed environment. In other words, the findings that might come from a more standard ethnography in which intervention and iteration is not central do not typically apply. As such, Barab, Thomas, Dodge, Carteaux, and Tuzun (in press) introduced the term “design ethnography.” Whereas, the basic ethnographer describes and interprets a particular culture and the applied or critical ethnographer (potentially) critiques and changes the local culture, the design ethnographer goes one step further; “actually reifying the social commitments and emergent critique into an artifact that can be used by people in contexts beyond those in which the initial ethnographic work was carried out.” It remains the shared responsibility of the design research community in education to help explicitly spell out the methodological strengths and weakness of the emerging design ethnography methods.

Problem of Meaningfulness

If design study methods are not yet developed sufficiently to construct generalizations that contribute to the problem of demarcation, does that mean that they have no value or can provide no generalizable statements? On the contrary, important scientific discoveries have not typically come about as a result of a scientist slavishly following the “scientific method” (Holton, 1998). Indeed, Medawar (1991) argued:

[T]he scientific article is a fraud in the sense that it does give a totally misleading narrative of the processes of thought that go into the making of scientific discoveries. The inductive format of the scientific article should be discarded. The discussion which in the traditional scientific article goes last should surely come at the beginning. The scientific facts and scientific acts should follow the discussion, and scientists should not be ashamed to admit, as many of them apparently *are* ashamed to admit, that hypotheses appear in their minds along uncharted byways of thought; that they are imaginative and inspirational in character; that they are indeed adventures of

²It should be noted that the criticism made by Shavelson et al. (2003) was weakened by its attack on the narrative methods preferred by Bruner, which do not describe adequately the data-analytic techniques used by, for example, Cobb and his colleagues (e.g., McClain, 2002; McClain & Cobb, 2001).

the mind. What, after all, is the good of scientists reproaching others for their neglect of, or indifference to, the scientific style of thinking they set such store by, if their own writings show that they themselves have no clear understanding of it? (p. 233)

In other words, design studies, particularly to the extent that they are hypothesis and framework *generating*, may be viewed as contributing to model *formulation* (rather than for model estimation or model validation, see Sloane & Gorard, 2003). Model and hypothesis generation is a crucial part of conducting a worthwhile scientific investigation. It does not represent some “pre-scientific” messing around that should be accorded little status. For example, Russell Hulse (Nobel Laureate in physics, 1993) noted at a talk at the National Science Foundation (Hulse, 2003), that an infinite number of hypotheses may be tested. This machination of testing hypotheses (i.e., the mechanical application of the “scientific method”) does not, by itself, create useful knowledge. It is important, rather, to test *powerful* (i.e., construct-advancing) hypotheses, which often emerge from studies directed primarily at the problem of meaningfulness.

In this sense, the recent National Research Council report (NRC, 2002) is clear about the function of good questions (hypotheses) in advancing science. But, apart from a footnote that, sometimes, good observations come “out of the blue,” there is little discussion or direction on how a researcher might *generate* one (p. 57). There is a suggestion in this report that good questions can be gleaned from a literature review, but this exhortation begs the question of how the good questions in the literature emerge. The report does not explain how one might go *beyond* any literature’s constructs. Indeed, the models guiding a given scientific literature (which are always tentative) may be inhibitors to progress. Müller, corecipient of the Nobel Prize for the discovery of high-temperature superconductors, decided after a thorough, but fruitless, literature review that, “I just don’t talk to the theoreticians. They just held me back” (cited in Holton, 1998, p. xxiv).

Note that I am, of course, *not* advocating dispensing with literature reviews; rather, agreeing with the authors in this special issue that the constructs that animate any competent literature review often emerge in design contexts and that the character of these studies may promote the identification and growth of new ideas and constructs.

In this way, design studies can contribute to the problem of meaningfulness, a problem that precedes and is foundational for later contributions to addressing the problem of demarcation. Indeed, how can one establish demarcations for claims if the scientific concepts under consideration are sparse or unclear?

Generalization of conceptual frameworks or articulation strategies. The contribution of design studies to generalization, as I see it, is primarily in their contribution to the thinking of other researchers who must conduct research in newly explored areas. Because the conceptual frameworks in design

studies grow authentically and are grounded in the lived experience of teachers and students, they often appeal to researchers who work in similar contexts. Thus, a mathematics education researcher may find that the concept of *socio-mathematical norms* attractive because it may guide observation, suggest variables to study, or help make sense of data. Later studies, using different methodologies may extend generalizations across actors, behaviors, and context, and extend the work into the purview of studies of diffusion of innovations (Rogers, 1995).

Use of Technical Language

If the discussions about design studies remain within a small community, then problems with terminology are not that important. However, when the work of a research community becomes the focus of attention of a larger community of researchers, then the use of terms that are transparent or at least understandable by the larger community is highly desirable.

One problematic word, for me, is “theory.” According to the National Academy of Sciences (NAS, 1998), “In science, a [theory is] a well-substantiated explanation of some aspect of the natural world that can incorporate facts, laws, inferences, and tested hypotheses” (p. 5). Scientists use the word, guardedly, to describe a hard-won consensus statement or model. In the case of the teaching of evolution (the topic of NAS, 1998), for example, the authors point out the difficulty in communicating with the public when the concept of “theory” has been debased in the vernacular to mean mere opinion.

The word “theory” surfaced in various forms in the writings of many authors in the *Educational Researcher* (Kelly, 2003) special issue, and reappears in this special issue. In fact, the article by diSessa and Cobb devotes space to explicating this issue. I find their discussion most important. Because my concern here is with the reading of educational research studies by those outside of educational research (particularly from the hard sciences), I prefer to use word “theory” and “theoretical” very sparingly and instead to use working words (as follows) so that the comprehensibility of the piece to those in other traditions is (hopefully) increased. I adopt this perspective from the experience of having run many educational research panels at NSF that involved scientists from outside of education.

Because design studies are in their (relative) infancy, and we still rely almost exclusively on text to convey meaning, I advocate the use of *working* words – words that have a task to perform and that have a clear intension, such as “hypothesis,” “conjecture,” “observation,” “model,” “framework,” “explanation,” “belief,” “rationale,” “logical or argumentative structure,” “perspective,” “analysis, or “synthesis.” Note, for example, the work Confrey and Lachance (2000) get out of the word “conjecture.”

Definition is particularly useful since it reminds us to apply concepts such as *extension* and *intension*. Thus, design studies should not only provide examples

(which contribute to extension), but also point to the inherent characteristics that cut across or distinguish exemplars (which contribute to intension). Thus, the conscious use of the connotative/denotative aspects of the word *definition* advise descriptive studies to avoid falling into Stokes' (1997) *Peterson's quadrant*,³ in which "interesting" or "fascinating" protocols are presented for their own sake (akin to distinctive markings on birds, which contributes solely to extension). Instead, by paying more attention to intension, the work can begin to grow into one of the more productive quadrants, for example, Pasteur's quadrant (see Martinez, Whang, & Hamilton, 2003).

Restrictions on Researcher Bias

Researcher bias is inevitable. It is not a moral flaw; it is an expression of human judgment (e.g., Gigerenzer, Todd, & ABC Research Group, 1999; Goldstein & Hogarth, 1997). Nevertheless, mature methodologies have guidelines for minimizing its occurrence. To illustrate how all-encompassing bias can be, consider the following statement from a prominent scientist who was convinced of the obviousness of an explanatory framework that proved to be nonviable: "Nothing of all this can be considered as hypothesis or private opinion; on the contrary, they are truths which, in order to be made clear, only require attention and the observation of facts." (Lamarck, 1809, defending Lamarckism, cited in NAS, 1998, p. 96).

For example, what operational caveats exist to guide the sensible use of design study methods? How does the researcher select episodes for analysis? How does he or she decide what aspects of these are meaningful? What about contrary instances? What about outliers? What is not being said about the miles of videotape left unwatched and student "artifacts" unread? What does triangulation mean in this context? And what is the function of inter-rater reliability?

Further, whose view of the "environment" should guide claims: the observer or the organism? A leading proponent of the need to take not only researchers' but actors' (e.g., students') constructions into account for advancing design studies is Lobato (2003). Her reconceptualization of the transfer of learning literature can guide the design and analysis of measures within a design study. Further, it advises against an uncritical acceptance of externally imposed standardized tests as the *sine qua non* of measures of "scientific" research in education (see also Confrey et al., 2001).

Stage-appropriate Use of Methods

I discussed, previously, the need to fit claims into the proper warrant category (i.e., claims about actors, behaviors or contexts). A related issue arises for design stud-

³A label I like despite Stokes's protest (p. 75).

ies: *stage-appropriate* use of research methods. If we consider an entire program of research as a design act (Bannan-Ritland, 2003), then different studies may be seen as belonging in stages within a larger trajectory that animates the program or portfolio of work. Thus, research methods and data-analytic techniques appropriate for claims about an emerging framework may be inappropriate for claims that are being subjected to a stringent test, or research claims about diffusion of innovations (see Zaritsky, Kelly, Flowers, Rogers, & O'Neill, 2003). What is being suggested here is that part of the methodological work for design studies is to clarify in which stage and for what purposes methods are appropriate and inappropriate.

Balancing Contingent With Necessary Claims

Consider the example of airplane design in diSessa and Cobb (this issue). It is true that there is much freedom in the design of a plane (it could be a Boeing 777 or a Fokker). What moves aeronautics from craft know-how, to engineering knowledge, to the level of a science is the investigation, discovery and modeling of the coupling between the *contingent* (the parts of the design that are to some extent arbitrary) and the *necessary* (e.g., the proper application of Bernoulli's principle to the design of the wing—or similar airfoil). Note that “beneath” Bernoulli's principles lie further necessary principles including fluid mechanics, gravity, and so on.

I believe that we must support a similar trajectory for design research in education. Design studies occur in real-world settings. They remain open to the vicissitudes of the actors, their behaviors and the complexity of the current context. In other words, the raw material of design studies is inescapably contingent: one action begets the next, and so on, often in unpredictable (and perhaps unrepeatable) ways. The researcher responds to this aleatory ebullience and tries to express its essence parsimoniously.

Until we can begin to discover and model the necessary components of learning (e.g., based in cognitive neuroscience: e.g., Dehaene, 1997; OECD, 2002), design study output is prone to be dismissed as mere “narrative”⁴ (e.g. Shavelson et al., 2003) – that the data are merely “anecdotal.” Our descriptions cannot, over the long haul, remain at the level of contingency. When we move toward identifying what is *necessary* in the situation, then the methodology becomes more mature, scientifically. It is in the connection of the contingent to the necessary where the real work of theory-building occurs, in my opinion.

As one strategy, other researchers can be recruited to explore the *necessary* aspects of the frameworks and findings of design studies. Three illuminating examples are provided in McCandliss et al., (2003). Second, design study methods may be re-construed so that they are able to “hand off” hypotheses, models, intuitions,

⁴Design study researchers need not be overly defensive about uses of “narrative.” The history and philosophy of science demonstrates that prospective scientific practice is shot through with narrative and the scientific imagination is, according to Holton (1998), fundamentally thematic.

and so forth, to researchers who want to develop other aspects of model building (Sloane & Gorard, 2003), or who work at later stages of the research-as-design process (Bannan-Ritland, 2003). Ultimately, if this set of methods is to advance, it must collaborate with other methodologists in the social sciences and build cross-disciplinary initiatives to the behavioral, cognitive and social sciences.

Producing Usable Knowledge

Educational researchers choose to answer to a variety of policymakers and stakeholders. Not only are their scientific claims difficult to substantiate (in no small part due to the limitations of randomized trials, survey and ethnographic methods in real settings), but they must “wrap” their principled insights in some curricular form so it can fit into the real work of teachers and learners.

Further, educational researchers act within and perturb ongoing systems, which greatly complexifies the scientific problem. In one case, even when a randomized field trial was anticipated, school-level resources contributed to the difficulties faced in bringing a systems-thinking curriculum to fruition and worked against an assessment of its effectiveness (Cline & Mandinach, 2000; see also, Mandinach & Cline, 2000). The laboratory scientist’s “error variance” is the educational researcher’s reality.

The existence of a research team in a classroom is always a perturbation. We can see this observation borne out in a multi-level sense in Fishman et al. (this issue). Further, the impact of these perturbations at the school system level can be detrimental even to the continuance of a research endeavor (e.g., Confrey, Bell & Carrejo, 2001).

What this implies, methodologically, is that design study researchers must consider criteria for success that exceed satisfying the criteria for “scientific” claims. Ultimately, design research may be seen as a practical endeavor. Can educational research produce knowledge artifacts that provide solutions for teachers and learners that are efficiently, workable, economical and do not entail a significant “switching cost” from current practice? We must consider later (and unsupported) adoption and adaptation of our research products in the set of evaluative criteria. Methodologically, we will have to study and borrow from other disciplines (e.g., Rogers, 1995; Zaitsky et al., 2003).

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